CRYSTAL STRUCTURE OF LOCALIZED QUANTUM UNIPOTENT COORDINATE CATEGORY

MASAKI KASHIWARA AND TOSHIKI NAKASHIMA

ABSTRACT. A localized quantum unipotent coordinate category $\widetilde{\mathscr{C}_w}$ associated with a Weyl group element w is a rigid monoidal category which is obtained by applying the localization process to the category $\mathscr{C}_w$, a subcategory of the category R-gmod of finite-dimensional graded modules of a quiver Hecke algebra R. We shall show that the family of the isomorphism classes (up to grading shifts) of simple objects in $\widetilde{\mathscr{C}_w}$ possesses a crystal structure and it is isomorphic to the cellular crystal associated with w. As an application of this result, we shall show the connectedness of the crystal graph of an arbitrary cellular crystal.

CONTENTS

Date: September 2, 2025.

2020 *Mathematics Subject Classification.* 05E10, 18M05, 18M15, 16D90, 16T20, 17B37, 81R10.

Key words and phrases. Localization, Monoidal categories, Quiver Hecke algebra, Crystals, Affinization, R-matrix.

The research of M. Kashiwara was supported by Grant-in-Aid for Scientific Research (B) 23K20206, Japan Society for the Promotion of Science.

The research of T.Nakashima is supported by Grant-in-Aid for Scientific Research (C) 20K03564, Japan Society for the Promotion of Science.

1. Introduction

One of the most potent tools in the crystal base theory ([7]) is the so-called "crystal operators" $\tilde{e}_i, \tilde{f}_i$. These operators act on $U_q(\mathfrak{g})$-modules or the nilpotent subalgebra $U_q^-(\mathfrak{g}) \subset U_q(\mathfrak{g})$, where $U_q(\mathfrak{g})$ is a quantum algebra over the field $\mathbb{Q}(q)$ associated with a symmetrizable Kac-Moody Lie algebra $\mathfrak{g}$. Let (L, B) be the crystal basis of a $U_q(\mathfrak{g})$-module M or $U_q^-(\mathfrak{g})$, where L is a free module over the local subring $\mathcal{A} \subset \mathbb{Q}(q)$ at $q = 0$ and B is a $\mathbb{Q}$-basis of L/qL. The operators $\tilde{e}_i$ and $\tilde{f}_i$ act on L, the quotient space L/qL, and even the basis B. This induces an oriented graph structure on the basis B, known as the "crystal graph". Once the crystal base theory is constructed, a fundamental and significant problem is to realize its crystal graph using various methods, such as tableaux realizations ([17]), path realizations of perfect crystals ([3, 4]), geometric realizations ([19]), categorical realizations ([24],[26]), etc.

On the other hand, in [11, 12, 14] the localization of a monoidal category is studied. It is analogous to the standard localization of a commutative ring by a multiplicative set. To be more precise, for a commutative ring $\mathbf{k}$ and a $\mathbf{k}$-linear monoidal category $(\mathcal{T}, \otimes)$, a left braider in $\mathcal{T}$ is defined as a (C, R^ℓ_C), consisting of an object C, and morphisms

$$\mathrm{R}^\ell_C(X)\colon C \otimes X \to X \otimes C$$

which are functorial in $X \in \mathcal{T}$ and satisfy certain compatibility conditions with the tensor product $\otimes$. A family $\{(C_i, \mathrm{R}^\ell_{C_i}\}_i$ of left braiders in $\mathcal{T}$ is called a real commuting family if $\mathrm{R}^\ell_{C_i}(C_i) \in \mathbf{k}^\times \,\mathsf{id}_{C_i \otimes C_i}$, and $\mathrm{R}^\ell_{C_j}(C_i) \circ \mathrm{R}^\ell_{C_i}(C_j) \in \mathbf{k}^\times \,\mathsf{id}_{C_i \otimes C_j}$ for any i, j. Then, one can construct a $\mathbf{k}$-linear monoidal category $(\widetilde{\mathcal{T}}, \otimes)$ and a monoidal functor $\Phi\colon \mathcal{T} \to \widetilde{\mathcal{T}}$ such that the objects $\Phi(C_i)$ are invertible, and the morphisms $\Phi(\mathrm{R}^\ell_{C_i}(X))\colon \Phi(C_i) \otimes \Phi(X) \to \Phi(X) \otimes \Phi(C_i)$ are isomorphisms for all i and all $X \in \mathcal{T}$.

Let R be a quiver Hecke algebra associated with a Kac-Moody Lie algebra $\mathfrak{g}$, and let R-gmod denote the category of finite-dimensional graded R-modules. For an element w of the Weyl group W, there exists a significant and intriguing subcategory $\mathscr{C}_w$ of R-gmod (see Section 5.1 for the precise definition). It has been established that the Grothendieck ring $K(\mathscr{C}_w)$ of $\mathscr{C}_w$ is isomorphic to the quantum unipotent coordinate ring $A_q(\mathfrak{n}(w))$ associated with w (see [6] and [10]). The algebra $A_q(\mathfrak{n}(w))$ can be interpreted as a q-deformation of the coordinate ring of the unipotent subgroup $N(w)$, and $A_q(\mathfrak{n}(w))$ possesses a quantum cluster algebra structure. It was demonstrated in [6] that the category $\mathscr{C}_w$ provides a monoidal categorification of $A_q(\mathfrak{n}(w))$ as a quantum cluster algebra when $\mathfrak{g}$ is symmetric.

In [11], it is shown that there exist graded left braiders in R-gmod:

$$(\mathsf{M}(w\Lambda_i, \Lambda_i), \mathrm{R}^\ell_{\mathsf{M}(w\Lambda_i,\Lambda_i)}, \phi^l_{\mathsf{M}(w\Lambda_i,\Lambda_i)}) \qquad \text{for any } i \in I$$

consisting of the *determinantial module* $\mathsf{M}(w\Lambda_i, \Lambda_i)$ (see § 4.6) and the morphism induced from *R-matrices*

$$\mathrm{R}^\ell_{\mathsf{M}(w\Lambda_i,\Lambda_i)}(X)\colon \mathsf{M}(w\Lambda_i, \Lambda_i) \circ X \longrightarrow X \circ \mathsf{M}(w\Lambda_i, \Lambda_i) \quad \text{for any } X \in R\text{-gmod}$$

of homogeneous degree $-(w\Lambda_i + \Lambda_i, \mathrm{wt}(X))$.

Under the categorification, the determinantial module $\mathsf{M}(w\Lambda_i, \Lambda_i)$ corresponds to the unipotent quantum minor $D(w\Lambda_i, \Lambda_i)$, which is a frozen variable of the quantum cluster algebra $A_q(\mathfrak{n}(w))$. Thus, the localization $\widetilde{\mathscr{C}_w}$ of $\mathscr{C}_w$ with respect to the aforementioned left braiders categorifies the localization $A_q(\mathfrak{n}(w))[D(w\Lambda_i, \Lambda_i)^{-1}; i \in I]$ of $A_q(\mathfrak{n}(w))$ at the frozen variables $D(w\Lambda_i, \Lambda_i)$ $(i \in I)$. Notably, through this categorification, the Grothendieck ring $K(\widetilde{\mathscr{C}_w})$ can be interpreted as a quantization of the coordinate ring of the unipotent cell N^w associated with w (see [22, Theorem 4.13]).

It turns out that the natural inclusion functor $\iota_w \colon \mathscr{C}_w \rightarrowtail R\text{-gmod}$ induces an equivalence of categories:

$$\widetilde{\iota}_w \colon \widetilde{\mathscr{C}_w} \xrightarrow{\sim} \big(R\text{-gmod}\big)[\mathsf{M}(w\Lambda_i, \Lambda_i)^{\circ -1}; i \in I] =: (R\text{-gmod})\widetilde{\ }[w].$$

It should be noted that a bunch of objects in R-gmod vanishes after the localization process. In [12, Proposition 3.3], the characterization of modules which vanish under the localization functor $\mathrm{Q}_w \colon R\text{-gmod} \to (R\text{-gmod})\widetilde{\ }[w]$ is provided. Recall that the self-dual simple modules in R-gmod correspond bijectively with the crystal $B(\infty)$ of $A_q(\mathfrak{n}_+)$ ([24]). It is established that a simple module M does not vanish under Q_w if and only if M corresponds to an element b in $B_w(\infty)$, where $B_w(\infty)$ is a subset of $B(\infty)$ introduced in [8], known as the Demazure crystal (see Proposition 2.4).

In [25], the second author constructed a geometric crystal structure on the Schubert cell X_w (resp. variety $\overline{X_w}$) associated with a Weyl group element w. It was shown that by tropicalizing this geometric crystal on the Langlands dual Schubert cell $X_w^\vee$ (resp. variety $\overline{X}_w^\vee$), one can obtain the crystal $B_{i_1} \otimes B_{i_2} \otimes \cdots \otimes B_{i_k}$, where $B_i := \{(m)_i \mid m \in \mathbb{Z}\}$ $(i \in I)$ is a crystal introduced in Example 2.6 and $\underline{w} := s_{i_1} s_{i_2} \cdots s_{i_k}$ is a reduced expression of w. Note that for a Weyl group element w, a reduced expression $s_{i_1} s_{i_2} \cdots s_{i_k}$ is often identified with a reduced word $i_1 i_2 \cdots i_k$. Here, the crystal $\mathbf{B}_{\underline{w}} := B_{i_1} \otimes B_{i_2} \otimes \cdots \otimes B_{i_k}$ is referred to as the *cellular crystal associated with* w. In [2], Kanakubo and the second author proved that when $\mathfrak{g}$ is semi-simple, the cellular crystal $\mathbf{B}_{\underline{w}} = B_{i_1} \otimes B_{i_2} \otimes \cdots \otimes B_{i_k}$ associated with a reduced word $\underline{w} = i_1 i_2 \cdots i_k$ is connected as a crystal graph. It is well-known that the crystal $B(\infty)$ is embedded in $\mathbf{B}_{\underline{w_0}}$, where $\underline{w_0} = i_1 i_2 \cdots i_N$ is a reduced word of the longest element in W ([2, 27]). As a set, $\mathbf{B}_{\underline{w_0}}$ can be identified with $\mathbb{Z}^N$, and for any $\mathbf{x}, \mathbf{y} \in \mathbf{B}_{\underline{w_0}}$, the summation $\mathbf{x} + \mathbf{y}$ is naturally defined in $\mathbb{Z}^N$.

For Λ in the set P_+ of dominant integral weights, set $\mathbf{h}_\Lambda := \tilde{f}_{i_1}^{m_1} \cdots \tilde{f}_{i_N}^{m_N}((0)_{i_1} \otimes \cdots (0)_{i_N})$, where $m_k := \langle h_{i_k}, s_{i_{k+1}} \cdots s_{i_N} \Lambda \rangle$. Then we have

$$\mathbf{B}_{\underline{w_0}} = \{-\mathbf{h}_\Lambda + x \mid \Lambda \in \mathsf{P}_+,\ x \in B(\infty)\},$$

which implies the connectedness of $\mathbf{B}_{\underline{w_0}}$ and consequently $\mathbf{B}_{\underline{w}}$ for any $w \in \mathsf{W}$ ([2]).

In the meanwhile, Lauda and Vazirani ([24]) showed that the family $\mathrm{Irr}(R\text{-gmod})$ of the isomorphism classes (up to grading shifts) of simple modules in R-gmod possesses a crystal structure and provided an isomorphism of the crystals

$$\Psi\colon \mathrm{Irr}(R\text{-gmod}) \xrightarrow{\sim} B(\infty).$$

For a semi-simple $\mathfrak{g}$, we have $R\text{-gmod} = \mathscr{C}_{w_0}$ and $\widetilde{R\text{-gmod}} = \widetilde{\mathscr{C}_{w_0}}$. Let $\Phi\colon R\text{-gmod} \to \widetilde{R\text{-gmod}}$ be the localization functor. In [11] it is shown that, for any simple object, $Y \in \widetilde{R\text{-gmod}}$ there exists $\Lambda \in \mathsf{P}_+$ and a simple object $X \in R\text{-gmod}$ such that $Y \simeq C_\Lambda^{\circ -1} \circ \Phi(X)$, where $C_\Lambda := \mathsf{M}(w\Lambda, \Lambda) = \widetilde{\mathrm{F}}_{i_1}^{m_1} \cdots \widetilde{\mathrm{F}}_{i_N}^{m_N} \mathbf{1}$ is a determinantial module. Thus, it is observed that

$$\mathrm{Irr}(\widetilde{R\text{-gmod}}) \simeq \mathbf{B}_{\underline{w_0}} \quad \text{by } C_\Lambda^{\circ -1} \circ X \longleftrightarrow -\mathbf{h}_\Lambda + \Psi(X),$$

where $\mathrm{Irr}(\widetilde{R\text{-gmod}})$ is the family of the isomorphism classes (up to grading shifts) of simple objects in $\widetilde{R\text{-gmod}}$.

Then, the second author proved ([26]) that the family of self-dual simple objects in the localized category possesses a crystal structure and it is isomorphic to the cellular crystal $\mathbf{B}_{\underline{w_0}} := B_{i_1} \otimes B_{i_2} \otimes \cdots \otimes B_{i_N}$. It implies that the family $\mathrm{Irr}(\widetilde{R\text{-gmod}})$ also possesses a crystal structure and it is isomorphic to the cellular crystal $\mathbf{B}_{\underline{w_0}}$ since for any simple module X in $\widetilde{R\text{-gmod}}$, there exists a unique integer n such that $q^n X$ is self-dual (see Lemma 4.4).

In [26] the following problem is presented: in the general Kac-Moody setting, $\mathrm{Irr}(\widetilde{\mathscr{C}_w})$ is endowed with a crystal structure and it is isomorphic to the cellular crystal $\mathbf{B}_{\underline{w}} := B_{i_1} \otimes B_{i_2} \otimes \cdots \otimes B_{i_k}$ associated with an arbitrary reduced word $\underline{w} = i_1 i_2 \cdots i_k$ of an arbitrary $w \in \mathsf{W}$. The main aim of this paper is to provide a positive answer to this problem.

Let us delve into more details. In [26], as mentioned above, the crystal structure of $\mathrm{Irr}(\widetilde{R\text{-gmod}})$ is established. However, the method employed there cannot be applied to our problem since $\mathrm{Irr}(\mathscr{C}_w)$ is not necessarily stable under the action of the crystal operator $\widetilde{\mathrm{E}}_i$ (see Remark 5.13). Additionally, an object in the localized category $\widetilde{\mathscr{C}_w}$ is no longer an R-module, and thus the action of $\widetilde{\mathrm{E}}_i$ on $\widetilde{\mathscr{C}_w}$ is not properly defined. Therefore, new tools are required to define the crystal structure on $\mathrm{Irr}(\widetilde{\mathscr{C}_w})$. They are "affinization", "R-matrix" and "rigidity".

In case R is symmetric, an affinization of $M \in R\text{-gmod}$ is simply given as $\mathbf{k}[z] \otimes_{\mathbf{k}} M$. However, in general, the existence of an affinization in R-gmod is not necessarily guaranteed. A general definition of affinization in a monoidal category introduced in

[16] (cf. [18]) is provided in Definition 3.9. An object M is called *affreal* if M is real and afford an affinization (see Definition 3.11). Then by Proposition 3.17 we find that if M is affreal and N is simple in a quasi-rigid category (see § 3.5), then there exists R-matrix $\mathbf{r}_{M,N}\colon M\otimes N\to N\otimes M$. As stated in Theorem 5.3, the category $\widetilde{\mathscr{C}_w}$ is rigid ([11], [12]), that is, any object X of $\widetilde{\mathscr{C}_w}$ holds the left dual $\mathscr{D}^{-1}(X)$ and the right dual $\mathscr{D}(X)$ (see Definition 3.4). These tools enable us to define a crystal structure on $\mathrm{Irr}(\widetilde{\mathscr{C}_w})$ as in Definition 6.11. Finally, we shall construct the isomorphism of crystals $\mathrm{K}_{\underline{w}}\colon \mathrm{Irr}(\widetilde{\mathscr{C}_w})\xrightarrow{\sim}\mathbf{B}_{\underline{w}}$ inductively using the map

$$\mathrm{K}'_{w',w}\colon \mathrm{Irr}(\widetilde{\mathscr{C}_w})\to \mathrm{Irr}(\tilde{\mathscr{C}}_{w'})\times\mathbb{Z}\quad\text{given by}\quad X\mapsto(\mathbf{E}^*_{w',w}(X),\varepsilon_i^*(X)),$$

where $w'=ws_i<w$ and see Definition 6.11 and § 6.3 for the definition of $\mathbf{E}^*_{w',w}$ and ε_i^*.

We note that Woo-Seok Jung and Euiyong Park ([1]) treats relevant topics, including the bijectivity of $\mathrm{Irr}(\widetilde{\mathscr{C}_w})\to\mathbf{B}_{\underline{w}}$.

The organization of the article is as follows. In § 2, we review the basics of the crystal base theory, in particular, the explicit structure of the cellular crystals which will be needed for later arguments. In § 3, the theory of affinization and R-matrix of monoidal category is provided following [15], which will play a crucial role to obtain the main theorem in the subsequent sections. § 4 is devoted to give the definition of the quiver Hecke algebra and the useful and important properties of its modules such as the head simplicity of convolution product, R-matrices, the invariants Λ, the shuffle lemmas and the determinantial modules. In § 5, several results on the localization of the category R-gmod and $\mathscr{C}_w$ are introduced and the relations of the localization functors and the convolution product are discussed. In § 6, we define the crystal structure on $\mathrm{Irr}(\widetilde{\mathscr{C}_w})$ and present some technical lemmas to prove the main theorems. As a corollary of the main theorem in § 7, we prove that there exists an isomorphism between $\mathrm{Irr}(\widetilde{\mathscr{C}_w})$ and the cellular crystal $\mathbf{B}_{\underline{w}}$. The proof of the main theorem will be presented in § 8. In the last section, we prove the connectedness of arbitrary cellular crystal as an application of the main theorem.

2. Crystal Bases and Crystals

2.1. **Convention.** In this paper. we use the following convention.

Convention. (i) A ring is a unital ring.
(ii) For a ring A, $A^\times$ denotes the group of invertible elements of A.
(iii) For a monoidal category $\mathcal{C}$ with a tensor product $\otimes$, we denote by $\mathcal{C}^{\mathrm{rev}}$ the monoidal category $\mathcal{C}$ with the reversed tensor product $\otimes_{\mathrm{rev}}$: $X\otimes_{\mathrm{rev}}Y:=Y\otimes X$.

2.2. **Crystal Bases.** A Cartan datum $\big(\mathsf{C}, \mathsf{P}, \Pi, \Pi^\vee, (\cdot,\cdot)\big)$ is a quintuple of a generalized Cartan matrix C, a free abelian group P, a set of simple roots, $\Pi = \{\alpha_i \mid i \in I\} \subset \mathsf{P}$, a set of simple coroots $\Pi^\vee = \{h_i \mid i \in I\} \subset \mathsf{P}^\vee := \mathrm{Hom}(\mathsf{P}, \mathbb{Z})$, and a $\mathbb{Q}$-valued symmetric bilinear form $(\cdot,\cdot)$ on P such that

(a) $\mathsf{C} = (\langle h_i, \alpha_j\rangle)_{i,j\in I}$,
(b) $(\alpha_i, \alpha_i) \in 2\mathbb{Z}_{>0}$ for any $i \in I$,
(c) $\langle h_i, \lambda\rangle = \dfrac{2(\alpha_i,\lambda)}{(\alpha_i,\alpha_i)}$ for $i \in I$ and $\lambda \in \mathsf{P}$,
(d) for each $i \in I$, there exists $\Lambda_i \in \mathsf{P}$ such that $\langle h_j, \Lambda_i\rangle = \delta_{ij}$ for any $j \in I$.

Let $\mathsf{Q} := \bigoplus_{i\in I} \mathbb{Z}\alpha_i$ and $\mathsf{Q}_+ := \bigoplus_{i\in I} \mathbb{Z}_{\geqslant 0}\alpha_i$ be the root lattice and the positive root lattice of the symmetrizable Kac-Moody algebra $\mathfrak{g} = \mathfrak{g}(\mathsf{C})$, respectively. Let $U_q(\mathfrak{g}) := \langle e_i, f_i, q^h \mid i \in I,\, h \in P^\vee\rangle$ (resp. $U_q^-(\mathfrak{g}) := \langle f_i \mid i \in I\rangle$) be the associated quantum group over $\mathbb{Q}(q)$ (resp. negative nilpotent subalgebra of $U_q(\mathfrak{g})$).

Let W be the Weyl group of $\mathfrak{g}$, the subgroup of $\mathrm{Aut}(\mathsf{P})$ generated by the simple reflections $\{s_i\}_{i\in I}$ where $s_i(\lambda) = \lambda - \langle h_i, \lambda\rangle\alpha_i$ for $\lambda \in \mathsf{P}$.

As in [7], there exists the crystal basis $(L(\infty), B(\infty))$ of $U_q^-(\mathfrak{g})$ defined by

$$\begin{aligned}
&L(\infty) := \sum_{k\geqslant 0, i_1,\cdots,i_k\in I} \mathbf{A}\tilde{f}_{i_1}\cdots\tilde{f}_{i_k}u_\infty,\\
&B(\infty) = \{\tilde{f}_{i_1}\cdots\tilde{f}_{i_k}u_\infty \bmod qL(\infty) \mid k \geqslant 0, i_1,\cdots,i_k \in I\} \setminus \{0\},\\
&\varepsilon_i(b) = \max\{k \mid \tilde{e}_i^k b \neq 0\}, \quad \varphi_i(b) = \varepsilon_i(b) + \langle h_i, \mathrm{wt}(b)\rangle,
\end{aligned}$$

where $u_\infty = 1 \in U_q(\mathfrak{g})$, $\tilde{e}_i$ and $\tilde{f}_i \in \mathrm{End}_{\mathbb{Q}(q)}(U_q^-(\mathfrak{g}))$ are the *crystal operators* ([7]) and $\mathbf{A}$ is the local subring of $\mathbb{Q}(q)$ at $q = 0$. We also obtain the crystal basis $(L(\lambda), B(\lambda))$ of the irreducible highest weight module $V(\lambda)$ with a highest weight vector u_λ for any dominant weight $\lambda \in \mathsf{P}_+ := \{\lambda \in \mathsf{P} \mid \langle h_i, \lambda\rangle \in \mathbb{Z}\}$ by a similar way.

2.3. **Crystals.** We define the notion of *crystal* as in [8], which is a combinatorial object abstracting the properties of crystal bases:

Definition 2.1 ([8])**.** A 6-tuple $\big(B, \mathrm{wt}, \{\varepsilon_i, \varphi_i, \tilde{e}_i, \tilde{f}_i\}_{i\in I}\big)$ is a *crystal* if B is a set and there exists a certain special element 0 outside of B and maps:

$$\mathrm{wt}\colon B \to P, \quad \varepsilon_i\colon B \to \mathbb{Z} \sqcup \{-\infty\}, \quad \varphi_i\colon B \to \mathbb{Z} \sqcup \{-\infty\} \quad (i \in I), \tag{2.1}$$

$$\tilde{e}_i\colon B \sqcup \{0\} \to B \sqcup \{0\}, \quad \tilde{f}_i\colon B \sqcup \{0\} \to B \sqcup \{0\} \ (i \in I), \tag{2.2}$$

satisfying :

(1) $\varphi_i(b) = \varepsilon_i(b) + \langle h_i, \mathrm{wt}(b)\rangle$.
(2) If $b, \tilde{e}_i b \in B$, then $\mathrm{wt}(\tilde{e}_i b) = \mathrm{wt}(b) + \alpha_i$, $\varepsilon_i(\tilde{e}_i b) = \varepsilon_i(b) - 1$, $\varphi_i(\tilde{e}_i b) = \varphi_i(b) + 1$.

(3) If $b, \tilde{f}_i b \in B$, then $\mathrm{wt}(\tilde{f}_i b) = \mathrm{wt}(b) - \alpha_i$, $\varepsilon_i(\tilde{f}_i b) = \varepsilon_i(b) + 1$, $\varphi_i(\tilde{f}_i b) = \varphi_i(b) - 1$.
(4) For $b, b' \in B$ and $i \in I$, one has $\tilde{f}_i b = b'$ if an only if $b = \tilde{e}_i b'$.
(5) If $\varphi_i(b) = -\infty$ for $b \in B$, then $\tilde{e}_i b = \tilde{f}_i b = 0$ and $\tilde{e}_i(0) = \tilde{f}_i(0) = 0$.

Definition 2.2. A *crystal graph* of a crystal B is an I-colored oriented graph defined by $b \xrightarrow{i} b' \Leftrightarrow \tilde{f}_i(b) = b'$ for $b, b' \in B$. A crystal B is *connected* if its crystal graph is a connected graph.

Definition 2.3 ([8])**.**

(i) Let $B = (B, \mathrm{wt}, \{\varepsilon_i\}, \{\varphi_i\}, \{\tilde{e}_i\}, \{\tilde{f}_i\})$ and $B' = (B', \mathrm{wt}', \{\varepsilon'_i\}, \{\varphi'_i\}, \{\tilde{e}'_i\}, \{\tilde{f}'_i\})$ be crystals. A morphism of crystal from B to B' is a map $\psi\colon B \sqcup \{0\} \to B' \sqcup \{0\}$ such that
  (a) $\psi(0) = 0$,
  (b) if $b \in B$ satisfies $\psi(b) \in B'$, then $\mathrm{wt}\big(\psi(b)\big) = \mathrm{wt}(b)$, $\varepsilon_i\big(\psi(b)\big) = \varepsilon_i(b)$, and $\varphi_i\big(\psi(b)\big) = \varphi_i(b)$,
  (c) if $b_1, b_2 \in B$ satisfy $\psi(b_1) \in B'$, $\psi(b_2) \in B'$ and $\widetilde{f}_i(b_1) = b_2$, then we have $\widetilde{f}'_i\big(\psi(b_1)\big) = \psi(b_2)$.

(ii) Let $B = (B, \mathrm{wt}, \{\varepsilon_i\}, \{\varphi_i\}, \{\tilde{e}_i, \}, \{\tilde{f}_i\})$ be a crystal, and let B' be a subset of B. Then B' is endowed with the crystal structure $B' = (B', \mathrm{wt}', \{\varepsilon'_i\}, \{\varphi'_i\}, \{\tilde{e}'_i\}, \{\tilde{f}'_i\})$ induced by B: wt', ε'_i, φ'_i are the restrictions of the corresponding functions on B and

$$\widetilde{e}'_i(b) = \begin{cases} \widetilde{e}_i(b) & \text{if } \widetilde{e}_i(b) \in B', \\ 0 & \text{otherwise,} \end{cases} \qquad \widetilde{f}'_i(b) = \begin{cases} \widetilde{f}_i(b) & \text{if } \widetilde{f}_i(b) \in B', \\ 0 & \text{otherwise} \end{cases} \qquad \text{for any } b \in B',$$

so that $B' \to B$ is a morphism of crystals. The crystal B' is called a *subcrystal* of B.

(iii) We say that a morphism $\Psi\colon B_1 \to B_2$ is *strict* if it satisfies $\widetilde{e}_i(\Psi(b)) = \Psi(\widetilde{e}_i b)$ and $\widetilde{f}_i(\Psi(b)) = \Psi(\widetilde{f}_i b)$ for any $b \in B_1$.

(iv) We say that a morphism $\Psi\colon B_1 \to B_2$ is an *embedding* if $B_1 \sqcup \{0\} \to B_2 \sqcup \{0\}$ is injective.

Proposition 2.4 ([8])**.** *For any $w \in \mathsf{W}$ there exists a unique subset $B_w(\infty)$ of $B(\infty)$ satisfying the following properties:*

(1) $B_w(\infty) = \{u_\infty\}$ *if* $w = 1$.
(2) *If* $s_i w < w$, *then* $B_w(\infty) = \displaystyle\bigcup_{k \geqslant 0} \tilde{f}_i^k B_{s_i w}(\infty)$.

Moreover, they satisfy

(3) $\tilde{e}_i B_w(\infty) \subset B_{s_i w}(\infty)$.

(4) *If* $w \geqslant w'$, *then* $B_w(\infty) \supset B_{w'}(\infty)$.

We call $B_w(\infty)$ *the* Demazure crystal *associated with* $w \in \mathsf{W}$ *with the crystal structure induced from* $B(\infty)$.

We can define the tensor product of crystals as follows ([7, 8]):

Proposition 2.5. *For crystals* B_1 *and* B_2, *set*

$$B_1 \otimes B_2 = \{b_1 \otimes b_2 \mid b_1 \in B_1,\, b_2 \in B_2\}$$

Then, $B_1 \otimes B_2$ *becomes a crystal by defining:*

$$\mathrm{wt}\,(b_1 \otimes b_2) = \mathrm{wt}\,(b_1) + \mathrm{wt}\,(b_2), \tag{2.3}$$

$$\varepsilon_i(b_1 \otimes b_2) = \max(\varepsilon_i(b_1), \varepsilon_i(b_2) - \langle h_i, \mathrm{wt}\,(b_1)\rangle), \tag{2.4}$$

$$\varphi_i(b_1 \otimes b_2) = \max(\varphi_i(b_2), \varphi_i(b_1) + \langle h_i, \mathrm{wt}\,(b_2)\rangle), \tag{2.5}$$

$$\tilde{e}_i(b_1 \otimes b_2) = \begin{cases} \tilde{e}_i b_1 \otimes b_2 \text{ if } \varphi_i(b_1) \geqslant \varepsilon_i(b_2), \\ b_1 \otimes \tilde{e}_i b_2 \text{ if } \varphi_i(b_1) < \varepsilon_i(b_2), \end{cases} \tag{2.6}$$

$$\tilde{f}_i(b_1 \otimes b_2) = \begin{cases} \tilde{f}_i b_1 \otimes b_2 \text{ if } \varphi_i(b_1) > \varepsilon_i(b_2), \\ b_1 \otimes \tilde{f}_i b_2 \text{ if } \varphi_i(b_1) \leqslant \varepsilon_i(b_2). \end{cases} \tag{2.7}$$

Example 2.6. (1) For $i \in I$, set $B_i := \{(n)_i \,|\, n \in \mathbb{Z}\}$ and

$$\begin{aligned} &\mathrm{wt}\,((n)_i) = n\alpha_i,\ \ \varepsilon_i((n)_i) = -n,\ \ \varphi_i((n)_i) = n, \\ &\varepsilon_j((n)_i) = \varphi_j((n)_i) = -\infty \ \ (i \neq j), \\ &\tilde{e}_i((n)_i) = (n+1)_i, \quad \tilde{f}_i((n)_i) = (n-1)_i, \\ &\tilde{e}_j((n)_i) = \tilde{f}_j((n)_i) = 0 \ \ (i \neq j). \end{aligned}$$

Then B_i $(i \in I)$ becomes a crystal. Note that $\tilde{e}_i$ and $\tilde{f}_i$ are inverse to each other.

(2) The crystal $T_\lambda := \{t_\lambda\}$ is defined by $\tilde{e}_i t_\lambda = \tilde{f}_i t_\lambda = 0$, $\varepsilon_i(t_\lambda) = \varepsilon_i(t_\lambda) = -\infty$ and $\mathrm{wt}\,(t_\lambda) = \lambda$.

Example 2.7.

(i) There exists a unique strict embedding $\Psi_i \colon B(\infty) \to B(\infty) \otimes B_i$ $(u_\infty \mapsto u_\infty \otimes (0)_i)$ for any $i \in I$ ([8]).

(ii) There is a unique embedding $\Psi_\lambda \colon B(\lambda) \to B(\infty) \otimes T_\lambda$ $(u_\lambda \mapsto u_\infty \otimes t_\lambda)$ for any $\lambda \in \mathsf{P}_+$. Here, note that $\varepsilon_i(b \otimes t_\lambda) = \varepsilon_i(b)$ and $\tilde{e}_i(b \otimes t_\lambda) = (\tilde{e}_i b) \otimes t_\lambda$. Thus, the map Ψ_λ induces an injective map $\Psi \colon B(\lambda) \to B(\infty)$ such that $\Psi \tilde{e}_i(b) = \tilde{e}_i \Psi(b)$ and $\varepsilon_i(b) = \varepsilon_i(\Psi(b))$ for any $b \in B(\lambda)$.

2.4. **Cellular Crystal $\mathbf{B}_{\underline{w}}$.** For a reduced expression $\underline{w} = s_{i_1}s_{i_2}\cdots s_{i_k}$ of some Weyl group element $w \in \mathsf{W}$, we call the crystal $\mathbf{B}_{\underline{w}} := B_{i_1} \otimes \cdots \otimes B_{i_k}$ the *cellular crystal* associated with a reduced expression $\underline{w}$. Note that it is obtained by applying the tropicalization functor to the geometric crystal on the Langlands-dual Schubert cell $X_w^\vee$ ([25]). It is immediate from the braid-type isomorphisms ([28]) that for any $w \in \mathsf{W}$ and its arbitrary reduced words $i_1 \cdots i_m$ and $j_1 \cdots j_m$, we get the following isomorphism of crystals:

$$B_{i_1} \otimes \cdots \otimes B_{i_m} \simeq B_{j_1} \otimes \cdots \otimes B_{j_m}. \tag{2.8}$$

Here, we shall investigate an explicit structure of the crystal $\mathbf{B}_{\underline{w}}$. Fix a reduced expression $\underline{w} = s_{i_1} \cdots s_{i_m}$ and write

$$(x_1, \cdots, x_m) := \tilde{f}_{i_1}^{x_1}(0)_{i_1} \otimes \cdots \otimes \tilde{f}_{i_m}^{x_m}(0)_{i_m} = (-x_1)_{i_1} \otimes \cdots \otimes (-x_m)_{i_m} \in \mathbf{B}_{\underline{w}},$$

where $\tilde{f}_i^n(0)_i$ means $\tilde{e}_i^{-n}(0)_i$ when $n < 0$.

By the tensor structure of crystals in Proposition 2.5, we can describe the explicit crystal structure on $\mathbf{B}_{\underline{w}} := B_{i_1} \otimes \cdots \otimes B_{i_m}$ as follows: For $x = (x_1, \cdots, x_m) \in \mathbf{B}_{\underline{w}}$ and $k \in [1, m]$, define

$$\sigma_k(x) := x_k + \sum_{1 \leqslant j < k} \langle h_{i_k}, \alpha_{i_j} \rangle x_j$$

and for $i \in I$ define

$$\widetilde{\sigma}^{(i)}(x) := \max\{\sigma_k(x) \mid 1 \leqslant k \leqslant m \,\text{and}\, i_k = i\}, \tag{2.9}$$

$$\widetilde{M}^{(i)} = \widetilde{M}^{(i)}(x) := \{k \mid 1 \leqslant k \leqslant m,\ i_k = i,\ \sigma_k(x) = \widetilde{\sigma}^{(i)}(x)\}, \tag{2.10}$$

$$\widetilde{m}_f^{(i)} = \widetilde{m}_f^{(i)}(x) := \max \widetilde{M}^{(i)}(x), \quad \widetilde{m}_e^{(i)} = \widetilde{m}_e^{(i)}(x) := \min \widetilde{M}^{(i)}(x). \tag{2.11}$$

Now, the actions of the crystal operators $\tilde{e}_i, \tilde{f}_i$ and the functions ε_i, φ_i and wt are written explicitly:

$$\tilde{f}_i(x)_k := x_k + \delta_{k, \widetilde{m}_f^{(i)}}, \qquad \tilde{e}_i(x)_k := x_k - \delta_{k, \widetilde{m}_e^{(i)}}, \tag{2.12}$$

$$\mathrm{wt}\,(x) := -\sum_{k=1}^{m} x_k \alpha_{i_k}, \quad \varepsilon_i(x) := \widetilde{\sigma}^{(i)}(x), \quad \varphi_i(x) := \langle h_i, \mathrm{wt}\,(x) \rangle + \varepsilon_i(x). \tag{2.13}$$

3. Generality of graded monoidal categories

In this section, we briefly review the notions and the results in [15] which axiomatize the contents of [5], [6], [10], etc. For more details, consult [15].

3.1. **Affine objects in** $\mathrm{Pro}(\mathcal{C})$**.**

Definition 3.1 (see e.g. [20, § 1.4, 3.1, 6.1])**.** (1) For a category $\mathcal{C}$, let $\mathcal{C}^\vee$ denote the category of functors from $\mathcal{C}^{\mathrm{op}}$ to $\mathbf{Set}^{\mathrm{op}}$. Then, we have a fully faithful functor $\mathcal{C} \to \mathcal{C}^\vee$ given by $\mathcal{C} \ni X \mapsto (\mathcal{C} \ni Y \mapsto \mathrm{Hom}_{\mathcal{C}}(X, Y) \in \mathbf{Set})$, called the *Yoneda functor*.

(2) A category I is *directed* if it satisfies the following conditions:
   (a) I is non-empty,
   (b) for any $i, j \in I$, there exist $k \in I$ and morphisms $i \to k$, $j \to k$,
   (c) for any parallel morphisms $f, g \colon i \rightrightarrows j$, there exists a morphism $h \colon j \to k$ such that $h \circ f = h \circ g$.

   If I^{op} is directed, we say that I is *co-directed*.

(3) A *pro-object* in $\mathcal{C}$ is an object of $\mathcal{C}^\vee$ which is isomorphic to $\varprojlim \beta$ for some functor $\beta \colon I \to \mathcal{C} \to \mathcal{C}^\vee$ with co-directed I.

(4) Let $\mathrm{Pro}(\mathcal{C})$ denote the full subcategory of $\mathcal{C}^\vee$ consisting of pro-objects.

(5) The co-directed projective limit in $\mathrm{Pro}(\mathcal{C})$ is denoted by "$\varprojlim$".

Let $\mathbf{k}$ be a base field. Let $\mathcal{C}$ be a $\mathbf{k}$-linear category such that

$$(3.1)\quad \begin{cases} \bullet\ \mathcal{C} \text{ is abelian,} \\ \bullet\ \mathcal{C} \text{ is } \mathbf{k}\text{-linear graded, namely } \mathcal{C} \text{ is endowed with a } \mathbf{k}\text{-linear auto-equivalence } q, \\ \bullet\ \text{any object has a finite length,} \\ \bullet\ \text{any simple object } S \text{ is absolutely simple, i.e., } \mathbf{k} \xrightarrow{\sim} \mathrm{E{\scriptstyle ND}}_{\mathcal{C}}(S). \end{cases}$$

Here, we denote

$$(3.2)\quad \begin{aligned} &\mathrm{H{\scriptstyle OM}}_{\mathcal{C}}(M,N) = \textstyle\bigoplus_{n\in\mathbb{Z}} \mathrm{H{\scriptstyle OM}}_{\mathcal{C}}(M,N)_n \ \text{ with } \mathrm{H{\scriptstyle OM}}_{\mathcal{C}}(M,N)_n = \mathrm{Hom}_{\mathcal{C}}(q^n M, N), \\ &\text{and } \mathrm{E{\scriptstyle ND}}_{\mathcal{C}}(M) := \mathrm{H{\scriptstyle OM}}_{\mathcal{C}}(M,M). \end{aligned}$$

It follows from (3.1) that $\mathrm{H{\scriptstyle OM}}_{\mathcal{C}}(M,N)$ is finite-dimensional over $\mathbf{k}$ for any $M, N \in \mathcal{C}$ and that $S \not\simeq q^k S$ for any $k \in \mathbb{Z} \setminus \{0\}$ and any non-zero $S \in \mathcal{C}$.

By [15, Lemma 2.1], the full subcategory $\mathcal{C}$ of $\mathrm{Pro}(\mathcal{C})$ is stable by taking subquotients. The grading shift functor is extended to $\mathrm{Pro}(\mathcal{C})$.

Let $A = \bigoplus_{k\in\mathbb{Z}} A_k$ be a commutative graded $\mathbf{k}$-algebra such that

$$(3.3)\quad \begin{cases} \bullet\ A_0 \simeq \mathbf{k}, \\ \bullet\ A_k = 0 \text{ for any } k < 0, \\ \bullet\ A \text{ is a finitely generated } \mathbf{k}\text{-algebra.} \end{cases}$$

In particular, A is a noetherian ring and $\dim_{\mathbf{k}} A_k < \infty$ for all k. For each $m \in \mathbb{Z}$, set

$$A_{\geqslant m} := \bigoplus\nolimits_{k \geqslant m} A_k,$$

which is an ideal of A.

Hereafter, we assume that $\mathcal{C}$ satisfies (3.1) *and A satisfies* (3.3)*.*

We denote by $\mathrm{Mod}_{\mathrm{gr}}(A, \mathrm{Pro}(\mathcal{C}))$ the category of graded A-modules in $\mathrm{Pro}(\mathcal{C})$, i.e., objects $\mathsf{M} \in \mathrm{Pro}(\mathcal{C})$ with a graded algebra homomorphism $A \to \mathrm{E{\scriptstyle ND}}_{\mathrm{Pro}(\mathcal{C})}(\mathsf{M})$.

For $\mathsf{M}, \mathsf{N} \in \mathrm{Mod}_{\mathrm{gr}}(A, \mathrm{Pro}(\mathcal{C}))$, we set $\mathrm{H{\scriptstyle OM}}_A(\mathsf{M}, \mathsf{N}) := \bigoplus_{k \in \mathbb{Z}} \mathrm{H{\scriptstyle OM}}_A(\mathsf{M}, \mathsf{N})_k$ with $\mathrm{H{\scriptstyle OM}}_A(\mathsf{M}, \mathsf{N})_k := \mathrm{Hom}_{\mathrm{Mod}_{\mathrm{gr}}(A, \mathrm{Pro}(\mathcal{C}))}(q^k \mathsf{M}, \mathsf{N})$. Then $\mathrm{H{\scriptstyle OM}}_A(\mathsf{M}, \mathsf{N})$ has a structure of a graded A-module.

Definition 3.2. We denote by $\mathrm{Pro}_{\mathrm{coh}}(A, \mathcal{C})$ the full subcategory of $\mathrm{Mod}_{\mathrm{gr}}(A, \mathrm{Pro}(\mathcal{C}))$ consisting of $\mathsf{M} \in \mathrm{Mod}_{\mathrm{gr}}(A, \mathrm{Pro}(\mathcal{C}))$ such that

(a) $\mathsf{M}/A_{>0}\mathsf{M} \in \mathcal{C}$,

(b) $\mathsf{M} \xrightarrow{\sim} \text{“}\varprojlim_k\text{”}\, \mathsf{M}/A_{\geqslant k}\mathsf{M}$, or equivalently $\text{“}\varprojlim_k\text{”}\, A_{\geqslant k}\mathsf{M} \simeq 0$.

Here $A_{\geqslant k}\mathsf{M} := \mathrm{Im}(A_{\geqslant k} \otimes_A \mathsf{M} \to A \otimes_A \mathsf{M}) \subset \mathsf{M}$. Note that $X \otimes_A \mathsf{M}$ belongs to $\mathrm{Mod}_{\mathrm{gr}}(A, \mathrm{Pro}(\mathcal{C}))$ for any $\mathsf{M} \in \mathrm{Mod}_{\mathrm{gr}}(A, \mathrm{Pro}(\mathcal{C}))$ and any finitely generated graded A-module X.

Definition 3.3. Let z be an indeterminate of homogeneous degree $d \in \mathbb{Z}_{>0}$. An object of $\mathrm{Pro}_{\mathrm{coh}}(\mathbf{k}[z], \mathcal{C})$ is nothing but a pair (M, z) such that

(a) $\mathsf{M} \in \mathrm{Pro}(\mathcal{C})$ and $z \in \mathrm{E{\scriptstyle ND}}_{\mathrm{Pro}_{\mathrm{coh}}(\mathcal{C})}(\mathsf{M})_d$,

(b) $\mathsf{M}/z\mathsf{M} \in \mathcal{C}$,

(c) $\mathsf{M} \xrightarrow{\sim} \text{“}\varprojlim_n\text{”}\, \mathsf{M}/z^n\mathsf{M}$.

If (M, z) satisfies further the following condition

(d) $z \in \mathrm{E{\scriptstyle ND}}_{\mathrm{Pro}_{\mathrm{coh}}(\mathcal{C})}(\mathsf{M})$ is a monomorphism,

then we say that (M, z) is an *affine object* (in $\mathcal{C}$).

We denote by $\mathrm{Aff}_z(\mathcal{C})$ the category of affine objects.

3.2. **Graded monoidal categories.** In the sequel, let $(\mathcal{C}, \otimes)$ be an abelian $\mathbf{k}$-linear graded monoidal category which satisfies

$$(3.4)\quad \begin{cases} \text{(a) } \mathcal{C} \text{ satisfies (3.1)}, \\ \text{(b) } \otimes \text{ is } \mathbf{k}\text{-bilinear and bi-exact}, \\ \text{(c) the unit object } \mathbf{1} \text{ is simple and satisfies } \mathrm{E{\scriptstyle ND}}_{\mathcal{C}}(\mathbf{1}) \simeq \mathbf{k}, \\ \text{(d) } \otimes \text{ is compatible with the grading shift functor } q, \\ \quad\ \text{i.e., } q(X \otimes Y) \simeq (qX) \otimes Y \simeq X \otimes (qY) \text{ (see [16, (1.8)])}. \end{cases}$$

For generalities on monoidal categories, we refer the reader to [20, Chapter 4].

By identifying q with the invertible central object $q\mathbf{1} \in \mathcal{C}$, we have $qX \simeq q \otimes X$.

The category $\mathrm{Pro}(\mathcal{C})$ has also a structure of monoidal category in which the tensor product $\otimes$ is bi-exact.

Definition 3.4. An object X (resp. Y) of a monoidal category $\mathcal{A}$ has a *right dual* (*resp. left dual*) Y (resp. X) if there exists an object $Y \in \mathcal{A}$ (resp. $X \in \mathcal{A}$) and morphisms $\varepsilon\colon X \otimes Y \to \mathbf{1}$ and $\eta\colon \mathbf{1} \to Y \otimes X$ such that the compositions

$$X \simeq X \otimes \mathbf{1} \xrightarrow{\mathsf{id}_X \otimes \eta} X \otimes Y \otimes X \xrightarrow{\varepsilon \otimes \mathsf{id}_X} \mathbf{1} \otimes X \simeq X,$$
$$Y \simeq \mathbf{1} \otimes Y \xrightarrow{\eta \otimes \mathsf{id}_Y} Y \otimes X \otimes Y \xrightarrow{\mathsf{id}_Y \otimes \varepsilon} Y \otimes \mathbf{1} \simeq Y$$

are the identities. We denote the right dual of X by $\mathscr{D}(X)$ and the left dual of Y by $\mathscr{D}^{-1}(Y)$. If any object in $\mathcal{A}$ has a right dual and a left dual, we say that $\mathcal{A}$ is a *rigid* category.

Definition 3.5 ([15, Definition 4.12])**.** We denote by $\mathrm{Raff}_z(\mathcal{C})$ the category with $\mathrm{Ob}\big(\mathrm{Pro}_{\mathrm{coh}}(\mathbf{k}[z], \mathcal{C})\big)$ as the set of objects and with the morphisms defined as follows. For $\mathsf{M}, \mathsf{N} \in \mathrm{Ob}\big(\mathrm{Raff}_z(\mathcal{C})\big)$,

$$\begin{aligned}\mathrm{Hom}_{\mathrm{Raff}_z(\mathcal{C})}(\mathsf{M}, \mathsf{N}) &= \varinjlim_{k \in \mathbb{Z}_{\geqslant 0}} \mathrm{Hom}_{\mathrm{Pro}_{\mathrm{coh}}(\mathbf{k}[z], \mathcal{C})}\big((z^k\mathbf{k}[z]) \otimes_{\mathbf{k}[z]} \mathsf{M}, \mathsf{N}\big) \\ &\simeq \varinjlim_{k \in \mathbb{Z}_{\geqslant 0}} \mathrm{Hom}_{\mathrm{Pro}_{\mathrm{coh}}(\mathbf{k}[z], \mathcal{C})}(\mathsf{M}, \mathbf{k}[z]z^{-k} \otimes_{\mathbf{k}[z]} \mathsf{N}).\end{aligned}$$

Note that we have

$$\mathrm{H{\scriptstyle OM}}_{\mathrm{Raff}_z(\mathcal{C})}(\mathsf{M}, \mathsf{N}) \simeq \mathbf{k}[z, z^{-1}] \otimes_{\mathbf{k}[z]} \mathrm{H{\scriptstyle OM}}_{\mathrm{Pro}_{\mathrm{coh}}(\mathbf{k}[z], \mathcal{C})}(\mathsf{M}, \mathsf{N}) \quad \text{for } \mathsf{M}, \mathsf{N} \in \mathrm{Pro}_{\mathrm{coh}}(\mathbf{k}[z], \mathcal{C}).$$

We call $\mathrm{Raff}_z(\mathcal{C})$ the category of rational affine objects of $\mathcal{C}$.

The category $\mathrm{Raff}_z(\mathcal{C})$ is an abelian category with $\mathbf{k}$-finite-dimensional hom. We have a faithful functor $\mathrm{Aff}_z(\mathcal{C}) \to \mathrm{Raff}_z(\mathcal{C})$, which is essentially surjective [15, Lemma 4.13].

3.3. **Affine objects.** Let $\mathcal{C}$ be an abelian graded $\mathbf{k}$-linear monoidal category which satisfies (3.4). The category $\mathrm{Aff}_z(\mathcal{C})$ has a monoidal category structure with $\otimes_z$ as a tensor product, where $\mathsf{M} \otimes_z \mathsf{N} := \mathrm{Coker}(\mathsf{M} \otimes \mathsf{N} \xrightarrow{z \otimes 1 - 1 \otimes z} \mathsf{M} \otimes \mathsf{N})$.

Proposition 3.6 ([15, Proposition 5.6])**.** *Assume that $\mathcal{C}$ is rigid. Then $\mathrm{Aff}_z(\mathcal{C})$ is a rigid monoidal category. We denote by ${\mathscr{D}_{\mathrm{aff}}}^{\pm 1}$ their right and left duality functors.*

Definition 3.7. Let $M, N \in \mathcal{C}$ be simple objects.

(i) If $\dim \mathrm{H{\scriptstyle OM}}(M \otimes N, N \otimes M) = 1$, then we say that the pair (M, N) is Λ-*definable*, and a non-zero morphism $\mathbf{r} \in \mathrm{H{\scriptstyle OM}}(M \otimes N, N \otimes M)$ is called the *R-matrix* between M and N, denoted by $\mathbf{r}_{M,N}$. It is well-defined up to a constant multiple. We set

$$\Lambda(M, N) := \deg(\mathbf{r}).$$

(ii) If (N, M) is also Λ-definable in addition, we say that (M, N) is $\mathfrak{d}$-*definable* and write

$$\mathfrak{d}(M, N) := \frac{1}{2}\big(\Lambda(M, N) + \Lambda(N, M)\big) \in \frac{1}{2}\mathbb{Z}.$$

3.4. **Rational centers and affinizations.** Let $\mathcal{C}$ be a $\mathbf{k}$-linear graded monoidal category which satisfies the following conditions:

$$\text{(3.5)}\quad \begin{cases} \mathcal{C} \text{ satisfies (3.4) and the following additional condition:} \\ \text{(e) } \mathcal{C} \text{ has a decomposition } \mathcal{C} = \bigoplus_{\lambda \in \mathsf{\Lambda}} \mathcal{C}_\lambda \text{ compatible with a monoidal} \\ \quad\ \text{structure where } \mathsf{\Lambda} \text{ is an abelian monoid, and } \mathbf{1} \in \mathcal{C}_\lambda \text{ with } \lambda = 0. \end{cases}$$

Let z be a homogeneous indeterminate with degree $d \in \mathbb{Z}_{>0}$.
Remark that we have also bi-exact bifunctors:

$$\otimes : \mathcal{C} \times \mathrm{Raff}_z(\mathcal{C}) \to \mathrm{Raff}_z(\mathcal{C}) \quad \text{and} \quad \otimes : \mathrm{Raff}_z(\mathcal{C}) \times \mathcal{C} \to \mathrm{Raff}_z(\mathcal{C}).$$

Definition 3.8. A *rational center* in $\mathcal{C}$ is a triple $(\mathsf{M}, \phi, \mathrm{R}_\mathsf{M})$ of $\mathsf{M} \in \mathrm{Aff}_z(\mathcal{C})$, an additive map $\phi \colon \mathsf{\Lambda} \to \mathbb{Z}$ and an isomorphism

$$\mathrm{R}_\mathsf{M}(X) \colon q^{\otimes \phi(\lambda)} \otimes \mathsf{M} \otimes X \xrightarrow{\ \sim\ } X \otimes \mathsf{M}$$

in $\mathrm{Raff}_z(\mathcal{C})$ functorial in $X \in \mathcal{C}_\lambda$ such that the diagrams

$$\begin{tikzcd} q^{\otimes \phi(\lambda+\mu)} \otimes \mathsf{M} \otimes X \otimes Y \arrow[r, "\mathrm{R}_\mathsf{M}(X) \otimes Y"] \arrow[dr, "\mathrm{R}_\mathsf{M}(X \otimes Y)"'] & q^{\otimes \phi(\mu)} \otimes X \otimes \mathsf{M} \otimes Y \arrow[d, "X \otimes \mathrm{R}_\mathsf{M}(Y)"] \\ & X \otimes Y \otimes \mathsf{M} \end{tikzcd}$$

and

$$\begin{tikzcd} \mathsf{M} \otimes \mathbf{1} \arrow[r, "\mathrm{R}_\mathsf{M}(\mathbf{1})"] \arrow[dr, "\sim"] & \mathbf{1} \otimes \mathsf{M} \arrow[d, "\wr"] \\ & \mathsf{M} \end{tikzcd}$$

commute in $\mathrm{Raff}_z(\mathcal{C})$ for any $X \in \mathcal{C}_\lambda$ and $Y \in \mathcal{C}_\mu$ $(\lambda, \mu \in \mathsf{\Lambda})$.

Note that the commutativity of the bottom diagram is a consequence of the one of the top.

In the sequel, we neglect grading shifts if there is no afraid of confusion. For example if we say that $f\colon M \to N$ is a morphism, it means that $f \in \mathrm{H{\scriptstyle OM}}(M,N)_d$ for some $d \in \mathbb{Z}$. If we want to emphasize d, we say that f is a morphism of homogeneous degree d.

Definition 3.9 ([15, Definition 6.7])**.** An *affinization* M of $M \in \mathcal{C}$ (of degree $d \in \mathbb{Z}_{>0}$) is an affine object $(\mathsf{M}, z_{\mathsf{M}})$ endowed with a rational center $(\mathsf{M}, \mathrm{R}_{\mathsf{M}})$ and an isomorphism $\mathsf{M}/z_{\mathsf{M}}\mathsf{M} \simeq M$ such that $\deg(z_{\mathsf{M}}) = d$.

We sometimes simply write M for affinization if no confusion arises.

Proposition 3.10 ([15, Proposition 6.9])**.** *Let* $(\mathsf{M}, \mathrm{R}_{\mathsf{M}})$ *be a rational center in* $\mathcal{C}$*, and let* $L \in \mathcal{C}$*. Assume that* M *and* L *do not vanish. Then there exist* $m \in \mathbb{Z}$ *and a morphism* $\mathrm{R}^{\mathrm{ren}}_{\mathsf{M},L}\colon \mathsf{M} \otimes L \to L \otimes \mathsf{M}$ *in* $\mathrm{Pro}_{\mathrm{coh}}(\mathbf{k}[z], \mathcal{C})$ *such that*

(a) $\mathrm{R}^{\mathrm{ren}}_{\mathsf{M},L}$ *is equal to* $z^m \mathrm{R}_{\mathsf{M}}(L)\colon \mathsf{M} \otimes L \to L \otimes \mathsf{M}$ *in* $\mathrm{Raff}_z(\mathcal{C})$*,*
(b) $\mathrm{R}^{\mathrm{ren}}_{\mathsf{M},L}|_{z=0}\colon (\mathsf{M}/z\mathsf{M}) \otimes L \to L \otimes (\mathsf{M}/z\mathsf{M})$ *does not vanish.*

Moreover such an integer m *and an* $\mathrm{R}^{\mathrm{ren}}_{\mathsf{M},L}$ *are unique.*

Similarly, there exist $m \in \mathbb{Z}$ *and a morphism* $\mathrm{R}^{\mathrm{ren}}_{L,\mathsf{M}}\colon L \otimes \mathsf{M} \to \mathsf{M} \otimes L$ *in* $\mathrm{Pro}_{\mathrm{coh}}(\mathbf{k}[z], \mathcal{C})$ *such that*

(c) $\mathrm{R}^{\mathrm{ren}}_{L,\mathsf{M}}$ *gives* $z^m \mathrm{R}_{\mathsf{M}}(L)^{-1}\colon L \otimes \mathsf{M} \to \mathsf{M} \otimes L$ *in* $\mathrm{Raff}_z(\mathcal{C})$*,*
(d) $\mathrm{R}^{\mathrm{ren}}_{L,\mathsf{M}}|_{z=0}\colon L \otimes (\mathsf{M}/z\mathsf{M}) \to (\mathsf{M}/z\mathsf{M}) \otimes L$ *does not vanish.*

When $\deg(z_{\mathsf{M}}) = \deg(z_{\mathsf{N}})$, since $\mathrm{R}^{\mathrm{ren}}_{\mathsf{M},\mathsf{N}}$ commutes with z_{M} and z_{N}, it induces a morphism in $\mathrm{Aff}_z(\mathcal{C})$

$$\mathsf{M} \otimes_z \mathsf{N} \xrightarrow{\overline{\mathrm{R}}^{\mathrm{ren}}_{\mathsf{M},\mathsf{N}}} \mathsf{N} \otimes_z \mathsf{M}\,, \tag{3.6}$$

which is denoted by $\overline{\mathrm{R}}^{\mathrm{ren}}_{\mathsf{M},\mathsf{N}}$. Here, z acts on M and N by z_{M} and z_{N}, respectively.

We say that a simple object M in a monoidal abelian category is called *real* if $M \otimes M$ is simple.

Definition 3.11. We say that a simple object $M \in \mathcal{C}$ is *affreal* if M is real and there is an affinization (M, z) of M. If $\deg(z) = d$, we say that M is *affreal of degree* d.

3.5. **Quasi-rigid Axiom.** Recall the duality in monoidal categories.

Lemma 3.12. *Assume that* $\mathcal{C}$ *is rigid. If* (M, N) *is a* Λ*-definable pair of simple modules, then* $(N, \mathscr{D}M)$ *is also* Λ*-definable and* $\Lambda(M, N) = \Lambda(N, \mathscr{D}M)$*. Moreover*

$\mathbf{r}_{N,\mathscr{D}M}$ *is obtained as the composition*

$$N \otimes \mathscr{D}M \to \mathscr{D}M \otimes M \otimes N \otimes \mathscr{D}M \xrightarrow{\mathbf{r}_{M,N}} \mathscr{D}M \otimes N \otimes M \otimes \mathscr{D}M \to \mathscr{D}M \otimes N.$$

Proof. It follows from

$$\mathrm{H{\scriptstyle OM}}(M \otimes N, N \otimes M) \simeq \mathrm{H{\scriptstyle OM}}(N, \mathscr{D}M \otimes N \otimes M) \simeq \mathrm{H{\scriptstyle OM}}(N \otimes \mathscr{D}M, \mathscr{D}M \otimes N).$$

□

Definition 3.13 ([15])**.** Let $(\mathcal{A}, \otimes)$ be a monoidal category. We say that $\mathcal{A}$ is a *quasi-rigid* monoidal category if it satisfies:

(a) $\mathcal{A}$ is abelian and $\otimes$ is bi-exact,
(b) for any $L, M, N \in \mathcal{A}$, $X \subset L \otimes M$ and $Y \subset M \otimes N$ such that $X \otimes N \subset L \otimes Y \subset L \otimes M \otimes N$, there exists $K \subset M$ such that $X \subset L \otimes K$ and $K \otimes N \subset Y$,
(c) for any $L, M, N \in \mathcal{A}$, $X \subset M \otimes N$ and $Y \subset L \otimes M$ such that $L \otimes X \subset Y \otimes N \subset L \otimes M \otimes N$, there exists $K \subset M$ such that $X \subset K \otimes N$ and $L \otimes K \subset Y$.

As for the following statements, see [14], [15, Sect.6], which will be needed in § 5:

Lemma 3.14. *Let M_k be non-zero object $(k = 1, 2, 3)$ of a quasi-rigid monoidal category $\mathcal{A}$, and let $\varphi_1 \colon L \to M_1 \otimes M_2$ and $\varphi_2 \colon M_2 \otimes M_3 \to L'$ be non-zero morphisms. Assume further that M_2 is simple. Then the composition*

$$L \otimes M_3 \xrightarrow{\varphi_1 \otimes M_3} M_1 \otimes M_2 \otimes M_3 \xrightarrow{M_1 \otimes \varphi_2} M_1 \otimes L'$$

does not vanish.

Lemma 3.15 ([15, Lemma 6.16])**.** *An abelian rigid monoidal category is quasi-rigid.*

Indeed, the notion "quasi-rigid" is rather natural since the monoidal categories appearing in this article are all quasi-rigid as seen in the example below.

Example 3.16. By [5, Lemma 3.1] we find that the monoidal categories R-gMod, R-gmod and $\mathscr{C}_w$ introduced later are quasi-rigid. By Lemma 3.15 above, we see that the category $\widetilde{\mathscr{C}_w}$ is also quasi-rigid since it is rigid [12].

Recall that for objects M, N in a monoidal category $(\mathcal{C}, \otimes)$, we denote by $M \nabla N$(resp. $M \Delta N$) the head (resp. socle) of $M \otimes N$.

Proposition 3.17 ([5], [15, Proposition 6.18, 6.19])**.** *Assume that $\mathcal{C}$ is quasi-rigid. Let M be an affreal object of $\mathcal{C}$ and N a simple object of $\mathcal{C}$. Then, we obtain*

(i) *$M \otimes N$ and $N \otimes M$ have simple heads and simple socles,*

(ii) *the pair* (M,N) *is* $\mathfrak{d}$*-definable* (*see Definition 3.7*) *and*

$$\mathrm{Hom}_{\mathcal{C}}(M\otimes N, N\otimes M)=\mathbf{k}\mathbf{r}_{M,N},\quad \mathrm{Hom}_{\mathcal{C}}(N\otimes M, M\otimes N)=\mathbf{k}\mathbf{r}_{N,M},$$
$$M\nabla N\simeq \mathrm{Im}(\mathbf{r}_{M,N})\simeq N\,\Delta\, M \quad \text{and} \quad N\nabla M\simeq \mathrm{Im}(\mathbf{r}_{N,M})\simeq M\,\Delta\, N$$

up to grading shifts,

(iii) $\Lambda(M, M\nabla N)=\Lambda(M,N)$ *and* $\Lambda(N\nabla M, M)=\Lambda(N,M)$,

(iv) *for any simple subquotient* S *of the radical* $\mathrm{Ker}(M\otimes N\to M\nabla N)$, *we have* $\Lambda(M,S)<\Lambda(M,N)$,

(v) *for any simple subquotient* S *of* $(M\otimes N)/(M\,\Delta\, N)$, *we have* $\Lambda(S,M)<\Lambda(N,M)$,

(vi) *for any simple subquotient* S *of the radical* $\mathrm{Ker}(N\otimes M\to N\nabla M)$, *we have* $\Lambda(S,M)<\Lambda(N,M)$,

(vii) *for any simple subquotient* S *of* $(N\otimes M)/(N\,\Delta\, M)$, *we have* $\Lambda(M,S)<\Lambda(M,N)$

(viii) *we have* $\mathrm{End}_{\,\mathcal{C}}(M\otimes N)=\mathbf{k}\,\mathsf{id}_{M\otimes N}$ *and* $\mathrm{End}_{\,\mathcal{C}}(N\otimes M)=\mathbf{k}\,\mathsf{id}_{N\otimes M}$,

(ix) $\mathfrak{d}(M, M^{\otimes n}\nabla N)\leqslant \max(\mathfrak{d}(M,N)-n,0)$ *for any* $n\in\mathbb{Z}_{\geqslant 0}$.

In particular, $M\nabla N$, *as well as* $M\,\Delta\, N$, *appears only once in the composition series of* $M\otimes N$ (*up to a grading*).

The analogous statement to the following lemma appears in [5, Corollary 3.13], [13, Lemma 2.7].

Lemma 3.18. *Assume that* $\mathcal{C}$ *is quasi-rigid. Let* L *be an affreal simple object of* $\mathcal{C}$. *Then for any simple* $M\in\mathcal{C}$, *we have*

$$(L\nabla M)\nabla \mathscr{D}L\simeq M \quad \text{and} \quad \mathscr{D}^{-1}L\nabla(M\nabla L)\simeq M.$$

Remark that this lemma is important to define the crystal structure on $\mathrm{Irr}(\widetilde{\mathscr{C}_w})$ later.

Proposition 3.19. *Assume that* $\mathcal{C}$ *is quasi-rigid. Let* L *be a real simple object in* $\mathcal{C}$ *with an affinization of degree* $2d\in\mathbb{Z}_{>0}$. *Then, we have*

(i) $\mathfrak{d}(L,X)\in d\mathbb{Z}_{\geqslant 0}$ *for any simple* $X\in\mathcal{C}$,

(ii) $\Lambda(L,M)+\Lambda(L,N)-\Lambda(L,S)\in 2d\mathbb{Z}_{\geqslant 0}$ *for any simple* M, N *and any simple subquotient* S *of* $M\otimes N$.

Proof. Let $(\mathsf{L},\mathrm{R}_{\mathsf{L}})$ be an affinization with a spectral parameter z with $\deg(z)=2d$. For a simple $X\in\mathcal{C}$, there exists $n_X\in\mathbb{Z}$ such that $z^{n_X}\mathrm{R}_{\mathsf{L}}(X)\colon \mathsf{L}\otimes X\to X\otimes\mathsf{L}$ is a morphism in $\mathrm{Aff}_z(\mathcal{C})$ and does not vanish at $z=0$. Its degree is $\Lambda(L,X)$. Similarly, there exists $m_X\in\mathbb{Z}$ such that $z^{m_X}\mathrm{R}_{\mathsf{L}}(X)^{-1}\colon X\otimes\mathsf{L}\to\mathsf{L}\otimes X$ is a morphism in $\mathrm{Aff}_z(\mathcal{C})$ and does not vanish at $z=0$. Its degree coincides with $\Lambda(X,L)$.

(i) The composition $z^{m_X+n_X}\,\mathsf{id}_{\mathsf{L}\otimes X}$ is an endomorphism of $\Lambda\otimes X$ in $\mathrm{Aff}_z(\mathcal{C})$ of degree $2(m_X+n_X)d \geqslant 0$, Since it coincides with $\Lambda(L,X)+\Lambda(X,L) = 2\mathfrak{d}(L,X)$. Hence $\mathfrak{d}(L,X)\in\mathbb{Z}_{\geqslant 0}d$.

(ii) $z^{n_M+n_N}\mathrm{R}_{\mathsf{L}}(M\otimes N)\colon \mathsf{L}\otimes(M\otimes N)\to(M\otimes N)\otimes\mathsf{L}$ is a morphism in $\mathrm{Aff}_z(\mathcal{C})$. Hence it induces a morphism $z^{n_M+n_N}\mathrm{R}_{\mathsf{L}}(S)\colon \mathsf{L}\otimes S\to S\otimes\mathsf{L}$ in $\mathrm{Aff}_z(\mathcal{C})$. Hence there exists $s\in\mathbb{Z}_{\geqslant 0}$ such that $z^{n_M+n_N-s}\mathrm{R}_{\mathsf{L}}(S)\colon \mathsf{L}\otimes S\to S\otimes\mathsf{L}$ is a morphism in $\mathrm{Aff}_z(\mathcal{C})$ and does not vanish. Since we have

$$\begin{aligned}\Lambda(L,S) &= \deg\bigl(z^{n_M+n_N-s}\mathrm{R}_{\mathsf{L}}(S)\bigr) = \deg\bigl(z^{n_M+n_N-s}\mathrm{R}_{\mathsf{L}}(M\otimes N)\bigr)\\ &= \Lambda(L,M)+\Lambda(L,N)-\deg(z^s),\end{aligned}$$

which implies the desired result. □

Remark 3.20. It follows from Example 3.16 that we can apply the results of this subsection to the categories R-gmod, $\mathscr{C}_w$ and $\widetilde{\mathscr{C}_w}$.

4. Quiver Hecke algebras and their modules

4.1. **Quiver Hecke algebras.** In the sequel, $\mathbf{k}$ is a base field. Let $\bigl(\mathsf{C},\mathsf{P},\Pi,\Pi^\vee,(\cdot,\cdot)\bigr)$ be a Cartan datum. A *quiver Hecke datum* is a family of polynomials $(Q_{i,j}(u,v)\in\mathbf{k}[u,v])_{i,j\in I}$ of the form

$$Q_{i,j}(u,v)=\begin{cases}\displaystyle\sum_{p(\alpha_i,\alpha_i)+q(\alpha_j,\alpha_j)=-2(\alpha_i,\alpha_j)} t_{i,j;p,q}u^pv^q & \text{if } i\neq j,\\ 0 & \text{if } i=j,\end{cases} \tag{4.1}$$

such that $t_{i,j;-a_{ij},0}\in\mathbf{k}^\times$ and $Q_{i,j}(u,v)=Q_{j,i}(v,u)$ for all $i,j\in I$. We set

$$\overline{Q}_{i,j}(u,v,w) := \frac{Q_{i,j}(u,v)-Q_{i,j}(w,v)}{u-w}\in\mathbf{k}[u,v,w].$$

For $\beta\in\mathsf{Q}_+$, the set $I^\beta := \Bigl\{\nu=(\nu_1,\ldots,\nu_n)\in I^n \bigm| \sum_{k=1}^n\alpha_{\nu_k}=\beta\Bigr\}$ is stable under the action of the symmetric group $\mathfrak{S}_n=\langle s_k \mid k=1,\ldots,n-1\rangle$ given by place permutations.

The height of $\beta=\sum_{i\in I}b_i\alpha_i\in\mathsf{Q}$ is given by $\mathrm{ht}(\beta):=\sum_{i\in I}|b_i|$.

Definition 4.1. Let $\beta\in\mathsf{Q}_+$ with $\mathrm{ht}(\beta)=n$. The *quiver Hecke algebra* $R(\beta)$ associated with a Cartan datum $\bigl(\mathsf{C},\Pi,\mathsf{P},\Pi^\vee,(\cdot,\cdot)\bigr)$ and a quiver Hecke datum $(Q_{i,j}(u,v))_{i,j\in I}$ is the $\mathbf{k}$-algebra generated by

$$\{e(\nu)\mid \nu\in I^\beta\},\ \{x_k\mid 1\leqslant k\leqslant n\},\ \text{and } \{\tau_l\mid 1\leqslant l\leqslant n-1\}$$

subject to the defining relations:

$$
\begin{aligned}
& e(\nu)e(\nu') = \delta_{\nu,\nu'}e(\nu),\ \sum_{\nu\in I^\beta} e(\nu) = 1,\ x_k e(\nu) = e(\nu)x_k,\ x_k x_l = x_l x_k,\\
& \tau_l e(\nu) = e(s_l(\nu))\tau_l,\ \tau_k\tau_l = \tau_l\tau_k \text{ if } |k-l|>1,\\
& \tau_k^2 = \sum_{\nu\in I^\beta} Q_{\nu_k,\nu_{k+1}}(x_k,x_{k+1})e(\nu),\\
& \tau_k x_l - x_{s_k(l)}\tau_k = \big(\delta(l=k+1)-\delta(l=k)\big)\sum_{\nu\in I^\beta,\ \nu_k=\nu_{k+1}} e(\nu),\\
& \tau_{k+1}\tau_k\tau_{k+1} - \tau_k\tau_{k+1}\tau_k = \sum_{\nu\in I^\beta,\ \nu_k=\nu_{k+2}} \overline{Q}_{\nu_k,\nu_{k+1}}(x_k,x_{k+1},x_{k+2})e(\nu).
\end{aligned}
$$

The algebra $R(\beta)$ is equipped with the $\mathbb{Z}$-grading given by

$$\deg(e(\nu)) = 0,\quad \deg(x_k e(\nu)) = (\alpha_{\nu_k},\alpha_{\nu_k}),\quad \deg(\tau_l e(\nu)) = -(\alpha_{\nu_l},\alpha_{\nu_{l+1}}). \tag{4.2}$$

We denote by $R(\beta)$-gMod the category of graded $R(\beta)$-modules with degree preserving homomorphisms. We write $R(\beta)$-gmod for the full subcategory of $R(\beta)$-gMod consisting of the graded modules which are finite-dimensional over $\mathbf{k}$, and $R(\beta)$-proj for the full subcategory of $R(\beta)$-gMod consisting of finitely generated projective graded $R(\beta)$-modules. We set R-gMod$:=\bigoplus_{\beta\in\mathsf{Q}_+} R(\beta)$-gMod, R-proj$:=\bigoplus_{\beta\in\mathsf{Q}_+} R(\beta)$-proj, and R-gmod$:=\bigoplus_{\beta\in\mathsf{Q}_+} R(\beta)$-gmod. The trivial $R(0)$-module $\mathbf{k}$ with degree 0 is denoted by $\mathbf{1}$. For simplicity, we write "a module" instead of "a graded module". We define the grading shift functor q by $(qM)_k = M_{k-1}$ for a graded module $M = \bigoplus_{k\in\mathbb{Z}} M_k$. For $M,N \in R(\beta)$-gMod, $\mathrm{Hom}_{R(\beta)}(M,N)$ denotes the space of degree preserving module homomorphisms. We define

$$\mathrm{H{\scriptsize OM}}_{R(\beta)}(M,N) := \bigoplus_{k\in\mathbb{Z}} \mathrm{H{\scriptsize OM}}_{R(\beta)}(M,N)_k \quad \text{with } \mathrm{H{\scriptsize OM}}_{R(\beta)}(M,N)_k = \mathrm{Hom}_{R(\beta)}(q^k M,N),$$

and set $\deg(f) := k$ for $f\in \mathrm{H{\scriptsize OM}}_{R(\beta)}(M,N)_k$. When $M=N$, we write $\mathrm{E{\scriptsize ND}}_{R(\beta)}(M) = \mathrm{H{\scriptsize OM}}_{R(\beta)}(M,M)$. We sometimes write R for $R(\beta)$ in $\mathrm{H{\scriptsize OM}}_{R(\beta)}(M,N)$ for simplicity.

For $\beta,\beta'\in\mathsf{Q}_+$, set $e(\beta,\beta') := \sum_{\nu\in I^\beta,\nu'\in I^{\beta'}} e(\nu*\nu')$, where $\nu*\nu'$ is the concatenation of ν and ν'. Then there is an injective ring homomorphism

$$R(\beta)\otimes R(\beta') \to e(\beta,\beta')R(\beta+\beta')e(\beta,\beta')$$

given by $e(\nu)\otimes e(\nu')\mapsto e(\nu,\nu')$, $x_k e(\beta)\otimes 1 \mapsto x_k e(\beta,\beta')$, $1\otimes x_k e(\beta')\mapsto x_{k+\mathrm{ht}(\beta)}e(\beta,\beta')$, $\tau_k e(\beta)\otimes 1\mapsto \tau_k e(\beta,\beta')$ and $1\otimes \tau_k e(\beta')\mapsto \tau_{k+\mathrm{ht}(\beta)}e(\beta,\beta')$. For $a\in R(\beta)$ and $a'\in R(\beta')$, the image of $a\otimes a'$ is sometimes denoted by $a\boxtimes a'$.

For $M \in R(\beta)$-gMod and $N \in R(\beta')$-gMod, we set

$$M \circ N := R(\beta+\beta')e(\beta,\beta') \otimes_{R(\beta)\otimes R(\beta')} (M \otimes N).$$

For $u \in M$ and $v \in N$, the image of $u \otimes v$ by the map $M \otimes N \to M \circ N$ is sometimes denoted by $u \boxtimes v$. We also write $M \boxtimes N \subset M \circ N$ for the image of $M \otimes N$ in $M \circ N$.

For $\alpha, \beta \in \mathsf{Q}_+$, let X be an $R(\alpha+\beta)$-module. Then $e(\alpha,\beta)X$ is an $R(\alpha) \otimes R(\beta)$-module. We denote it by $\mathrm{Res}_{\alpha,\beta}X$. We have

$$\begin{aligned} &\mathrm{Hom}_{R(\alpha)\otimes R(\beta)}\big(M \otimes N, \mathrm{Res}_{\alpha,\beta}(X)\big) \simeq \mathrm{Hom}_{R(\alpha+\beta)}(M \circ N, X), \\ &\mathrm{Hom}_{R(\alpha)\otimes R(\beta)}\big(\mathrm{Res}_{\alpha,\beta}(X), M \otimes N\big) \simeq \mathrm{Hom}_{R(\alpha+\beta)}(X, q^{(\alpha,\beta)} N \circ M) \end{aligned} \tag{4.3}$$

for any $R(\alpha)$-module M, any $R(\beta)$-module N and any $R(\alpha+\beta)$-module X.

Recall that we denote by $M \nabla N$ the head of $M \circ N$ and by $M \Delta N$ the socle of $M \circ N$. We say that simple R-modules M and N *strongly commute* if $M \circ N$ is simple. Recall that a simple R-module L is *real* if L strongly commutes with itself, that is, $L \circ L$ is simple. Note that if M and N strongly commute, then M and N commute, i.e., $M \circ N \simeq N \circ M$ up to a grading shift.

For $i \in I$, the functors E_i and F_i are defined by

$$\begin{aligned} &\mathrm{E}_i(M) = e(\alpha_i, \beta-\alpha_i)M \in R(\beta-\alpha_i)\text{-gMod} \\ &\mathrm{F}_i(M) = R(\alpha_i) \circ M \in R(\beta+\alpha_i)\text{-gMod} \end{aligned} \qquad \text{for any } R(\beta)\text{-module } M.$$

For $i \in I$ and $n \in \mathbb{Z}_{>0}$, let $L(i)$ be the simple $R(\alpha_i)$-module concentrated on degree 0 and $P(i^n)$ the indecomposable projective $R(n\alpha_i)$-module whose head is isomorphic to $L(i^n) = \langle i^n \rangle := q_i^{\frac{n(n-1)}{2}} L(i)^{\circ n}$, where $q_i := q^{(\alpha_i,\alpha_i)/2}$. We use the both notations $L(i^n)$ and $\langle i^n \rangle$ in the sequel.

Then, for $M \in R(\beta)$-gMod, we define

$$\begin{aligned} &\mathrm{E}_i^{(n)} M := \mathrm{H{\scriptstyle OM}}_{R(n\alpha_i)}\big(P(i^n),\, e(n\alpha_i, \beta-n\alpha_i)M\big) \in R(\beta-n\alpha_i)\text{-gMod}, \\ &\mathrm{F}_i^{(n)} M := P(i^n) \circ M \in R(\beta+n\alpha_i)\text{-gMod}. \end{aligned}$$

For $i \in I$ and a non-zero $M \in R(\beta)$-gMod, we define

$$\begin{aligned} \mathrm{wt}(M) = -\beta, \qquad &\varepsilon_i(M) = \max\{k \geqslant 0 \mid E_i^k M \not\simeq 0\}, \\ &\varphi_i(M) = \varepsilon_i(M) + \langle h_i, \mathrm{wt}(M) \rangle. \end{aligned} \tag{4.4}$$

We can also define E_i^*, F_i^*, ε_i^*, etc. in the same manner as above if we replace $e(\alpha_i, \beta-\alpha_i)$, $R(\alpha_i) \circ -$, etc. with $e(\beta-\alpha_i, \alpha_i)$, $- \circ R(\alpha_i)$, etc.

We denote by $K(R\text{-proj})$ and $K(R\text{-gmod})$ the Grothendieck groups of R-proj and R-gmod, respectively.

Let $\mathrm{Irr}(R\text{-gmod})$ be the set of the equivalence classes of simple modules in R-gmod up to grading shifts. Namely, for simple modules M and N, we identify M and N in $\mathrm{Irr}(R\text{-gmod})$ if $M \simeq q^n N$ for some $n \in \mathbb{Z}$.

For a simple module $M \in R$-gmod, define

$$\widetilde{\mathrm{E}}_i(M) := q_i^{\varepsilon_i(M)-1}\mathrm{hd}\,(\mathrm{E}_i M) \simeq q_i^{1-\varepsilon_i(M)}\mathrm{soc}\,(\mathrm{E}_i M) \quad \text{and} \quad \widetilde{\mathrm{F}}_i(M) := q_i^{\varepsilon_i(M)}\langle i\rangle \mathbin{\nabla} M. \tag{4.5}$$

We set

$$\widetilde{\mathrm{E}}_i^{\max}(M) := \widetilde{\mathrm{E}}_i^{\varepsilon_i(M)} M \simeq \mathrm{E}_i^{(\varepsilon_i(M))} M.$$

The following theorem is due to Lauda-Vazirani.

Theorem 4.2 ([24])**.** *The 6-tuple* $\big(\mathrm{Irr}(R\text{-gmod}), \mathrm{wt}, \{\varepsilon_i\}_{i\in I}, \{\varphi_i\}_{i\in I}, \{\widetilde{\mathrm{E}}_i\}_{i\in I}, \{\widetilde{\mathrm{F}}_i\}_{i\in I}\big)$ *is a crystal and there exists an isomorphism of crystals:*

$$\Psi\colon \mathrm{Irr}(R\text{-gmod}) \xrightarrow{\sim} B(\infty) \quad (\mathbf{1} \mapsto u_\infty).$$

Remark 4.3 (cf. [23])**.** Let $\beta \in \mathsf{Q}_+$ and let S be a subset of I^β satisfying the following conditions:

(a) if $\mu = (\mu_1, \dots, \mu_m) \in S$, then $\mu_k \neq \mu_{k+1}$ for any k such that $1 \leqslant k < n$,
(b) for any $\mu = (\mu_1, \dots, \mu_m) \in S$ and k such that $1 \leqslant k < n$, $s_k\mu \in S$ if $(\alpha_{\mu_k}, \alpha_{\mu_{k+1}}) = 0$,
(c) if $\mu = (\mu_1, \dots, \mu_m) \in S$ and $\mu_k = \mu_{k+2}$ for k such that $1 \leqslant k \leqslant n-2$, we have $\langle h_{\mu_k}, \alpha_{\mu_{k+1}}\rangle < -1$.

Then we can easily check that the vector space $M := \bigoplus_{\mu\in S} \mathbf{k}\, u_\mu$ with basis $\{u_\mu\}_{\mu\in S}$ has a structure of $R(\beta)$-module such that $\deg(u_\mu) = 0$, $e(\mu)u_\mu = u_\mu$ $x_k|_M = 0$ and
$\tau_k u_\mu = \begin{cases} u_{s_k\mu} & \text{if } (\alpha_{\mu_k}, \alpha_{\mu_{k+1}}) = 0, \\ 0 & \text{otherwise.}\end{cases}$
Here we assume that $Q_{i,j}(u,v) = 1$ when $(\alpha_i, \alpha_j) = 0$ without loss of generality. If S is a minimal one among the non-empty subsets satisfying the above conditions, then M is a simple module. Sometimes, M is denoted by $\langle \nu_1, \dots, \nu_n\rangle$ taking a member $\langle \nu_1, \dots, \nu_n\rangle$ of S. With this terminology, $\langle i\rangle$ and $\langle i, j\rangle$ $(i \neq j)$ are such examples. We have $\langle i\rangle = L(i)$, Thus, in the sequel, we use frequently $\langle i\rangle$ instead of $L(i)$ for an abbreviation.

4.2. **Duality in** R**-gmod.** For $M \in R(\beta)$-gmod, we set $M^\star := \mathrm{H{\scriptstyle OM}}_{\mathbf{k}}(M, \mathbf{k})$ with the $R(\beta)$-action given by

$$(a \cdot f)(u) := f(\rho(a)u), \quad \text{for } f \in M^\star,\ a \in R(\beta) \text{ and } u \in M,$$

where ρ is the antiautomorphism of $R(\beta)$ which fixes the generators $e(\nu)$, x_k, τ_l. We say that $M \in R$-gmod is *self-dual* if $M \simeq M^\star$ in R-gmod. The following lemma is well-known.

Lemma 4.4 (cf. [11])**.** *For any simple $M \in R$-gmod, there exists a unique $n \in \mathbb{Z}$ such that $q^n M$ is self-dual.*

It means that any simple module in R-gmod is isomorphic to its dual up to a grading shift.

We have

$$(M \circ N)^\star \simeq q^{(\operatorname{wt} M, \operatorname{wt} N)} N^\star \circ M^\star \quad \text{functorially in } M, N \in R\text{-gmod}.$$

Lemma 4.5. *Let M and N be simple modules in R-gmod.*

(i) *Any simple quotient of $M \circ N$ appears as a simple submodule of $N \circ M$ (up to grading shifts). Conversely, any simple submodule of $M \circ N$ appears as a simple quotient of $N \circ M$.*

(ii) *There exists a morphism $f\colon M \circ N \to N \circ M$ such that $\operatorname{Im}(f)$ is simple.*

(iii) *If (M, N) is a Λ-definable pair (see Definition 3.7), then $M \circ N$ has a simple head and $N \circ M$ has a simple socle. Moreover, we have*

$$M \nabla N \simeq \operatorname{Im}(\mathbf{r}_{M,N}) \simeq N \Delta M.$$

Proof. If $M \circ N \twoheadrightarrow S$ is a simple quotient of $M \circ N$, then, applying *, we obtain

$$S \simeq S^\star \rightarrowtail (M \circ N)^\star \simeq N^\star \circ M^\star \simeq N \circ M.$$

Hence S appears as a simple submodule of $N \circ M$. Other statements are now obvious. □

4.3. **R-matrices.** Let $\beta \in \mathsf{Q}_+$ with $m = \operatorname{ht}(\beta)$. For $k = 1, \ldots, m-1$ and $\nu \in I^\beta$, the *intertwiner* $\varphi_k \in R(\beta)$ is defined by

$$\varphi_k e(\nu) = \begin{cases} \big(\tau_k(x_k - x_{k+1}) + 1\big) e(\nu) & \text{if } \nu_k = \nu_{k+1}, \\ \tau_k e(\nu) & \text{otherwise.} \end{cases} \tag{4.6}$$

Since $\{\varphi_k\}_{1 \leqslant k < m}$ satisfies the braid relation, we can define φ_w for any element w of the symmetric group $\mathfrak{S}_m$ of degree m.

For $m, n \in \mathbb{Z}_{\geqslant 0}$, we set $w[m, n]$ to be the element of $\mathfrak{S}_{m+n}$ such that

$$w[m, n](k) := \begin{cases} k + n & \text{if } 1 \leqslant k \leqslant m, \\ k - m & \text{if } m < k \leqslant m + n. \end{cases}$$

For $M \in \mathrm{Mod}_{\mathrm{gr}}\big(R(\beta)\big)$ and $N \in \mathrm{Mod}_{\mathrm{gr}}\big(R(\gamma)\big)$, the map $M \otimes N \to N \circ M$ given by

$$u \otimes v \mapsto \varphi_{w[\mathrm{ht}(\gamma),\mathrm{ht}(\beta)]}(v \boxtimes u)$$

is $R(\beta) \otimes R(\gamma)$-linear, and hence it extends to an $R(\beta+\gamma)$-module homomorphism (neglecting a grading shift)

$$\mathrm{R}^{\mathrm{univ}}_{M,N} \colon M \circ N \longrightarrow N \circ M.$$

We call it the *universal R-matrix*.

For $\beta \in \mathsf{Q}_+$ and $i \in I$, let $\mathfrak{p}_{i,\beta}$ be an element in the center $Z(R(\beta))$ of $R(\beta)$ given by

$$\mathfrak{p}_{i,\beta} := \sum_{\nu \in I^\beta} \Big(\prod_{a \in \{1,\ldots,\mathrm{ht}(\beta)\},\ \nu_a = i} x_a \Big) e(\nu) \in Z(R(\beta)). \tag{4.7}$$

Definition 4.6 ([18])**.** For a simple module $M \in R(\beta)$-gmod, an *affinization of* M is a pair $(\mathsf{M}, z_{\mathsf{M}})$ of an $R(\beta)$-module M and an endomorphism z_{M} of M with degree $d_{\mathsf{M}} \in \mathbb{Z}_{>0}$ such that

(4.8)
(i) $\mathsf{M}/z_{\mathsf{M}}\mathsf{M} \simeq M$,
(ii) M is a finitely generated free module over the polynomial ring $\mathbf{k}[z_{\mathsf{M}}]$,
(iii) $\mathfrak{p}_{i,\beta}\mathsf{M} \neq 0$ for all $i \in I$.

We recall that a real simple module is *affreal* if it admits an affinization.

Note that, if $(\mathsf{M}, z_{\mathsf{M}})$ is an affinization of M, then $(\mathsf{M}, \mathrm{R}^{\mathrm{univ}}_{\mathsf{M}})$ is an affinization of M in the sense of Definition 3.9 ([18]). Here, $\mathrm{R}^{\mathrm{univ}}_{\mathsf{M}}(X) \colon \mathsf{M} \circ X \to X \circ \mathsf{M}$ is the universal R-matrix, and we identify M with "$\varprojlim_k$" $\mathsf{M}/z_{\mathsf{M}}^k\mathsf{M} \in \mathrm{Aff}_z(R\text{-gmod})$. Hence we can apply the results in § 3.5, in particular Proposition 3.17.

Example 4.7. The simple R-module $L(i) = \langle i \rangle$ is affreal.

4.4. **Normal sequences.** Recall the definition of $\Lambda(M,N)$ and $\mathfrak{d}(M,N)$ in Definition 3.7. We define

$$\widetilde{\Lambda}(M,N) := \frac{1}{2}\big(\Lambda(M,N) + (\mathrm{wt}(M), \mathrm{wt}(N))\big)$$

for a Λ-definable pair (M,N) of simple modules in R-gmod.

Note that $\mathfrak{d}(M,N)$ and $\widetilde{\Lambda}(M,N)$ are non-negative integers ([11, Lemma 3.11]) as soon as one of M and N is affreal.

Let $\mathcal{C}$ be a quasi-rigid monoidal category satisfying (3.4).

Definition 4.8. Let $(M_1, \ldots, M_r)$ be a sequence of simple objects in $\mathcal{C}$.

(i) We say that $(M_1, \dots, M_r)$ is *almost affreal* if M_k is affreal except for possibly one k.

(ii) A sequence $(M_1, \dots, M_r)$ of simple objects is called a *normal sequence* if (M_j, M_k) is Λ-definable for any $1 \leqslant j < k \leqslant r$ and the composition of r-matrices

$$\begin{aligned}\mathbf{r}_{M_1,\dots,M_r} &:= (\mathbf{r}_{M_{r-1},M_r}) \circ \cdots \circ (\mathbf{r}_{M_2,M_r} \circ \cdots \circ \mathbf{r}_{M_2,M_3}) \circ (\mathbf{r}_{M_1,M_r} \circ \cdots \circ \mathbf{r}_{M_1,M_2}) \\ &: M_1 \circ \cdots \circ M_r \longrightarrow M_r \circ \cdots \circ M_1\end{aligned}$$

does not vanish.

Lemma 4.9 ([12, § 2.2])**.** *If $(M_1, \dots, M_r)$ be an almost affreal normal sequence in $\mathcal{C}$, then $\operatorname{Im}(\mathbf{r}_{M_1,\dots,M_r})$ is simple, and it is isomorphic to the head of $M_1 \circ \cdots \circ M_r$ and to the socle of $M_r \circ \cdots \circ M_1$.*

Lemma 4.10 ([9, Lemma 2.7, Lemma 2.8])**.** *Let $(L_1, \dots, L_r)$ be an almost affreal sequence of simple modules in $\mathcal{C}$. Then we have*

(i) *Assume that L_1 is affreal. Then $(L_1, \dots, L_r)$ is a normal sequence if and only if $(L_2, \dots, L_r)$ is a normal sequence and*

$$\Lambda(L_1, \operatorname{hd}(L_2 \circ \cdots \circ L_r)) = \sum_{2 \leqslant j \leqslant r} \Lambda(L_1, L_j).$$

(ii) *Assume that L_r is affreal. Then, $(L_1, \dots, L_r)$ is a normal sequence if and only if $(L_1, \dots, L_{r-1})$ is a normal sequence and*

$$\Lambda(\operatorname{hd}(L_1 \circ \cdots \circ L_{r-1}), L_r) = \sum_{1 \leqslant j \leqslant r-1} \Lambda(L_j, L_r).$$

Lemma 4.11. *Let (L, M, N) be an almost affreal sequence of simples in $\mathcal{C}$.*

(i) *If L and M commute, then (L, M, N) is a normal sequence.*

(ii) *Assume that $\mathcal{C}$ is rigid. Then, (L, M, N) is a normal sequence if and only if $(M, N, \mathscr{D}L)$ is normal.*

(iii) *Assume that $\mathcal{C}$ is rigid and that $\mathscr{D}L$ and N commute. Then, (L, M, N) is a normal sequence.*

Proof. (i) follows from $\Lambda(L \nabla M, N) = \Lambda(L, N) + \Lambda(M, N)$ and Lemma 4.10 (ii).

(ii) Let

$$\mathbf{r}_{L,M,N} : L \otimes M \otimes N \xrightarrow{\mathbf{r}_{M,N}} L \otimes N \otimes M \xrightarrow{\mathbf{r}_{L,N}} N \otimes L \otimes M \xrightarrow{\mathbf{r}_{L,M}} N \otimes M \otimes L$$

be the R-matrix. We see by Lemma 3.12 that $\mathbf{r}_{M,N,\mathscr{D}L}$ is obtained as the composition

$$\begin{tikzcd}
M \otimes N \otimes \mathscr{D}L \arrow[rr, "\mathbf{r}_{M,N,\mathscr{D}L}"] \arrow[d] & & \mathscr{D}L \otimes N \otimes M \\
\mathscr{D}L \otimes L \otimes M \otimes N \otimes \mathscr{D}L \arrow[rr, "\mathscr{D}L \otimes \mathbf{r}_{L,M,N} \otimes \mathscr{D}L"] & & \mathscr{D}L \otimes N \otimes M \otimes L \otimes \mathscr{D}L. \arrow[u]
\end{tikzcd}$$

Hence if $\mathbf{r}_{M,N,\mathscr{D}L}$ does not vanish, then $\mathbf{r}_{L,M,N}$ does not vanish. Similarly $\mathbf{r}_{L,M,N}$ is obtained as the composition

$$\begin{tikzcd}
L \otimes M \otimes N \arrow[rr, "\mathbf{r}_{L,M,N}"] \arrow[d] & & N \otimes M \otimes L \\
L \otimes M \otimes N \otimes \mathscr{D}L \otimes L \arrow[rr, "L \otimes \mathbf{r}_{M,N,\mathscr{D}L} \otimes L"] & & L \otimes \mathscr{D}L \otimes N \otimes M \otimes L. \arrow[u]
\end{tikzcd}$$

Hence if $\mathbf{r}_{L,M,N}$ does not vanish, then $\mathbf{r}_{M,N,\mathscr{D}L}$ does not vanish.

(iii) follows from (i) and (ii).

□

Proposition 4.12 ([12, Proposition 2.9]). *Let (L, M, N) be an almost affreal sequence of simple modules in R-gmod. If $\widetilde{\Lambda}(L, N) = 0$ and one of L and N is affreal, then (L, M, N) is a normal sequence.*

4.5. **Head simplicity of convolutions with $L(i)$.**

Definition 4.13. For $i \in I$, $\beta \in \mathsf{Q}_+$ and a simple $R(\beta)$-module M, define

$$\mathfrak{d}_i(M) := \varepsilon_i(M) + \varepsilon_i^*(M) + \langle h_i, \mathrm{wt}(M) \rangle.$$

Recall the following lemma.

Lemma 4.14 ([10, Corollary 3.8]). *For $i \in I$, $\beta \in \mathsf{Q}_+$ and a simple module $R(\beta)$-module M, we have*

$$\begin{aligned}
\Lambda(L(i), M) &= (\alpha_i, \alpha_i)\varepsilon_i(M) + (\alpha_i, \mathrm{wt}(M)), \\
\Lambda(M, L(i)) &= (\alpha_i, \alpha_i)\varepsilon_i^*(M) + (\alpha_i, \mathrm{wt}(M)), \\
\widetilde{\Lambda}(L(i), M) &= \frac{(\alpha_i, \alpha_i)}{2}\varepsilon_i(M), \qquad \widetilde{\Lambda}(M, L(i)) = \frac{(\alpha_i, \alpha_i)}{2}\varepsilon_i^*(M), \\
\mathfrak{d}(L(i), M) &= \frac{(\alpha_i, \alpha_i)}{2}\,\mathfrak{d}_i(M).
\end{aligned}$$

Proposition 4.15 (cf. [24]). *Let $i \in I$, $n \in \mathbb{Z}_{\geqslant 0}$ and let M be a simple module.*

(i) *If* $\mathfrak{d}_i(M) = 0$*, then we have* $L(i) \nabla M \simeq L(i) \circ M \simeq M \circ L(i) \simeq M \nabla L(i)$ *(up to grading shifts) and* $\mathfrak{d}_i(L(i) \circ M) = 0$*.*
If $\mathfrak{d}_i(M) > 0$*, then we have*

$$\mathfrak{d}_i(L(i) \nabla M) = \mathfrak{d}_i(M \nabla L(i)) = \mathfrak{d}_i(M) - 1,$$
$$\varepsilon_i(M \nabla L(i)) = \varepsilon_i(M), \quad \varepsilon_i^*(L(i) \nabla M) = \varepsilon_i^*(M).$$

(ii) *We have*

$$\begin{aligned} \mathfrak{d}_i(L(i^n) \nabla M) &= \max\bigl(\mathfrak{d}_i(M) - n, 0\bigr), \\ \mathfrak{d}_i(M \nabla L(i^n)) &= \max\bigl(\mathfrak{d}_i(M) - n, 0\bigr). \end{aligned} \tag{4.9}$$

(iii) *We have*

$$\varepsilon_i(M \nabla L(i^n)) = \max\bigl(\varepsilon_i(M),\ n - \mathrm{wt}_i(M) - \varepsilon_i^*(M)\bigr),$$
$$\varepsilon_i^*(L(i^n) \nabla M) = \max\bigl(\varepsilon_i^*(M),\ n - \mathrm{wt}_i(M) - \varepsilon_i(M)\bigr).$$

4.6. **Determinantial modules.** We define the partial order $\preccurlyeq$ on P as follows: $\lambda \preccurlyeq \mu$ for $\lambda, \mu \in \mathsf{P}$ if there exists a sequence of positive real roots $\beta_1, \ldots, \beta_r$ such that $(\beta_k, s_{\beta_{k+1}} s_{\beta_{k+2}} \cdots s_{\beta_r} \mu) > 0$ for all $1 \leqslant k \leqslant r$ and $\lambda = s_{\beta_1} s_{\beta_2} \cdots s_{\beta_r} \mu$. We have $\mu - \lambda \in \mathsf{Q}_+$ if $\lambda \preccurlyeq \mu$. Hence $\preccurlyeq$ is a partial order on P.

Assume that $\lambda, \mu \in \mathsf{W}\Lambda$ for some $\Lambda \in \mathsf{P}_+$. Then $\lambda \preccurlyeq \mu$ if and only if there exist $w, v \in \mathsf{W}$ such that $\lambda = v\Lambda$, $\mu = v\Lambda$ and $w \geqslant v$.

For $\Lambda \in \mathsf{P}_+$ and $\lambda, \mu \in \mathsf{W}\Lambda$ such that $\lambda \preccurlyeq \mu$, there exists a self-dual simple module $\mathsf{M}(\lambda, \mu)$ in $R(\mu - \lambda)$-gmod, called the *determinantial module*. (See [11, Section 3.3] for the precise definition and more properties of them.)

Proposition 4.16 ([10, Lemma 1.7, Proposition 4.2], [11, Lemma 3.23, Theorem 3.26])**.** *Let* $\Lambda \in \mathsf{P}_+$*, and* $\lambda, \mu \in \mathsf{W}\Lambda$ *with* $\lambda \preceq \mu$*.*

(i) $\mathsf{M}(\lambda, \mu)$ *is an affreal simple module.*
(ii) *If* $\langle h_i, \lambda \rangle \leqslant 0$ *and* $s_i\lambda \preceq \mu$*, then*

$$\varepsilon_i(\mathsf{M}(\lambda, \mu)) = -\langle h_i, \lambda \rangle \quad \textit{and} \quad E_i^{(-\langle h_i, \lambda \rangle)} \mathsf{M}(\lambda, \mu) \simeq \mathsf{M}(s_i\lambda, \mu).$$

(iii) *If* $\langle h_i, \mu \rangle \geqslant 0$ *and* $\lambda \preceq s_i\mu$*, then*

$$\varepsilon_i^*(\mathsf{M}(\lambda, \mu)) = \langle h_i, \mu \rangle \quad \textit{and} \quad E_i^{*(\langle h_i, \mu \rangle)} \mathsf{M}(\lambda, \mu) \simeq \mathsf{M}(\lambda, s_i\mu).$$

For $w \in \mathsf{W}$, we say that $\lambda \in \mathsf{P}$ is *w-dominant* if $(\beta, \lambda) \geqslant 0$ for any $\beta \in \Delta_+ \cap w^{-1}\Delta_-$. In this case, we have $\lambda - w\lambda \in \mathsf{Q}_+$.

Theorem 4.17 ([12, Theorem 2.18]). *Let $w \in W$ and let $\lambda \in \mathsf{P}$. Assume that λ is w-dominant. Then there exists a self-dual simple $R(\lambda - w\lambda)$-module $\mathsf{M}_w(w\lambda, \lambda)$ which satisfies the following conditions.*

(a) *If $i \in I$ satisfies $\langle h_i, w\lambda \rangle \geqslant 0$, then $\varepsilon_i\big(\mathsf{M}_w(w\lambda, \lambda)\big) = 0$.*
(b) *If $i \in I$ satisfies $\langle h_i, \lambda \rangle \leqslant 0$, then $\varepsilon_i^*\big(\mathsf{M}_w(w\lambda, \lambda)\big) = 0$.*
(c) *If $i \in I$ satisfies $s_i w \prec w$, then $\mathsf{M}_w(w\lambda, \lambda) \simeq L(i^m) \nabla \mathsf{M}_{s_i w}(s_i w\lambda, \lambda)$ where $m = \langle h_i, s_i w\lambda \rangle = \varepsilon_i(\mathsf{M}_w(w\lambda, \lambda)) \in \mathbb{Z}_{\geqslant 0}$.*
(d) *If $i \in I$ satisfies $ws_i \prec w$, then $\mathsf{M}_w(w\lambda, \lambda) \simeq \mathsf{M}_{ws_i}(w\lambda, s_i\lambda) \nabla L(i^m)$ where $m = \langle h_i, \lambda \rangle = \varepsilon_i^*(\mathsf{M}_w(w\lambda, \lambda)) \in \mathbb{Z}_{\geqslant 0}$.*
(e) *For any $\mu \in \mathsf{P}_+$ such that $\lambda + \mu \in \mathsf{P}_+$, we have*

$$\mathsf{M}(w\mu, \mu) \nabla \mathsf{M}_w(w\lambda, \lambda) \simeq \mathsf{M}\big(w(\lambda + \mu), \lambda + \mu\big)$$

up to a grading shift. Moreover such an $\mathsf{M}_w(w\lambda, \lambda)$ is unique up to an isomorphism. In particular, $\mathsf{M}_w(w\Lambda, \Lambda) \simeq \mathsf{M}(w\Lambda, \Lambda)$ for $\Lambda \in \mathsf{P}_+$.

We call $\mathsf{M}_w(w\lambda, \lambda)$ a *generalized determinantial module.*

Lemma 4.18 ([12, Lemma 2.21]). *Let $w \in W$ and let $\lambda \in \mathsf{P}$ be a w-dominant weight. Then, for any $\mu \in \mathsf{P}_+$, we have*

$$\mathsf{M}(w\mu, \mu) \nabla \mathsf{M}_w(w\lambda, \lambda) \simeq \mathsf{M}_w\big(w(\lambda + \mu), \lambda + \mu\big)$$

up to a grading shift.

4.7. **Shuffle lemma.** The shuffle lemma is useful to calculate explicitly the actions of E_i and E_i^* on a convolution product of R-modules. Let Δ be the set of roots, and Δ^{real} the set of real roots. For $\alpha \in \Delta^{\mathrm{real}}$, set $\mathsf{d}_\alpha := (\alpha, \alpha)/2$, $\alpha^\vee := (\mathsf{d}_\alpha)^{-1}\alpha$. We set also $\mathsf{d}_i := \mathsf{d}_{\alpha_i}$ for $i \in I$.

Lemma 4.19 ([6, Proposition 10.1.5]). *Let $M \in \mathrm{Mod}(R(\beta))$, $N \in \mathrm{Mod}(R(\gamma))$, $i \in I$ and $n \in \mathbb{Z}_{\geqslant 0}$. Then $\mathrm{E}_i^{(n)}(M \circ N)$ has an increasing filtration $\{F_k\}_{k \in \mathbb{Z}}$ of $R(\beta+\gamma-n\alpha_i)$-submodules such that*

(a) *$F_k = 0$ for $k < 0$,*
(b) *$F_k = \mathrm{E}_i^{(n)}(M \circ N)$ for $k \geqslant n$,*
(c) *$F_k/F_{k-1} \simeq q^{k(n-k)\mathsf{d}_i - k(\alpha_i, \beta)} \mathrm{E}_i^{(n-k)}(M) \circ \mathrm{E}_i^{(k)}(N)$.*

It is a categorification of the formula of the comultiplication:

$$\Delta(e_i^{(n)}) = \sum_{k=0}^{n} q^{k(n-k)\mathsf{d}_i} e_i^{(n-k)} t_i^k \otimes e_i^{(k)}.$$

Lemma 4.20 ([6, Proposition 10.1.5])**.** *Let $M \in \mathrm{Mod}(R(\beta))$, $N \in \mathrm{Mod}(R(\gamma))$, $i \in I$ and $n \in \mathbb{Z}_{\geqslant 0}$. Then $\mathrm{E}_i^{*(n)}(M \circ N)$ has an increasing filtration $\{F_k\}_{k \in \mathbb{Z}}$ of $R(\beta+\gamma-n\alpha_i)$-submodules such that*

(a) *$F_k = 0$ for $k < 0$,*
(b) *$F_k = \mathrm{E}_i^{*(n)}(M \circ N)$ for $k \geqslant n$,*
(c) *$F_k/F_{k-1} \simeq q^{k(n-k)\mathsf{d}_i - k(\alpha_i,\gamma)} \mathrm{E}_i^{*(k)}(M) \circ \mathrm{E}_i^{*(n-k)}(N)$.*

Lemma 4.21 ([6, Proposition 10.1.5])**.** *Let $Y, Z \in R\text{-gmod}$, and $y, z \in \mathbb{Z}_{\geqslant 0}$. Then we have*

(i) *If $\mathrm{E}_i^{y+1} Y \simeq 0$, then there exists a monomorphism of degree $yz\mathsf{d}_i + z\big(\alpha_i, \mathrm{wt}(Y)\big)$:*

$$\mathrm{E}_i^{(y)} Y \circ \mathrm{E}_i^{(z)} Z \rightarrowtail \mathrm{E}_i^{(y+z)}(Y \circ Z). \tag{4.10}$$

(ii) *If $\mathrm{E}_i^{z+1} Z \simeq 0$, then there exists an epimorphism of degree $-yz\mathsf{d}_i - z\big(\alpha_i, \mathrm{wt}(Y)\big)$:*

$$\mathrm{E}_i^{(y+z)}(Y \circ Z) \twoheadrightarrow \mathrm{E}_i^{(y)} Y \circ E_i^{(z)} Z. \tag{4.11}$$

(iii) *If $E_i^{y+1} Y \simeq 0$ and $E_i^{z+1} Z \simeq 0$, then (4.10) and (4.11) are isomorphisms, inverse to each other.*
(iv) *If $\mathrm{E}_i^{y+1} Y \simeq 0$ and $\mathrm{E}_i^{z+1} Z \simeq 0$, Then there exists an exact sequence*

$$0 \longrightarrow E_i^{(y)} Y \circ E_i^{(z-1)} Z \xrightarrow{\ f\ } E_i^{(y+z-1)}(Y \circ Z) \xrightarrow{\ g\ } E_i^{(y-1)} Y \circ E_i^{(z)} Z \longrightarrow 0.$$

Here f is of degree $y(z-1)\mathsf{d}_i + (z-1)\big(\alpha_i, \mathrm{wt}(Y)\big)$ and g is of degree $-(y-1)z\mathsf{d}_i - z\big(\alpha_i, \mathrm{wt}(Y)\big)$.

Proof. It follows from Lemma 4.19. □

Lemma 4.22 ([6, Proposition 10.1.5])**.** *Let $Y, Z \in R\text{-gmod}$, and $y, z \in \mathbb{Z}_{\geqslant 0}$. Then we have*

(i) *If $\mathrm{E}_i^{*\,z+1} Z \simeq 0$, then there exists a monomorphism of degree $yz\mathsf{d}_i + y\big(\alpha_i, \mathrm{wt}(Z)\big)$:*

$$\mathrm{E}_i^{*(y)} Y \circ \mathrm{E}_i^{*(z)} Z \rightarrowtail \mathrm{E}_i^{*(y+z)}(Y \circ Z). \tag{4.12}$$

(ii) *If $\mathrm{E}_i^{*\,y+1} Y \simeq 0$, then there exists an epimorphism of degree $-yz\mathsf{d}_i - y\big(\alpha_i, \mathrm{wt}(Z)\big)$:*

$$\mathrm{E}_i^{*(y+z)}(Y \circ Z) \twoheadrightarrow \mathrm{E}_i^{*(y)} Y \circ \mathrm{E}_i^{*(z)} Z. \tag{4.13}$$

(iii) *If $\mathrm{E}_i^{*\,y+1} Y \simeq 0$ and $\mathrm{E}_i^{*\,z+1} Z \simeq 0$, then (4.12) and (4.13) are isomorphisms, inverse to each other.*

(iv) *If* $\mathrm{E}_i^{*\,y+1}Y \simeq 0$ *and* $\mathrm{E}_i^{*\,z+1}Z \simeq 0$*, Then there exists an exact sequence*

$$0 \longrightarrow \mathrm{E}_i^{*\,(y-1)}Y \circ \mathrm{E}_i^{*\,(z)}Z \xrightarrow{f} \mathrm{E}_i^{*\,(y+z-1)}(Y \circ Z) \xrightarrow{g} \mathrm{E}_i^{*\,(y)}Y \circ \mathrm{E}_i^{*\,(z-1)}Z \longrightarrow 0.$$

Here, f *is of degree* $(y-1)z\mathsf{d}_i + (y-1)\big(\alpha_i, \mathrm{wt}(Z)\big)$*, and* g *is of degree* $-y(z-1)\mathsf{d}_i - y\big(\alpha_i, \mathrm{wt}(Z)\big)$*.*

Proposition 4.23. *Let* $M \in R$-gmod *and* $L \in R$-gmod *be simple modules, and assume that one of them is affreal. Set* $L' = \widetilde{\mathrm{E}}_i^{*\,\max}(L)$ *and* $M' = \widetilde{\mathrm{E}}_i^{*\,\max}(M)$*. Then we have*

(i) $\varepsilon_i^*(L \nabla M) \leqslant \varepsilon_i^*(L) + \varepsilon_i^*(M)$*.*

(ii) $\Lambda(L', M') \leqslant \Lambda(L, M) + \varepsilon_i^*(L)\big(\alpha_i, \mathrm{wt}(M)\big) - \varepsilon_i^*(M)\big(\alpha_i, \mathrm{wt}(L)\big)$ *and*

$$\widetilde{\Lambda}(L', M') \leqslant \widetilde{\Lambda}(L, M) + \varepsilon_i^*(L)\big(\alpha_i, \mathrm{wt}(M)\big) + \varepsilon_i^*(L)\varepsilon_i^*(M)\mathsf{d}_i.$$

If $\varepsilon_i^*(L \nabla M) = \varepsilon_i^*(L) + \varepsilon_i^*(M)$*, then the equality holds.*

(iii) *Assume either* $\varepsilon_i^*(L\nabla M) = \varepsilon_i^*(L) + \varepsilon_i^*(M)$ *or* $\Lambda(L', M') = \Lambda(L, M) + \varepsilon_i^*(L)\big(\alpha_i, \mathrm{wt}(M)\big) - \varepsilon_i^*(M)\big(\alpha_i, \mathrm{wt}(L)\big)$*. Then we have* $\widetilde{\mathrm{E}}_i^{*\,\max}(L \nabla M) \simeq L' \nabla M'$ *up to grading shifts,*

Proof. The statement (i) follows immediately from Proposition 3.19.

(ii) Let (L, z) be an affinization of L and set $\varepsilon_i^*(L) = n$ and $\varepsilon_i^*(M) = m$. Then $\mathsf{L}' := \mathrm{E}_i^{*\,(n)}\mathsf{L}$ is an affinization of L'. Let $R\colon \mathsf{L} \circ M \to M \circ \mathsf{L}$ be a renormalized R-matrix. By applying $\mathrm{E}_i^{*\,(n+m)}$, we obtain a commutative diagram:

$$\begin{tikzcd}
\mathrm{E}_i^{*\,(n)}(\mathsf{L}) \circ \mathrm{E}_i^{*\,(m)}M \arrow[r, "R'"] \arrow[d, "f"', "\wr"] & \mathrm{E}_i^{*\,(m)}M \circ \mathrm{E}_i^{*\,(n)}\mathsf{L} \arrow[d, "g"', "\wr"] \\
\mathrm{E}_i^{*\,(n+m)}(\mathsf{L} \circ M) \arrow[r, "\mathrm{E}_i^{*\,(n+m)}(R)"] & \mathrm{E}_i^{*\,(n+m)}(M \circ \mathsf{L}),
\end{tikzcd}$$

where the isomorphisms f and g are obtained by Lemma 4.22, $\deg(f) = nm\mathsf{d}_i + n(\alpha_i, \mathrm{wt}\, M)$ and $\deg(g) = nm\mathsf{d}_i + m(\alpha_i, \mathrm{wt}\, L)$, which follow from Lemma 4.19. Hence $R'\colon \mathsf{L}' \circ M' \to M' \circ \mathsf{L}'$ has degree $\Lambda(L, M) + n(\alpha_i, \mathrm{wt}\, M) - m(\alpha_i, \mathrm{wt}\, L)$. Since $R' = z^s \mathrm{R}^{\mathrm{ren}}_{\mathsf{L}.M'}$, we have $\Lambda(L', M') \leqslant \deg(R')$. If $\varepsilon_i^*(L\nabla M) = \varepsilon_i^*(L) + \varepsilon_i^*(M)$, then $\mathrm{E}_i^{*\,(n+m)}(L \circ M) \to \mathrm{E}_i^{*\,(n+m)}(L \nabla M)$ is an isomorphism, and hence $R'|_{z=0}$ does not vanish, which implies that $\Lambda(L', M') = \deg(R')$. Thus we obtain (ii). (iii) is obvious. □

Lemma 4.24 ([21, Lemma 3.13])**.** *Let* $M \in R$-gmod *be a simple module. Then for any* $n \in \mathbb{Z}_{\geqslant 0}$ *such that* $n \leqslant \varepsilon_i(M)$*,* $\mathrm{E}_i^{(n)}(M)$ *has a simple socle and a simple head. Moreover they are isomorphic to* $\widetilde{\mathrm{E}}_i^n M$ *(up to grading shifts).*

5. The categories $\mathscr{C}_w$ and $\widetilde{\mathscr{C}_w}$

We shall see several basic properties of the category $\mathscr{C}_w$ and its localization $\widetilde{\mathscr{C}_w}$, which is the main target of this article.

5.1. **Categories $\mathscr{C}_w$.** In this subsection, we recall the category $\mathscr{C}_w$ associated with $w \in \mathsf{W}$ (see [10]). For more details, see [11], [12], [14].

For $M \in R(\beta)$-gMod we define

$$\mathsf{W}(M) := \{\gamma \in \mathsf{Q}_+ \cap (\beta - \mathsf{Q}_+) \mid e(\gamma, \beta-\gamma)M \neq 0\},$$
$$\mathsf{W}^*(M) := \{\gamma \in \mathsf{Q}_+ \cap (\beta - \mathsf{Q}_+) \mid e(\beta-\gamma, \gamma)M \neq 0\}.$$

For any simple module M, we have ([29])

$$\mathsf{W}(M) \subset \operatorname{span}_{\mathbb{R}_{\geqslant 0}}(\mathsf{W}(M) \cap \Delta_+). \tag{5.1}$$

Here, for a subset A of $\mathbb{R} \otimes \mathsf{Q}$, we denote by $\operatorname{span}_{\mathbb{R}_{\geqslant 0}}(A)$ the smallest convex cone containing $A \cup \{0\}$.

For $w \in \mathsf{W}$, we define the full monoidal subcategory of R-gmod by

$$\mathscr{C}_w := \left\{ M \in R\text{-gmod} \;\middle|\; \mathsf{W}(M) \subset \mathsf{Q}_+ \cap w\mathsf{Q}_- \right\}. \tag{5.2}$$

An ordered pair (M, N) of R-modules is called *unmixed* if

$$\mathsf{W}^*(M) \cap \mathsf{W}(N) \subset \{0\}.$$

Lemma 5.1 ([12, Corollary 2.3])**.** *If (M, N) is unmixed, then (M, N) is Λ-definable and $\widetilde{\Lambda}(M, N) = 0$,*

For $\Lambda \in \mathsf{P}_+$ and $w, v \in \mathsf{W}$ such that $v \leqslant w$, we have

$$\mathsf{M}(w\Lambda, v\Lambda) \in \mathscr{C}_w. \tag{5.3}$$

Remark 5.2. Even if λ is w-dominant, $\mathsf{M}_w(w\lambda, \lambda)$ may not belong to $\mathscr{C}_w$. For example, take $\mathfrak{g} = A_2$, $w = s_1 s_2$ and $\lambda = s_1\Lambda_1$. Then $\mathsf{M}_w(w\lambda, \lambda) \simeq \langle 2 \rangle$ does not belong to $\mathscr{C}_w$, since $\alpha_2 \notin \Delta_+ \cap w\Delta_-$.

5.2. **Localizations of $\mathscr{C}_w$.** In this subsection we briefly recall the localizations of the categories $\mathscr{C}_w$ via left braiders studied in [11, 12].

Let $w \in \mathsf{W}$. Then there exists a real commuting family of graded left braiders in R-gmod

$$\left\{ \left(\mathsf{M}(w\Lambda_i, \Lambda_i), \mathrm{R}^{\ell}_{\mathsf{M}(w\Lambda_i,\Lambda_i)}, \phi_{\mathsf{M}(w\Lambda_i,\Lambda_i)}\right) \right\}_{i \in I} \tag{5.4}$$

(see [11, Proposition 5.1] for notations). Here $\mathrm{R}^{\ell}_{\mathsf{M}(w\Lambda_i,\Lambda_i)}(X) \colon \mathsf{M}(w\Lambda_i, \Lambda_i) \circ X \to X \circ \mathsf{M}(w\Lambda_i, \Lambda_i)$ is a morphism of degree $\phi_{\mathsf{M}(w\Lambda_i,\Lambda_i)}(\operatorname{wt}(X))$ functorially in $X \in R$-gmod.

Moreover it is a family of central objects in the category $\mathscr{C}_w$, i.e. $\mathrm{R}^{\ell}_{\mathsf{M}(w\Lambda_i,\Lambda_i)}(X)$ is an isomorphism for any $X \in \mathscr{C}_w$. Note that

$$\phi_{\mathsf{M}(w\Lambda_i,\Lambda_i)}(\beta) = -(w\Lambda_i + \Lambda_i, \beta) \quad \text{for any} \quad \beta \in \mathsf{Q}. \tag{5.5}$$

We denote by $\big(R\text{-gmod}\big)[\mathsf{M}(w\Lambda_i,\Lambda_i)^{\circ -1}; i \in I]$ the localization of R-gmod via the real commuting family of graded left braiders (5.4), and by $\widetilde{\mathscr{C}_w}$ the localization of $\mathscr{C}_w$ by the same left braiders. We have a quasi-commutative diagram

$$\begin{tikzcd} \mathscr{C}_w \arrow[r, tail, "\iota_w"'] \arrow[d, "\Phi_w"'] & R\text{-gmod} \arrow[d, "\mathrm{Q}_w"'] \\ \widetilde{\mathscr{C}_w} \arrow[r, tail, dashed, "\widetilde{\iota}_w"'] & \big(R\text{-gmod}\big)[\mathsf{M}(w\Lambda_i,\Lambda_i)^{\circ -1}; i \in I] \end{tikzcd}$$

where Φ_w and Q_w denote the localization functors, and $\widetilde{\iota}_w$ is the induced functor from the inclusion functor ι_w.

Theorem 5.3 ([11, Theorem 5.9, Theorem 5.11], [12, Theorem 3.9])**.**

(i) *The functor* $\widetilde{\iota}_w \colon \widetilde{\mathscr{C}_w} \to \big(R\text{-gmod}\big)[\mathsf{M}(w\Lambda_i,\Lambda_i)^{\circ -1}; i \in I]$ *is an equivalence of categories.*

(ii) *For any simple* $M \in R$-gmod, $\mathrm{Q}_w(M) \in \widetilde{\mathscr{C}_w}$ *is simple or zero.*

(iii) *The monoidal category* $\widetilde{\mathscr{C}_w}$ *is rigid, that is, every object of* $\widetilde{\mathscr{C}_w}$ *has a left dual and a right dual. Moreover* $\widetilde{\mathscr{C}_w}$ *satisfies condition* (3.1)*.*

Theorem 5.4 ([15, Theorem 8.3, Corollary 8.4])**.**

(i) $\Phi_w \colon \mathscr{C}_w \to \widetilde{\mathscr{C}_w}$ *is fully faithful.*

(ii) *The full subcategory* $\mathscr{C}_w$ *of* $\widetilde{\mathscr{C}_w}$ *is stable by taking subquotients.*

For $w \in \mathsf{W}$, let us take a reduced expression $\underline{w} = s_{i_1} \cdots s_{i_\ell}$ of w. We set

$$I_w := \{ i \in I \mid w\Lambda_i \neq \Lambda_i \} = \{ i_1, \dots, i_\ell \}.$$

Then the localization functor Q_w. factors as

$$R\text{-gmod} \longrightarrow R_{I_w}\text{-gmod} \to \widetilde{\mathscr{C}_w}.$$

Here, R_{I_w} denotes the quiver Hecke algebra defined by using I_w instead of I, and R-gmod $\longrightarrow R_{I_w}$-gmod is the exact monoidal functor given by

$$R(\alpha_i) \mapsto R_{I_w}(\alpha_i) \text{ or } R(\alpha_i) \mapsto 0 \text{ whether } i \in I_w \text{ or not.}$$

Remark 5.5. *Although the composition* $\mathscr{C}_w \to R$-gmod $\to \widetilde{\mathscr{C}_w}$ *is fully faithful* ([15, Theorem 8.3])*, we do not regard an object of* $\mathscr{C}_w$ *as an object of* $\widetilde{\mathscr{C}_w}$*.* For $M \in \mathscr{C}_w$, we distinguish $M \in \mathscr{C}_w$ and $\mathrm{Q}_w(M) \in \widetilde{\mathscr{C}_w}$.

For $\Lambda \in \mathsf{P}_+$, we set

$$\mathsf{C}_\Lambda := \mathsf{M}(w\Lambda, \Lambda) \in \mathscr{C}_w, \quad \text{and} \quad \widetilde{\mathsf{C}}_\Lambda := \mathrm{Q}_w(\mathsf{C}_\Lambda) \in \widetilde{\mathscr{C}_w}.$$

For $\lambda \in \mathsf{P}$, define

$$\widetilde{\mathsf{C}}_\lambda := (\widetilde{\mathsf{C}}_\mu)^{-1} \circ \widetilde{\mathsf{C}}_{\lambda+\mu} \in \widetilde{\mathscr{C}_w} \tag{5.6}$$

with $\mu \in \mathsf{P}_+$ such that $\lambda+\mu \in \mathsf{P}_+$ (up to grading shifts). Hence, we have $\widetilde{\mathsf{C}}_\lambda \circ \widetilde{\mathsf{C}}_\mu \simeq \widetilde{\mathsf{C}}_{\lambda+\mu}$ for any $\lambda, \mu \in \mathsf{P}$. If we want to emphasize w, we write $\widetilde{\mathsf{C}}^w_\lambda$ for $\widetilde{\mathsf{C}}_\lambda$.

Note that

$$\mathrm{Q}_w\big(\mathsf{M}_w(w\lambda, \lambda)\big) \simeq \widetilde{\mathsf{C}}_\lambda \quad \text{for any } w\text{-dominant } \lambda \in \mathsf{P}. \tag{5.7}$$

Indeed, by Lemma 4.18 we have

$$\mathsf{M}(w\mu, \mu) \nabla \mathsf{M}_w(w\lambda, \lambda) \simeq \mathsf{M}(w(\lambda+\mu), \lambda+\mu)$$

for any $\mu \in \mathsf{P}_+$ such that $\lambda + \mu \in \mathsf{P}_+$.

Proposition 5.6. *Let $L \in R$-gmod be a simple module, and let (L, z) be an affinization of L in the sense of* Definition 4.6*. Then $(\mathrm{Q}_w(\mathsf{L}), z)$ is an affinization of $\mathrm{Q}_w(L)$ in the sense of* Definition 3.9*.*

Proof. Let $\mathcal{A}$ be the abelian monoidal category consisting of pairs (X, R_X) where $X \in \widetilde{\mathscr{C}_w}$ and $R_X\colon \mathrm{Q}_w(\mathsf{L}) \circ X \xrightarrow{\sim} X \circ \mathrm{Q}_w(\mathsf{L})$ is an isomorphism in $\mathrm{Raff}_z(\widetilde{\mathscr{C}_w})$ (see Definition 3.5). The morphisms are obvious ones. The monoidal category structure of $\mathcal{A}$ is given by: $(X, R_X) \otimes (Y, R_Y) = (X \circ Y, R_{X\circ Y})$ where $R_{X\circ Y}$ is the composition $\mathrm{Q}_w(\mathsf{L}) \circ X \circ Y \xrightarrow{R_X \circ Y} X \circ \mathrm{Q}_w(\mathsf{L}) \circ Y \xrightarrow{X \circ R_Y} X \circ Y \circ \mathrm{Q}_w(\mathsf{L})$.

Then we can define the monoidal functor $\Psi\colon \mathscr{C}_w \to \mathcal{A}$ by $M \mapsto \big(\mathrm{Q}_w(M), \mathrm{R}^{\mathrm{univ}}_{\mathsf{L},M}\big)$. In order to see that Ψ extends to $\widetilde{\Psi}\colon \widetilde{\mathscr{C}_w} \to \mathcal{A}$, it is enough to show that $\Psi(\mathsf{C}_\Lambda)$ is invertible for any $\Lambda \in \mathsf{P}_+$. Set $\widetilde{C} = \mathrm{Q}_w(\mathsf{C}_\Lambda)$ and $\widetilde{\mathsf{L}} = \mathrm{Q}_w(\mathsf{L})$. Then, we have an isomorphism $\widetilde{\mathsf{L}} \circ \widetilde{C} \xrightarrow{\sim} \widetilde{C} \circ \widetilde{\mathsf{L}}$ in $\mathrm{Raff}_z(\widetilde{\mathscr{C}_w})$. By applying $\widetilde{C}^{-1} \circ \ \bullet\ \circ \widetilde{C}^{-1}$, we obtain an isomorphism $\widetilde{C}^{-1} \circ \widetilde{\mathsf{L}} \xrightarrow{\sim} \widetilde{\mathsf{L}} \circ \widetilde{C}^{-1}$. Let $R_{\widetilde{C}^{-1}}$ be its inverse. Then we can see easily that $(\widetilde{C}^{-1}, R_{\widetilde{C}^{-1}})$ is an inverse object of $\Psi(\mathsf{C}_\Lambda)$.

Thus, we see that Ψ extends to $\widetilde{\Psi}\colon \widetilde{\mathscr{C}_w} \to \mathcal{A}$. For $X \in \widetilde{\mathscr{C}_w}$, set $(X, R_X) = \widetilde{\Psi}(X)$, and let us define $R_{\mathrm{Q}_w(\mathsf{L})}(X)$ by $R_X\colon \mathrm{Q}_w(\mathsf{L}) \otimes X \to X \circ \mathrm{Q}_w(\mathsf{L})$. Then $\big(\mathrm{Q}_w(\mathsf{L}), R_{\mathrm{Q}_w(\mathsf{L})}\big)$ is a rational center (see Definition 3.8). Thus $\big(\mathrm{Q}_w(\mathsf{L}), R_{\mathrm{Q}_w(\mathsf{L})}\big)$ is an affinization of $\mathrm{Q}_w(L)$ in the sense of Definition 3.9. $\square$

Proposition 5.7. *Let M and N be simple modules in $\mathscr{C}_w$. Assume that $\big(\mathrm{Q}_w(M), \mathrm{Q}_w(N)\big)$ is a Λ-definable pair of simple objects in $\widetilde{\mathscr{C}_w}$. Then (M, N) is a Λ-definable pair. In particular, $M \circ N$ has a simple head, $N \circ M$ has a simple socle, and we have*

$$M \nabla N \simeq \mathrm{Im}(\mathbf{r}_{M,N}) \simeq N \Delta M.$$

Proof. It is an immediate consequence of Theorem 5.4 and Lemma 4.5. □

5.3. **Equivalence between $(\widetilde{\mathscr{C}_w})^{\mathrm{rev}}$ and $\widetilde{\mathscr{C}}_{w^{-1}}$.** Recall that, for a monoidal category $(\mathcal{C}, \otimes)$, we denote by $\mathcal{C}^{\mathrm{rev}}$ the monoidal category $\mathcal{C}$ with the tensor product $\otimes_{\mathrm{rev}}$: $M \otimes_{\mathrm{rev}} N := N \otimes M$.

Let $\psi\colon R(\beta) \xrightarrow{\sim} R(\beta)$ be the ring automorphism

$$\psi: \quad \begin{aligned} & e(\nu_1, \ldots, \nu_n) \mapsto e(\nu_n, \ldots, \nu_1), \\ & x_k \mapsto x_{n+1-k} \quad (1 \leqslant k \leqslant n), \\ & \tau_l \mapsto -\tau_{n-l} \quad (1 \leqslant l < n), \end{aligned} \tag{5.8}$$

where $n = \mathrm{ht}(\beta)$.

Then ψ induces an equivalence of monoidal categories ([12, § 3.2.])

$$\psi_*\colon (R\text{-gmod})^{\mathrm{rev}} \simeq R\text{-gmod}. \tag{5.9}$$

Lemma 5.8 ([12, Lemma2.23])**.** *We have*

$$\psi_*\big(\mathsf{M}_w(w\lambda, \lambda)\big) \simeq \mathsf{M}_{w^{-1}}(-\lambda, -w\lambda).$$

for any $w \in W$ and any w-dominant $\lambda \in \mathsf{P}$

Theorem 5.9 ([12, Theorem 3.7])**.** *There is an equivalence of monoidal categories $(\widetilde{\mathscr{C}_w})^{\mathrm{rev}}$ and $\widetilde{\mathscr{C}}_{w^{-1}}$. More precisely, we have a quasi-commutative diagram*

$$\begin{tikzcd} (R\text{-gmod})^{\mathrm{rev}} \arrow[r, "\sim", "\psi_*"'] \arrow[d, "\mathrm{Q}_w"'] & R\text{-gmod} \arrow[d, "\mathrm{Q}_{w^{-1}}"] \\ (\widetilde{\mathscr{C}_w})^{\mathrm{rev}} \arrow[r, "\sim", "\psi_*"'] & \widetilde{\mathscr{C}}_{w^{-1}}. \end{tikzcd}$$

5.4. **The category $\mathfrak{B}_w$.** In this subsection, we shall see the basic properties of the localization functor Q_w and the category $\mathfrak{B}_w$.

Recall that $S(b)$ denotes the self-dual simple R-module corresponding to $b \in B(\infty)$.

In Proposition 2.4, we introduced a subset $B_w(\infty)$ of $B(\infty)$ for each $w \in \mathsf{W}$.

Proposition 5.10 ([12, Proposition 3.1.]). *Let $w \in \mathsf{W}$ and let $\underline{w} = s_{i_1} s_{i_2} \cdots s_{i_\ell}$ be a reduced expression of w. Let M be a simple R-module. Then the following conditions are equivalent.*

(a) $e(i_1^{n_1}, \ldots, i_l^{n_l})M \not\simeq 0$ *for some* $(n_1, \ldots, n_\ell) \in \mathbb{Z}_{\geqslant 0}^\ell$,
(b) $\widetilde{\mathrm{E}}_{i_\ell}^{\max} \widetilde{\mathrm{E}}_{i_{\ell-1}}^{\max} \cdots \widetilde{\mathrm{E}}_{i_2}^{\max} \widetilde{\mathrm{E}}_{i_1}^{\max} M \simeq \mathbf{1}$,
(c) $\widetilde{\mathrm{E}}_{i_1}^{*\max} \cdots \widetilde{\mathrm{E}}_{i_\ell}^{*\max} M \simeq \mathbf{1}$,
(d) M *is a quotient of* $\langle i_1^{n_1} \rangle \circ \cdots \circ \langle i_\ell^{n_\ell} \rangle$ *for some* $(n_1, \ldots, n_\ell) \in \mathbb{Z}_{\geqslant 0}^\ell$,
(e) $M \simeq S(b)$ *up to a grading shift for some* $b \in B_w(\infty)$,
(f) $\mathrm{Q}_w(M)$ *is a simple module,*
(g) $\mathrm{Q}_w(M) \not\simeq 0$,
(h) $\Lambda(\mathsf{C}_\Lambda^w, M) = -\big(w\Lambda + \Lambda, \mathrm{wt}(M)\big)$ *for any* $\Lambda \in \mathsf{P}_+$,
(i) $\mathsf{C}_\Lambda^w \nabla M \in \mathscr{C}_w$ *for some* $\Lambda \in \mathsf{P}_+$.

Let $\mathfrak{B}_w$ be the full subcategory of R-gmod consisting of modules M such that any simple subquotient S of M satisfies the equivalent conditions above.

Let $\mathrm{Irr}(\mathscr{C}_w)$ be the set of equivalence classes of simple modules in $\mathscr{C}_w$ up to grading shifts. Namely, for simple modules M and N in $\mathscr{C}_w$, we identify M and N in $\mathrm{Irr}(\mathscr{C}_w)$ if $M \simeq q^n N$ for some $n \in \mathbb{Z}$. Similarly, let $\mathrm{Irr}(\mathfrak{B}_w)$ be the set of equivalence classes of simple modules in $\mathfrak{B}_w$ up to grading shifts. Hence, we have $\mathrm{Irr}(\mathscr{C}_w) \subset \mathrm{Irr}(\mathfrak{B}_w)$.

We have the following isomorphism of crystals in the sense of Definition 2.3 (ii)

$$\mathrm{Irr}(\mathfrak{B}_w) \simeq B_w(\infty).$$

Lemma 5.11. $\mathrm{Irr}(\mathfrak{B}_w) \sqcup \{0\}$ *is stable by* $\widetilde{\mathrm{E}}_i$ *and* $\widetilde{\mathrm{E}}_i^*$.

Proof. Since the case of $\widetilde{\mathrm{E}}_i^*$ is shown similarly to the case of $\widetilde{\mathrm{E}}_i$, we shall show the stability only for $\widetilde{\mathrm{E}}_i$. Take $S \in \mathrm{Irr}(\mathfrak{B}_w)$. The assertion is trivial when $\widetilde{\mathrm{E}}_i S \simeq 0$, we may assume that $\widetilde{\mathrm{E}}_i S$ is simple. By the condition (d) above, there exists an epimorphism $\langle i_1^{n_1} \rangle \circ \cdots \circ \langle i_\ell^{n_\ell} \rangle \twoheadrightarrow S$ for some $(n_1, \ldots, n_\ell) \in \mathbb{Z}_{\geqslant 0}^\ell$. Since E_i is an exact functor, we have an epimorphism

$$\pi\colon \mathrm{E}_i(\langle i_1^{n_1} \rangle \circ \cdots \circ \langle i_\ell^{n_\ell} \rangle) \twoheadrightarrow \mathrm{E}_i S \twoheadrightarrow \widetilde{\mathrm{E}}_i S.$$

By applying Lemma 4.19 (i), there exists a filtration

$$0 = F_0 \subset F_1 \subset \cdots \subset F_{\ell-1} \subset F_\ell = \mathrm{E}_i(\langle i_1^{n_1} \rangle \circ \cdots \circ \langle i_\ell^{n_\ell} \rangle)$$

such that F_k/F_{k-1} is isomorphic to $\langle i_1^{n_1} \rangle \circ \cdots \circ \mathrm{E}_i(\langle i_k^{n_k} \rangle) \circ \cdots \circ \langle i_\ell^{n_\ell} \rangle$. Let $k_0 \in \{1, 2, \cdots, \ell\}$ be the minimum among $k's$ such that $\pi_{|F_k}\colon F_k \twoheadrightarrow \widetilde{\mathrm{E}}_i S$ does not vanish.

Then, $\pi|_{F_{k_0}}\colon F_{k_0} \to \widetilde{\mathrm{E}}_i S$ is an epimorphism since $\widetilde{\mathrm{E}}_i S$ is simple. Hence we obtain $\mathrm{E}_i(\langle i_{k_0}^{n_{k_0}}\rangle) \not\simeq 0$, which implies $i = i_{k_0}$, $n_{k_0} > 0$, and obtain

$$\tilde{\pi}\colon F_{k_0}/F_{k_0-1} \simeq \langle i_1^{n_1}\rangle \circ \cdots \circ \langle i_{k_0}^{n_{k_0}-1}\rangle \circ \cdots \circ \langle i_\ell^{n_\ell}\rangle \twoheadrightarrow \widetilde{\mathrm{E}}_i S.$$

□

Lemma 5.12. $\mathrm{Irr}(\mathscr{C}_w) \sqcup \{0\}$ *is stable by* $\widetilde{\mathrm{E}}_i^*$.

Proof. This follows from $\mathsf{W}(\widetilde{\mathrm{E}}_i^* S) \subset \mathsf{W}(S)$. □

Remark 5.13. $\mathrm{Irr}(\mathscr{C}_w) \sqcup \{0\}$ is not necessarily stable by $\widetilde{\mathrm{E}}_i$. For example, for $\mathfrak{g} = A_2$, $w = s_1 s_2$, we have $\langle 12\rangle \in \mathscr{C}_w$ but $\widetilde{\mathrm{E}}_1\langle 12\rangle \simeq \langle 2\rangle \notin \mathscr{C}_w$. However $\langle 2\rangle \in \mathfrak{B}_w$.

The following lemma immediately follows from the definitions and Theorem 4.17.

Lemma 5.14. *For any w-dominant $\lambda \in \mathsf{P}$, we have*

$$\mathsf{M}_w(w\lambda, \lambda) \in \mathfrak{B}_w.$$

Lemma 5.15. *Let $w \in \mathsf{W}$ and $i \in I$. Assume that $w' := ws_i < w$.*

(i) *If a simple $S \in \mathfrak{B}_w$ satisfies $\varepsilon_i^*(S) = 0$, then $S \in \mathfrak{B}_{w'}$.*
(ii) *If $S \in \mathfrak{B}_w$ is simple, then $(\widetilde{\mathrm{F}}_i^*)^m(S) \simeq S \mathbin{\nabla} \langle i^m\rangle \in \mathfrak{B}_w$ for any $m \in \mathbb{Z}_{\geqslant 0}$*

Proof. Take a reduced expression $s_{i_1} \cdots s_{i_\ell}$ of w with $i = i_\ell$. By the assumption there exists an epimorphism

$$\langle i_1^{m_1}\rangle \circ \cdots \circ \langle i_\ell^{m_\ell}\rangle \twoheadrightarrow S.$$

(i) Since $\varepsilon_i^*(S) = 0$, we have $m_\ell = 0$, which implies that $S \in \mathfrak{B}_{w'}$.

(ii) We have an epimorphism

$$\langle i_1^{m_1}\rangle \circ \cdots \circ \langle i_{\ell-1}^{m_{\ell-1}}\rangle \circ \langle i_\ell^{m+m_\ell}\rangle \twoheadrightarrow S \mathbin{\nabla} \langle i^m\rangle.$$

□

Lemma 5.16. *Let $M, N \in R$-gmod be simple modules. If there exits a simple subquotient S of $M \circ N$ such that $S \in \mathfrak{B}_w$, then M and N belong to $\mathfrak{B}_w$.*

Proof. Since $\mathrm{Q}_w(S) \not\simeq 0$ is a subquotient of $\mathrm{Q}_w(M \circ N) \simeq \mathrm{Q}_w(M) \circ \mathrm{Q}_w(N)$, we have $\mathrm{Q}_w(M) \not\simeq 0$ and $\mathrm{Q}_w(N) \not\simeq 0$. □

Lemma 5.17. *Let M be a simple module in $\mathscr{C}_w$ and let N be a simple module in $\mathfrak{B}_w$. Then any simple quotient of $M \circ N$ belongs to $\mathfrak{B}_w$.*

Proof. Let S be a simple quotient of $M \circ N$. We shall show that S belongs to $\mathfrak{B}_w$.

By Lemma 4.5, S is a simple submodule of $N \circ M$. Take $\Lambda \in \mathsf{P}_+$ such that $\mathsf{C}_\Lambda \nabla N$ belongs to $\mathscr{C}_w$ (see Proposition 5.28). Then the composition

$$\mathsf{C}_\Lambda \circ S \rightarrowtail \mathsf{C}_\Lambda \circ N \circ M \twoheadrightarrow (\mathsf{C}_\Lambda \nabla N) \circ M$$

does not vanish by Lemma 3.14. Since $(\mathsf{C}_\Lambda \nabla N) \circ M \in \mathscr{C}_w$, we conclude that $\mathsf{C}_\Lambda \circ S$ has a simple quotient in $\mathscr{C}_w$, which implies the desired result by Proposition 5.10. □

Lemma 5.18. *Let M and N be simple modules in R-gmod, and let $f \colon M \circ N \to N \circ M$ be a morphism. Assume that*

(a) *$S := \mathrm{Im}(f)$ is simple and belongs to $\mathfrak{B}_w$,*

(b) *$\big(\mathrm{Q}_w(M), \mathrm{Q}_w(N)\big)$ is a Λ-definable pair (see* Definition 3.7*).*

Then we have

(i) *$M, N \in \mathfrak{B}_w$,*

(ii) *$\Lambda\big(\mathrm{Q}_w(M), \mathrm{Q}_w(N)\big) = \deg(f)$.*

Proof. We have a diagram in $\widetilde{\mathscr{C}_w}$:

$$\mathrm{Q}_w(f) \colon \mathrm{Q}_w(M) \circ \mathrm{Q}_w(N) \twoheadrightarrow \mathrm{Q}_w(S) \rightarrowtail \mathrm{Q}_w(N) \circ \mathrm{Q}_w(M).$$

Since $\mathrm{Q}_w(S)$ is simple, $\mathrm{Q}_w(M)$ and $\mathrm{Q}_w(N)$ do not vanish, and hence we obtain (i). Since $\mathrm{Q}_w(f)$ does not vanish, it coincides with the R-matrix $\mathbf{r}_{\mathrm{Q}_w(M), \mathrm{Q}_w(N)}$, and hence we obtain (ii). □

The following lemma is an immediate consequence of the lemma above.

Lemma 5.19. *Let M, N be simple modules in R-gmod. Assume that (M, N) is Λ-definable. If $M \nabla N \in \mathfrak{B}_w$, then we have*

(i) *$M, N \in \mathfrak{B}_w$,*

(ii) *$\mathrm{Q}_w(M) \nabla \mathrm{Q}_w(N) \simeq \mathrm{Q}_w(M \nabla N)$,*

(iii) *$\Lambda(M, N) = \Lambda\big(\mathrm{Q}_w(M), \mathrm{Q}_w(N)\big)$.*

Proposition 5.20. *Let $L \in \mathscr{C}_w$ and $M \in \mathfrak{B}_w$ be simple modules. Assume that (L, M) is Λ-definable. Then we have*

(i) *$L \nabla M \in \mathfrak{B}_w$,*

(ii) *$\mathrm{Q}_w\big(L \nabla M\big) \simeq \mathrm{Q}_w(L) \nabla \mathrm{Q}_w(M)$,*

(iii) *$\Lambda(L, M) = \Lambda\big(\mathrm{Q}_w(L), \mathrm{Q}_w(M)\big)$.*

Proof. (i) follows from Lemma 5.17. The remaining statements follow from Lemma 5.19. □

Lemma 5.21. *Let M and N be simple modules in $\mathfrak{B}_w$. Assume that one of them is affreal. Then, we have*

$$\Lambda\big(\mathrm{Q}_w(M), \mathrm{Q}_w(N)\big) \leqslant \Lambda(M, N).$$

Proof. The proof is similar to the one of Lemma 5.19. Assume that M is affreal. Let (M, z) be an affinization of M, and let $R\colon \mathsf{M} \circ N \to N \circ \mathsf{M}$ be the renormalized R-matrix. Then $\Lambda(M, N)$ is the homogeneous degree of R. Applying Q_w we obtain a morphism in $\mathrm{Aff}_z(\widetilde{\mathscr{C}_w})$:

$$\mathrm{Q}_w(R)\colon \mathrm{Q}_w(\mathsf{M}) \circ \mathrm{Q}_w(N) \to \mathrm{Q}_w(N) \circ \mathrm{Q}_w(\mathsf{M}).$$

Take the largest $c \in \mathbb{Z}_{\geqslant 0}$ such that the image of $\mathrm{Q}_w(R)$ is contained in $\mathrm{Q}_w(N) \circ z^c\mathrm{Q}_w(\mathsf{M})$. Then $\Lambda\big(\mathrm{Q}_w(M), \mathrm{Q}_w(N)\big)$ is the homogeneous degree of $z^{-c}\mathrm{Q}_w(R)$. Hence we obtain the desired result.

We can similarly prove the assertion when N is affreal. □

Remark 5.22. For affreal $L \in \mathscr{C}_w$ and $M \in \mathfrak{B}_w$, the simple module $M \nabla L$ may not belong to $\mathfrak{B}_w$. For example, take $\mathfrak{g} = A_2$, $w = s_1s_2$, $L = \langle 1\rangle$, $M = \langle 2\rangle$. Then $L \nabla M \simeq \langle 12\rangle \in \mathscr{C}_w$ and $M \nabla L \simeq \langle 21\rangle \notin \mathfrak{B}_w$.

For a Λ-definable pair (X, Y) of simple objects of $\widetilde{\mathscr{C}_w}$, we set

$$\widetilde{\Lambda}(X, Y) = \Big(\Lambda(X, Y) + \big(\mathrm{wt}(X), \mathrm{wt}(Y)\big)\Big)/2.$$

If one of X and Y is affreal, then $\widetilde{\Lambda}(X, Y) \in \mathbb{Z}$ (cf. [11, Lemma 3.11]).

The following lemma is obtained from [11, 5.1].

Lemma 5.23. *For any $\lambda \in \mathsf{P}$ and any simple $X \in \widetilde{\mathscr{C}_w}$, we have*

$$\begin{aligned} &\Lambda(\widetilde{\mathsf{C}}_\lambda, X) = -\big(w\lambda + \lambda, \mathrm{wt}(X)\big) \quad \textit{and} \quad \Lambda(X, \widetilde{\mathsf{C}}_\lambda) = \big(\mathrm{wt}(X), w\lambda + \lambda\big), \\ &\widetilde{\Lambda}(\widetilde{\mathsf{C}}_\lambda, X) = -\big(\lambda, \mathrm{wt}(X)\big) \quad \textit{and} \quad \widetilde{\Lambda}(X, \widetilde{\mathsf{C}}_\lambda) = \big(\mathrm{wt}(X), w\lambda\big). \end{aligned} \tag{5.10}$$

Note that they are possibly negative integers.

Proof. Since $\Lambda(\widetilde{\mathsf{C}}_{\lambda-\mu}, X) = \Lambda(\widetilde{\mathsf{C}}_\lambda, X) - \Lambda(\widetilde{\mathsf{C}}_\mu, X)$, etc., we may assume that $\lambda \in \mathsf{P}_+$. For any $X \in \widetilde{\mathscr{C}_w}$, we can find $\mu \in \mathsf{P}_+$ and a simple $M \in \mathscr{C}_w$ such that $X \simeq \widetilde{\mathsf{C}}_{-\mu} \circ \mathrm{Q}_w(M)$. We also get $\Lambda(\mathsf{C}_\lambda, M) = -(w\lambda + \lambda, \mathrm{wt}(M))$ by [11, Proposition 5.1]. Thus, by [11, Lemma 3.16 (ii), (5.1)] and Proposition 5.20 (iii), we obtain

$$\begin{aligned} \Lambda(\widetilde{\mathsf{C}}_\lambda, X) &= -\Lambda(\widetilde{\mathsf{C}}_\lambda, \widetilde{\mathsf{C}}_\mu) + \Lambda(\widetilde{\mathsf{C}}_\lambda, \mathrm{Q}_w(M)) = -\Lambda(\mathsf{C}_\lambda, \mathsf{C}_\mu) + \Lambda(\mathsf{C}_\lambda, M) \\ &= -(w\lambda + \lambda, \mu - w\mu) - (w\lambda + \lambda, \mathrm{wt}(M)) = -(w\lambda + \lambda, \mathrm{wt}(X)), \end{aligned}$$

where also note that $\Lambda(\mathsf{C}_\lambda, \mathsf{C}_\mu) = (\lambda + w\lambda, \mu - w\mu)$ as in [11, (5.1)]. □

Corollary 5.24. *For any $i \in I_w$ and any simple $M \in \mathscr{C}_w$, we have*
$$\widetilde{\Lambda}\big(\mathrm{Q}_w(M), \mathrm{Q}_w(\langle i\rangle)\big) = \widetilde{\Lambda}(M, \langle i\rangle) = \mathsf{d}_i\varepsilon_i^*(M).$$

Proof. By Lemma 4.14, $\widetilde{\Lambda}(M, \langle i\rangle) = \mathsf{d}_i\varepsilon_i^*(M)$, and by Example 4.7, Proposition 5.20 implies that
$$\widetilde{\Lambda}\big(\mathrm{Q}_w(M), \mathrm{Q}_w(\langle i\rangle)\big) = \widetilde{\Lambda}(M, \langle i\rangle).$$
□

Proposition 5.25. *Let $i \in I_w$ and let $M \in \mathfrak{B}_w$ be a simple module.*

(i) *Assume that $\widetilde{\mathrm{F}}_i(M) \in \mathfrak{B}_w$. Then we have*
$$\mathrm{Q}_w(\widetilde{\mathrm{F}}_i(M)) \simeq \mathrm{Q}_w(\langle i\rangle) \mathbin{\nabla} \mathrm{Q}_w(M).$$

(ii) *Assume that $\widetilde{\mathrm{E}}_i(M) \not\simeq 0$. Then we have*
$$\mathrm{Q}_w(\widetilde{\mathrm{E}}_i(M)) \simeq \mathrm{Q}_w(M) \mathbin{\nabla} \mathscr{D}\big(\mathrm{Q}_w(\langle i\rangle)\big).$$

Proof. (i) follows from Lemma 5.19.

(ii) Since $M \simeq \widetilde{\mathrm{F}}_i\widetilde{\mathrm{E}}_iM = \langle i\rangle \mathbin{\nabla} \widetilde{\mathrm{E}}_iM$, it follows from (i) that $\mathrm{Q}_w(M) \simeq \mathrm{Q}_w(\langle i\rangle) \mathbin{\nabla} \mathrm{Q}_w(\widetilde{\mathrm{E}}_i(M))$, which implies that $\mathrm{Q}_w(\widetilde{\mathrm{E}}_i(M)) \simeq \mathrm{Q}_w(M) \mathbin{\nabla} \mathscr{D}\big(\mathrm{Q}_w(\langle i\rangle)\big)$ by Lemma 3.18. □

Note that this proposition motivates us to define the crystal operators on $\mathrm{Irr}(\widetilde{\mathscr{C}_w})$ in § 6.

5.5. **Convolutions with C_Λ.** In this subsection, we shall prove Proposition 5.28, which is a more precise statement than Proposition 5.10 (i).

Lemma 5.26. *Let M be a simple module in R-gmod such that $\mathrm{Q}_w(M)$ is simple and $\mathsf{W}^*(M) \in \mathsf{Q}_+ \cap w\mathsf{Q}_+$. Assume that M commutes with C_{Λ_i} for any $i \in I_w$ such that $ws_i > w$. Then $M \simeq \mathbf{1}$.*

Proof. We may assume that $I_w = I$. Set $K = \{i \in I \mid ws_i > w\}$. Let $\beta = -\mathrm{wt}(M) \in \mathsf{Q}_+$. For any $\Lambda \in \mathsf{P}_+$, we have $\Lambda(\mathsf{C}_\Lambda, M) = (w\Lambda + \Lambda, \beta)$. On the other hand, since $\mathsf{C}_\Lambda \in \mathscr{C}_w$, the pair (M, C_Λ) is unmixed, and hence Lemma 5.1 implies that
$$\Lambda(M, \mathsf{C}_\Lambda) = -(w\Lambda - \Lambda, -\beta) = (w\Lambda - \Lambda, \beta).$$
Therefore, we obtain
$$\mathfrak{d}(\mathsf{C}_\Lambda, M) = (w\Lambda, \beta).$$
Hence we obtain
$$(w\Lambda_i, \beta) = 0 \quad \text{for any } i \in K,$$

which means that $w^{-1}\beta \in \sum_{i\in I\setminus K} \mathbb{Z}\alpha_i$. Since $\beta \in w\mathsf{Q}_+$, we have $\beta \in \sum_{i\in I\setminus K} \mathbb{Z}_{\geqslant 0} w\alpha_i$. Since $w\alpha_i \in \mathsf{Q}_-$ for any $i \in I \setminus K$, we obtain $\beta \in \mathsf{Q}_-$. Since $\beta \in \mathsf{Q}_+$, we conclude that $\beta = 0$. □

Proposition 5.27. *Let $M \in R$-gmod be a simple module R-gmod such that $\mathrm{Q}_w(M)$ is simple. If M commutes with C_{Λ_i} for any $i \in I_w$ such that $ws_i > w$, then $M \in \mathscr{C}_w$.*

Proof. We may assume that $I_w = I$. We can write ([29])

$$M \simeq X \nabla Y$$

with a simple X such that $\mathsf{W}^*(X) \subset \mathsf{Q}_+ \cap w\mathsf{Q}_+$ and a simple $Y \in \mathscr{C}_w$. Since $\mathrm{Q}_w(M)$ is simple, the objects $\mathrm{Q}_w(X)$ and $\mathrm{Q}_w(Y)$ are also simple. For any $\Lambda \in \mathsf{P}_+$, we have (see Proposition 5.10)

$$\begin{aligned}\Lambda(\mathsf{C}_\Lambda, M) = -\big(w\Lambda + \Lambda, \mathrm{wt}(M)\big) &= -\big(w\Lambda + \Lambda, \mathrm{wt}(X)\big) - \big(w\Lambda + \Lambda, \mathrm{wt}(Y)\big) \\ &= \Lambda(\mathsf{C}_\Lambda, X) + \Lambda(\mathsf{C}_\Lambda, Y).\end{aligned}$$

On the other hand, since (X, C_Λ) is unmixed, $(X, Y, \mathsf{C}_\Lambda)$ is a normal sequence by Proposition 4.12 and Lemma 5.1, and hence we have

$$\Lambda(M, \mathsf{C}_\Lambda) = \Lambda(X, \mathsf{C}_\Lambda) + \Lambda(Y, \mathsf{C}_\Lambda).$$

Hence we have

$$\mathfrak{d}(\mathsf{C}_\Lambda, M) = \mathfrak{d}(\mathsf{C}_\Lambda, X) + \mathfrak{d}(\mathsf{C}_\Lambda, Y) = \mathfrak{d}(\mathsf{C}_\Lambda, X).$$

Hence X commutes with C_{Λ_i} for any $i \in I_w$ such that $ws_i > w$. Then $X \simeq \mathbf{1}$ by Lemma 5.26, which implies that $M \in \mathscr{C}_w$. □

We need the following proposition in § 8 to prove the main theorem.

Proposition 5.28. *Let M be a simple module in R-gmod such that $\mathrm{Q}_w(M)$ is simple. Set $K = \{i \in I_w \mid ws_i > w\}$. Then there exists $\Lambda \in \sum_{i\in K} \mathbb{Z}_{\geqslant 0}\Lambda_i$ such that $\mathsf{C}_\Lambda \nabla M \in \mathscr{C}_w$.*

Proof. Set $\lambda = \sum_{i\in K} \Lambda_i$. Proposition 3.17 (ix) implies that $\mathfrak{d}(\mathsf{C}_\lambda, (\mathsf{C}_\lambda)^{\circ n} \nabla M) = 0$ for some $n > 0$. Hence setting $\Lambda = n\lambda \in \sum_{i\in K} \mathbb{Z}_{\geqslant 0}\Lambda_i$, the module $\mathsf{C}_\Lambda \nabla M$ commutes with C_{Λ_i} for any $i \in K$. Since $\mathrm{Q}_w(\mathsf{C}_\Lambda \nabla M)$ is simple, Proposition 5.27 implies that $\mathsf{C}_\Lambda \nabla M \in \mathscr{C}_w$. □

6. Crystal structure on $\widetilde{\mathscr{C}_w}$

Let $\mathrm{Irr}(\widetilde{\mathscr{C}_w})$ be the set of the equivalence classes of simple objects in $\widetilde{\mathscr{C}_w}$ up to grading shifts. In this section, we introduce a crystal structure on $\mathrm{Irr}(\widetilde{\mathscr{C}_w})$, which is one of our main aims.

6.1. **Root objects in the localization $\widetilde{\mathscr{C}}_w$.** Recall that an object X in a monoidal category $\mathcal{C}$ is *invertible* if there exists $X' \in \mathcal{C}$ such that $X \otimes X' \simeq X' \otimes X \simeq \mathbf{1}$. If $\mathcal{C}$ is rigid, then $X' \simeq \mathscr{D}X \simeq \mathscr{D}^{-1}X$.

Lemma 6.1. *Let $L \in \widetilde{\mathscr{C}}_w$ be a simple object and assume that L is invertible. Then any simple $X \in \widetilde{\mathscr{C}}_w$ strongly commutes with L.*

Proof. We have $X \simeq X \circ (L \circ \mathscr{D}L) \simeq (X \circ L) \circ \mathscr{D}L$. Hence, $(X \circ L) \circ \mathscr{D}L$ is simple, which implies that $X \circ L$ is simple. $\square$

Lemma 6.2. *An affreal simple $L \in \widetilde{\mathscr{C}}_w$ is invertible if and only if $\mathfrak{d}(L, \mathscr{D}L) = 0$.*

Proof. It is obvious that the invertibility of L implies $\mathfrak{d}(L, \mathscr{D}L) = 0$. Assume that $\mathfrak{d}(L, \mathscr{D}L) = 0$. Then $L \circ \mathscr{D}L \simeq L \circ \mathscr{D}L$ is simple, and hence $L \circ \mathscr{D}L \to 1$ is an isomorphism, which implies that L is invertible. $\square$

Definition 6.3. Let $L \in \widetilde{\mathscr{C}}_w$ be a real simple object. We say that L is a *root object* (of degree $d_L \in \mathbb{Z}_{>0}$) if

(a) L has an affinization (L, z) with $\deg(z) = 2d_L$ (see Definition 3.9),
(b) $\mathfrak{d}(L, \mathscr{D}^{-1}L) = d_L$.

Lemma 6.4. *If $L \in \widetilde{\mathscr{C}}_w$ is a root object, then $\mathscr{D}^{\pm 1}L$ is also a root object.*

Proof. By [15, Proposition 5.6], $\mathscr{D}^{\pm 1}L$ has an affinization of degree $2d_L$. $\square$

Lemma 6.5. *If $L \in \widetilde{\mathscr{C}}_w$ is a root object, then $\Lambda(L, \mathscr{D}L) = 0$ and $\Lambda(\mathscr{D}L, L) = 2d_L$.*

Proof. We have $2d_L = 2\,\mathfrak{d}(L, \mathscr{D}L) = \Lambda(L, \mathscr{D}L) + \Lambda(\mathscr{D}L, L)$ and $\Lambda(L, \mathscr{D}L) = \Lambda(L, L) = 0$ by Lemma 3.12. Hence we obtain $\Lambda(\mathscr{D}L, L) = 2d_L$. $\square$

Proposition 6.6 (cf. Proposition 3.17)**.** *Let L be a root object in $\widetilde{\mathscr{C}}_w$ of degree d_L, and M a simple object in $\widetilde{\mathscr{C}}_w$. If $\mathfrak{d}(L, M) > 0$, then we have*

(i)

$$\Lambda(L, L \nabla M) = \Lambda(L, M) \quad \textit{and} \quad \Lambda(L \nabla M, L) = \Lambda(M, L) - 2d_L,$$
$$\Lambda(L, M \nabla L) = \Lambda(L, M) - 2d_L \quad \textit{and} \quad \Lambda(M \nabla L, L) = \Lambda(M, L),$$
$$\mathfrak{d}(L \nabla M, L) = \mathfrak{d}(M \nabla L, L) = \mathfrak{d}(M, L) - \mathsf{d}_L.$$

(ii) *Assume further that $\big(\mathrm{wt}(L), \mathrm{wt}(L)\big) = 2d_L$. Then, we have*

$$\widetilde{\Lambda}(L, M \nabla L) = \widetilde{\Lambda}(L, M) \quad \textit{and} \quad \widetilde{\Lambda}(L \nabla M, L) = \widetilde{\Lambda}(M, L).$$

Proof. Since $\Lambda(L \nabla M, L) < \Lambda(M, L)$ by Proposition 3.17, we have $\Lambda(L \nabla M, L) \leqslant \Lambda(M, L) - 2d_L$ by Proposition 3.19. On the other hand, the same proposition implies that

$$\Lambda(L \nabla M, L) + \Lambda(\mathscr{D}L, L) - \Lambda(M, L) \geqslant 0$$

since $(L \nabla M) \nabla \mathscr{D}L \simeq M$. Since $\Lambda(\mathscr{D}L, L) = 2\mathsf{d}_L$, we have $\Lambda(M, L) - 2d_L \leqslant \Lambda(L \nabla M, L)$. Therefore, we obtain $\Lambda(L \nabla M, L) = \Lambda(M, L) - 2d_L$. The other assertions are obtained similarly. $\square$

Proposition 6.7. *Let L be a root object of $\widetilde{\mathscr{C}}_w$, and M a simple object in $\widetilde{\mathscr{C}}_w$. If $\mathfrak{d}(L, M) \neq d_L$, then we have*

$$(L \nabla M) \nabla L \simeq L \nabla (M \nabla L).$$

Proof. If $\mathfrak{d}(L, M) = 0$, then L and M commute, and hence $L \circ M \circ L$ is simple. Thus, we may assume that $\mathfrak{d}(L, M) \geqslant 2d_L$ by Proposition 3.19. Let K be the kernel of $M \circ L \to M \nabla L$, i.e., the radical of $M \circ L$. Then we have a commutative diagram with an exact row:

$$\begin{tikzcd}
0 \arrow[r] & L \circ K \arrow[r, "\iota"] \arrow[dr, "f"'] & L \circ M \circ L \arrow[r, "p"] \arrow[d, "\pi'"'] & L \circ (M \nabla L) \arrow[r] \arrow[dl, dashed, "\pi"] & 0 \\
& & (L \nabla M) \nabla L & &
\end{tikzcd}$$

Let us show that the composition $f\colon L \circ K \to (L \nabla M) \nabla L$ vanishes. Indeed, if it holds, we obtain an epimorphism $\pi\colon L \circ (M \nabla L) \twoheadrightarrow (L \nabla M) \nabla L$, which shows the claim.

In order to see the vanishing of f, it is enough to show that $L \nabla S \not\simeq (L \nabla M) \nabla L$ for any simple subquotient S of K since $L \nabla S$ covers the composition series of $L \circ K$. We have

$$\Lambda(S, L) < \Lambda(M, L) = \Lambda\big(L \nabla M, L\big) + 2d_L = \Lambda\big((L \nabla M) \nabla L, L\big) + 2d_L.$$

Here, the first inequality follows from Proposition 3.17, and the equalities follow from Proposition 6.6. If $\mathfrak{d}(L, S) > 0$, then we have $\Lambda(L \nabla S, L) = \Lambda(S, L) - 2d_L$ by Proposition 6.6. we have $L \nabla S \not\simeq (L \nabla M) \nabla L$. Hence we may assume that $\mathfrak{d}(L, S) = 0$ or equivalently, L commutes with S. Then, $S \nabla L \simeq (L \nabla M) \nabla L$ implies that $S \simeq L \nabla M$, which derives a contradiction $0 = \mathfrak{d}(L, S) = \mathfrak{d}(L, L \nabla M) = \mathfrak{d}(L, M) - d_L \geqslant 2d_L - d_L > 0$. $\square$

The following proposition will be used in this section and § 8.

Proposition 6.8. *Assume that $i \in I$ satisfies either $s_i w < w$ or $w s_i < w$. Then, $\mathrm{Q}_w(\langle i \rangle)$ is a root object or invertible.*

Proof. Since $\mathrm{Q}_w(\langle i\rangle)$ has an affinization of degree $2\mathsf{d}_i$ by Proposition 5.6, it is enough to show that

$$\mathfrak{d}\big(\mathrm{Q}_w(\langle i\rangle), \mathscr{D}^{-1}\mathrm{Q}_w(\langle i\rangle)\big) \leqslant \mathsf{d}_i. \tag{6.1}$$

Set $M = \mathsf{M}(w\Lambda_i, \Lambda_i)$. Then $\mathscr{D}^{-1}\mathrm{Q}_w(\langle i\rangle) \simeq \widetilde{\mathsf{C}}_{-\Lambda_i} \circ \mathrm{Q}_w(\mathrm{E}_i^* M)$ by Theorem 4.17(d). Hence we obtain

$$\mathfrak{d}\big(\mathrm{Q}_w(\langle i\rangle), \mathscr{D}^{-1}\mathrm{Q}_w(\langle i\rangle)\big) \leqslant \mathfrak{d}\big(\langle i\rangle, \mathrm{E}_i^* M\big) = \mathsf{d}_i\, \mathfrak{d}_i(\mathrm{E}_i^* M)$$

by Lemma 5.21.

By Proposition 4.16, we have $\varepsilon_i^*(\mathrm{E}_i^* M) = 0$ and $\varepsilon_i(\mathrm{E}_i^* M) \leqslant \max\big(0, -\langle h_i, w\Lambda_i\rangle\big)$. Hence, we obtain

$$\begin{aligned}\mathfrak{d}_i(\mathrm{E}_i^* M) &= \varepsilon_i(\mathrm{E}_i^* M) + \varepsilon_i^*(\mathrm{E}_i^* M) + \langle h_i, \mathrm{wt}\,(\mathrm{E}_i^* M)\rangle \\ &\leqslant \max\big(0, -\langle h_i, w\Lambda_i\rangle\big) + \langle h_i, w\Lambda_i - s_i\Lambda_i\rangle = \max\big(\langle h_i, w\Lambda_i - s_i\Lambda_i\rangle, 1\big).\end{aligned}$$

If $ws_i < w$, then we have $w\Lambda_i - s_i\Lambda_i \in \mathsf{Q}_-$ and hence $\langle h_i, w\Lambda_i - s_i\Lambda_i\rangle \leqslant 0$. Hence we obtain (6.1).

If $s_i w < w$, then we have $w\alpha_i \in \Delta_-$ and hence $\langle h_i, w\Lambda_i - s_i\Lambda_i\rangle = \langle w^{-1}h_i, \Lambda_i\rangle + 1 \leqslant 1$, which implies (6.1). □

Lemma 6.9. *Assume that $i \in I$ satisfies $w' := ws_i < w$ and $i \notin I_{w'}$. Then $\mathrm{Q}_w(\langle i\rangle)$ is invertible.*

Proof. Set $\lambda = \Lambda_i + s_i\Lambda_i = 2\Lambda_i - \alpha_i \in \mathsf{P}_+$. Then, $\langle h_j, s_i\Lambda_i\rangle = \langle h_j, \lambda\rangle \in \mathbb{Z}_{\geqslant 0}$ for any $j \in I_{w'}$, and we have

$$\mathsf{M}(w's_i\Lambda_i, s_i\Lambda_i) \simeq \mathsf{M}(w'\lambda, \lambda) \simeq \mathsf{M}(w\lambda, \lambda).$$

Hence, we have $\mathsf{M}(w\lambda, \lambda)\nabla\langle i\rangle \simeq \mathsf{M}(w\Lambda_i, \Lambda_i)$, which implies that $\mathrm{Q}_w(\langle i\rangle) \simeq \widetilde{\mathsf{C}}_\lambda^{\circ -1} \circ \widetilde{\mathsf{C}}_{\Lambda_i} \simeq \widetilde{\mathsf{C}}_{-s_i\Lambda_i}$ is invertible. □

Remark 6.10. In general, $\mathrm{Q}_w(\langle i\rangle)$ may be neither a root object nor an invertible object. For example, take $\mathfrak{g} = A_3$, $w = s_2 w_0 = s_1 s_3 s_2 s_1 s_3$, and $i = 2$. Then $\mathrm{Q}_w(\langle i\rangle) \simeq \langle 132\rangle^{-1} \circ \langle 12\rangle \circ \langle 32\rangle$ and $\mathscr{D}^{-1}\mathrm{Q}_w(\langle i\rangle) \simeq \langle 1\rangle \circ \langle 3\rangle \circ \langle 132\rangle^{-1}$. Hence $\mathfrak{d}\big(\mathrm{Q}_w(\langle i\rangle), \mathscr{D}^{-1}\mathrm{Q}_w(\langle i\rangle)\big) = 2$.

6.2. **Simple root operators.** Now we shall define the simple root operators on the localized category. In this case, the simple root operators are bijective.

Definition 6.11. For any $i \in I_w$ and a simple $X \in \widetilde{\mathscr{C}_w}$, we define

$$\begin{aligned}
\varepsilon_i(X) &= \mathsf{d}_i^{-1}\widetilde{\Lambda}(\mathrm{Q}_w(\langle i\rangle), X) \in \mathbb{Z},\\
\varepsilon_i^*(X) &= \mathsf{d}_i^{-1}\widetilde{\Lambda}(X, \mathrm{Q}_w(\langle i\rangle)) \in \mathbb{Z},\\
\varphi_i(X) &= \varepsilon_i(X) + \langle h_i, \mathrm{wt}\, X\rangle \in \mathbb{Z},\\
\varphi_i^*(X) &= \varepsilon_i^*(X) + \langle h_i, \mathrm{wt}\, X\rangle \in \mathbb{Z},\\
\mathfrak{d}_i(X) &= \mathsf{d}_i^{-1}\mathfrak{d}(\mathrm{Q}_w(\langle i\rangle), X) = \varepsilon_i(X) + \varepsilon_i^*(X) + \langle h_i, \mathrm{wt}\, X\rangle \in \mathbb{Z}_{\geqslant 0},\\
\widetilde{\mathrm{F}}_i X &= q_i^{\varepsilon_i(X)}\mathrm{Q}_w(\langle i\rangle) \nabla X \in \widetilde{\mathscr{C}_w},\\
\widetilde{\mathrm{F}}_i^* X &= q_i^{\varepsilon_i^*(X)} X \nabla \mathrm{Q}_w(\langle i\rangle) \in \widetilde{\mathscr{C}_w},\\
\widetilde{\mathrm{E}}_i X &= q_i^{\varphi_i(X)+1} X \nabla \mathscr{D}\mathrm{Q}_w(\langle i\rangle) \in \widetilde{\mathscr{C}_w},\\
\widetilde{\mathrm{E}}_i^* X &= q_i^{\varphi_i^*(X)+1} \mathscr{D}^{-1}\mathrm{Q}_w(\langle i\rangle) \nabla X \in \widetilde{\mathscr{C}_w}.
\end{aligned}$$

For $i \in I \setminus I_w$, we understand $\widetilde{\mathrm{F}}_i X = \widetilde{\mathrm{E}}_i X = 0$, $\varepsilon_i(X) = \varepsilon_i^*(X) = \mathfrak{d}_i(X) = -\infty$.

The following proposition is immediate from the definition above and Lemma 3.18.

Proposition 6.12. *$\widetilde{\mathrm{F}}_i$ and $\widetilde{\mathrm{E}}_i$ are inverse to each other. Similarly, $\widetilde{\mathrm{F}}_i^*$ and $\widetilde{\mathrm{E}}_i^*$ are inverse to each other.*

One of the main aims is to show that these data define a crystal structure on $\widetilde{\mathscr{C}_w}$ the set of equivalence classes of simples in $\widetilde{\mathscr{C}_w}$ up to grading shifts.

Theorem 6.13. *The data in Definition 6.11 defines a crystal structure on $\mathrm{Irr}(\widetilde{\mathscr{C}_w})$.*

Proof. We have all the conditions in Definition 2.1 except (2) and (3). Thus, it suffices to show that if $X \in \mathrm{Irr}(\widetilde{\mathscr{C}_w})$, then $\mathrm{wt}(\widetilde{\mathrm{F}}_i X) = \mathrm{wt}(X) - \alpha_i$, $\varepsilon_i(\widetilde{\mathrm{F}}_i X) = \varepsilon_i(X) + 1$, $\varphi_i(\widetilde{\mathrm{F}}_i X) = \varphi_i(X) - 1$, which implies Definition 2.1 (3). The condition (2) is obtained by the fact $\widetilde{\mathrm{E}}_i = \widetilde{\mathrm{F}}_i^{-1}$.

Now, by Proposition 6.6(i) we have $\Lambda\big(\mathrm{Q}_w(\langle i\rangle), \mathrm{Q}_w(\langle i\rangle) \nabla X\big) = \Lambda\big(\mathrm{Q}_w(\langle i\rangle), X\big)$ and then

$$\begin{aligned}
&2\widetilde{\Lambda}\big(\mathrm{Q}_w(\langle i\rangle), \mathrm{Q}_w(\langle i\rangle) \nabla X\big)\\
&\qquad = \Lambda(\mathrm{Q}_w(\langle i\rangle, X) + (\mathrm{wt}(\mathrm{Q}_w(\langle i\rangle), \mathrm{wt}(\mathrm{Q}_w(\langle i\rangle) + \mathrm{wt}(X))\\
&\qquad = 2\widetilde{\Lambda}(\mathrm{Q}_w(\langle i\rangle, X) + (\mathrm{wt}(\mathrm{Q}_w(\langle i\rangle)), \mathrm{wt}(\mathrm{Q}_w(\langle i\rangle))) = 2\widetilde{\Lambda}(\mathrm{Q}_w(\langle i\rangle, X) + 2\mathsf{d}_i,
\end{aligned}$$

which means $\varepsilon_i(\widetilde{\mathrm{F}}_i X) = \varepsilon_i(X) + 1$ and the other assertions are obtained by the definitions. $\square$

Proposition 6.14. *For any $i \in I_w$ and any simple $X \in \widetilde{\mathscr{C}_w}$, we have*

(i) $\varphi_i(X) = \mathsf{d}_i^{-1}\widetilde{\Lambda}\big(X, \mathscr{D}\mathrm{Q}_w(\langle i\rangle)\big)$,
(ii) $\varphi_i^*(X) = \mathsf{d}_i^{-1}\widetilde{\Lambda}\big(\mathscr{D}^{-1}\mathrm{Q}_w(\langle i\rangle), X\big)$.

Proof. This follows from $\widetilde{\Lambda}(M,N) = \widetilde{\Lambda}(N,\mathscr{D}M) + (\mathrm{wt}\, M, \mathrm{wt}\, N) = \widetilde{\Lambda}(\mathscr{D}^{-1}N, M) + (\mathrm{wt}\, M, \mathrm{wt}\, N)$ for simple M, $N \in \widetilde{\mathscr{C}_w}$ such that one of them is affreal (see Lemma 3.12). ☐

Lemma 6.15. *Let $M \in \mathfrak{B}_w$ be a simple module and $i \in I_w$. Then we have*
(i) $\widetilde{\mathrm{E}}_i M$, $\widetilde{\mathrm{E}}_i^* M \in \mathfrak{B}_w$,
(ii) *If $\widetilde{\mathrm{E}}_i M \not\simeq 0$, then $\mathrm{Q}_w(\widetilde{\mathrm{E}}_i M) \simeq \widetilde{\mathrm{E}}_i\big(\mathrm{Q}_w(M)\big)$,*
(iii) *If $\widetilde{\mathrm{E}}_i^* M \not\simeq 0$, then $\mathrm{Q}_w(\widetilde{\mathrm{E}}_i^* M) \simeq \widetilde{\mathrm{E}}_i^*\big(\mathrm{Q}_w(M)\big)$.*

Proof. We prove only the statements for $\widetilde{\mathrm{E}}_i$. We may assume $\widetilde{\mathrm{E}}_i M \not\simeq 0$. Then we have

$$\langle i\rangle \circ \widetilde{\mathrm{E}}_i M \twoheadrightarrow M \rightarrowtail \widetilde{\mathrm{E}}_i M \circ \langle i\rangle.$$

Applying Q_w, we obtain

$$\mathrm{Q}_w(\langle i\rangle) \circ \mathrm{Q}_w(\widetilde{\mathrm{E}}_i M) \twoheadrightarrow \mathrm{Q}_w(M) \rightarrowtail \mathrm{Q}_w(\widetilde{\mathrm{E}}_i M) \circ \mathrm{Q}_w(\langle i\rangle).$$

Hence $\mathrm{Q}_w(\widetilde{\mathrm{E}}_i M) \not\simeq 0$, and then we have $\mathrm{Q}_w(\langle i\rangle) \nabla \mathrm{Q}_w(\widetilde{\mathrm{E}}_i M) \simeq \mathrm{Q}_w(M)$, or equivalently, $\widetilde{\mathrm{F}}_i\big(\mathrm{Q}_w(\widetilde{\mathrm{E}}_i M)\big) \simeq \mathrm{Q}_w(M)$. It implies that $\widetilde{\mathrm{E}}_i \mathrm{Q}_w(M) \simeq \mathrm{Q}_w(\widetilde{\mathrm{E}}_i M)$. ☐

Now, we can describe $\mathscr{D}\big(\mathrm{Q}_w(\langle i\rangle)$ explicitly by using the generalized determinantial modules.

Lemma 6.16. *For any $i \in I_w$, we have*

$$\mathscr{D}\big(\mathrm{Q}_w(\langle i\rangle)\big) \simeq \mathrm{Q}_w\big(\mathrm{E}_i \mathsf{M}_w(-\Lambda_i, -w^{-1}\Lambda_i)\big) \circ \widetilde{\mathsf{C}}_{w^{-1}\Lambda_i}.$$

Proof. Note that Lemma 5.8 implies

$$\mathrm{E}_i \mathsf{M}_w(-\Lambda_i, -w^{-1}\Lambda_i) \simeq \psi_*\Big(\mathrm{E}_i^* \mathsf{M}(w^{-1}\Lambda_i, \Lambda_i)\Big) \simeq \psi_*\Big(\mathsf{M}(w^{-1}\Lambda_i, s_i\Lambda_i)\Big).$$

The assertion follows from $\langle i\rangle \circ \big(\mathrm{E}_i \mathsf{M}_w(-\Lambda_i, -w^{-1}\Lambda_i)\big) \twoheadrightarrow \mathsf{M}_w(-\Lambda_i, -w^{-1}\Lambda_i)$ and $\mathrm{Q}_w\big(\mathsf{M}_w(-\Lambda_i, -w^{-1}\Lambda_i)\big) \simeq \widetilde{\mathsf{C}}_{-w^{-1}\Lambda_i}$ by (5.7). ☐

Lemma 6.17. *Let $i, j \in I_w$ such that $i \neq j$. Assume further that either $s_i w < w$ or $w s_j < w$. Then $\mathscr{D}\mathrm{Q}_w(\langle i\rangle)$ and $\mathrm{Q}_w(\langle j\rangle)$ commute.*

Proof. Assume first $s_i w < w$. Then, we have $\mathsf{W}(\langle i\rangle) \setminus \{0\} = \{\alpha_i\} \subset \Delta_+ \cap w\Delta_-$ and hence $\langle i\rangle \in \mathscr{C}_w$. Then $\mathsf{M}(w\Lambda_j, \Lambda_j)$ commutes with $\langle i\rangle$. Hence $\widetilde{\mathrm{E}}_j^{*\,\max} \mathsf{M}(w\Lambda_j, \Lambda_j) \simeq$

$\mathsf{M}(w\Lambda_j, s_j\Lambda_j)$ commutes with $\widetilde{\mathrm{E}}_j^{*\ \max}\langle i\rangle \simeq \langle i\rangle$, which implies that $\mathscr{D}^{-1}\mathrm{Q}_w(\langle j\rangle) \simeq \widetilde{\mathsf{C}}_j^{-1} \circ \mathrm{Q}_w\big(\widetilde{\mathrm{E}}_j^{*\ \max}\mathsf{M}(w\Lambda_j, \Lambda_j)\big)$ commutes with $\mathrm{Q}_w(\langle i\rangle)$.

Now assume that $ws_j < w$. Then $s_j w^{-1} < w^{-1}$. Hence $\mathscr{D}\big(\mathrm{Q}_{w^{-1}}(\langle j\rangle)\big)$ and $\mathrm{Q}_{w^{-1}}(\langle i\rangle)$ commute. By applying ψ_*, we obtain that $\mathscr{D}^{-1}\big(\mathrm{Q}_w\langle j\rangle\big)$ and $\mathrm{Q}_w(\langle i\rangle)$ commute by Theorem 5.9. □

The following lemma is an immediate consequence of Proposition 6.6, Proposition 6.7 and Proposition 6.8.

Lemma 6.18. *Assume that $i \in I$ satisfies $ws_i < w$. Let X be a simple object of $\widetilde{\mathscr{C}}_w$. Then, we have*

(i) *if $\mathfrak{d}_i(X) > 0$, then $\varepsilon_i\big((\widetilde{\mathrm{E}}_i^*)^n(X)\big) = \varepsilon_i(X)$ for any $n \in \mathbb{Z}_{\geqslant 0}$ and $\widetilde{\mathrm{F}}_i\widetilde{\mathrm{E}}_i^*(X) \simeq \widetilde{\mathrm{E}}_i^*\widetilde{\mathrm{F}}_i(X)$,*
(ii) *$\varepsilon_i(\widetilde{\mathrm{E}}_i^* X) = \varepsilon_i(X)$ or $\varepsilon_i(\widetilde{\mathrm{E}}_i^* X) = \varepsilon_i(X) - 1$.*

Note that in (i), $\mathrm{Q}_w(\langle i\rangle)$ is a root object.

Lemma 6.19. *Let $i, j \in I_w$ such that $i \neq j$. Assume further that either $s_i w < w$ or $ws_j < w$. Then we have*

(i) *$\widetilde{\mathrm{F}}_i$ and $\widetilde{\mathrm{F}}_j^*$ commute as operators on $\mathrm{Irr}(\widetilde{\mathscr{C}}_w)$,*
(ii) *$\varepsilon_i(\widetilde{\mathrm{F}}_j^* X) = \varepsilon_i(X)$ and $\varepsilon_j^*(\widetilde{\mathrm{F}}_i X) = \varepsilon_j^*(X)$ for any $X \in \mathrm{Irr}(\widetilde{\mathscr{C}}_w)$.*

Proof. By Lemma 6.17 and Lemma 4.11, $(\mathrm{Q}_w(\langle i\rangle), X, \mathrm{Q}_w(\langle j\rangle))$ is an almost affreal normal sequence. Hence we have

$$(\mathrm{Q}_w(\langle i\rangle) \nabla X) \nabla \mathrm{Q}_w(\langle j\rangle) \simeq \mathrm{Q}_w(\langle i\rangle) \nabla (X \nabla \mathrm{Q}_w(\langle j\rangle))$$

and

$$\widetilde{\Lambda}(\mathrm{Q}_w(\langle i\rangle), X \nabla \mathrm{Q}_w(\langle j\rangle)) = \widetilde{\Lambda}(\mathrm{Q}_w(\langle i\rangle), X) + \widetilde{\Lambda}(\mathrm{Q}_w(\langle i\rangle), \mathrm{Q}_w(\langle j\rangle)).$$

Note that $\widetilde{\Lambda}(\mathrm{Q}_w(\langle i\rangle), \mathrm{Q}_w(\langle j\rangle)) = \widetilde{\Lambda}(\langle i\rangle, \langle j\rangle) = 0$ since $\langle i\, j\rangle \in \mathfrak{B}_w$. □

Lemma 6.20. *Let $M \in \mathfrak{B}_w$ be a simple module. Then we have*

(i) *if $s_i w < w$, then we have $\widetilde{\mathrm{F}}_i(\mathrm{Q}_w(M)) \simeq \mathrm{Q}_w(\widetilde{\mathrm{F}}_i(M))$ and $\varepsilon_i(\mathrm{Q}_w(M)) = \varepsilon_i(M)$,*
(ii) *if $ws_i < w$, then we have $\widetilde{\mathrm{F}}_i^*(\mathrm{Q}_w(M)) \simeq \mathrm{Q}_w(\widetilde{\mathrm{F}}_i^*(M))$ and $\varepsilon_i^*(\mathrm{Q}_w(M)) = \varepsilon_i^*(M)$.*

Proof. (i) Since $\langle i\rangle \in \mathscr{C}_w$, it follows from Proposition 5.20.

(ii) follows from (i) for w^{-1} by applying ψ_* (see Theorem 5.9). □

Remark 6.21. In general, $\varepsilon_i(\mathrm{Q}_w(M))$ can be negative for $M \in \mathscr{C}_w$ (cf. Corollary 7.3). The equality $\varepsilon_i\big(\mathrm{Q}_w(M)\big) = \varepsilon_i(M)$ *does not hold* for every $i \in I_w$ and every simple $M \in \mathscr{C}_w$. For example, take $\mathfrak{g} = A_2$, $w = s_1 s_2$, $i = 2$ and $M = \langle 1 \rangle$. Then, $\mathscr{C}_w$ is generated by its central objects $\langle 1 \rangle$ and $\langle 12 \rangle$. We have $\langle 1 \rangle \nabla \langle 2 \rangle \simeq \langle 12 \rangle$ in R-gmod as in Remark 5.22 and then we have $\mathrm{Q}_w(\langle 1 \rangle) \circ \mathrm{Q}_w(\langle 2 \rangle) \simeq \mathrm{Q}_w(\langle 12 \rangle)$ in R-gmod$[C_i^{\circ -1}; i \in I] \simeq \widetilde{\mathscr{C}}_w$ (see Theorem 5.3). Thus, we have $\mathrm{Q}_w(\langle 2 \rangle) = \mathrm{Q}_w(\langle 1 \rangle)^{-1} \circ \mathrm{Q}_w(\langle 12 \rangle)$ and then

$$\begin{aligned}
\varepsilon_2\big(\mathrm{Q}_w(\langle 1 \rangle)\big) &= \mathsf{d}_2^{-1}\widetilde{\Lambda}(\mathrm{Q}_w(\langle 1 \rangle)^{-1} \circ \mathrm{Q}_w(\langle 12 \rangle),\, \mathrm{Q}_w(\langle 1 \rangle)) \\
&= -\mathsf{d}_2^{-1}\widetilde{\Lambda}(\mathrm{Q}_w(\langle 1 \rangle), \mathrm{Q}_w(\langle 1 \rangle)) + \mathsf{d}_2^{-1}\widetilde{\Lambda}(\mathrm{Q}_w(\langle 12 \rangle), \mathrm{Q}_w(\langle 1 \rangle)) \\
&= -\widetilde{\Lambda}(\langle 1 \rangle, \langle 1 \rangle) + \widetilde{\Lambda}(\langle 12 \rangle, \langle 1 \rangle) = -1 + 0 = -1,
\end{aligned}$$

where $\mathsf{d}_2 = 1$. However, we have $\varepsilon_2(\langle 1 \rangle) = 0$.

Another example is $\mathfrak{g} = A_3$, $w = s_1 w_0 = s_2 s_1 s_3 s_2 s_1$, $i = 1$ and $M = \langle 23 \rangle$. Then we have $\mathscr{C}_w = \{Z \in R\text{-gmod} \mid \mathrm{E}_1(M) \simeq 0\}$, $\mathrm{Q}_w(\langle 1 \rangle) = \langle 23 \rangle^{\circ -1} \circ \langle 231 \rangle$. We have $\varepsilon_1(M) = 0$ and $\varepsilon_1\big(\mathrm{Q}_w(M)\big) = \widetilde{\Lambda}(\langle 231 \rangle, \langle 23 \rangle) - \widetilde{\Lambda}(\langle 23 \rangle, \langle 23 \rangle) = -1$. Indeed, since the pair $(\langle 231 \rangle, \langle 23 \rangle)$ is unmixed, [12, Corollary 2.3]) implies that $\widetilde{\Lambda}(\langle 231 \rangle, \langle 23 \rangle) = 0$.

6.3. **The map $\mathbf{E}^*_{w',w}$.** *In this subsection, let $w \in \mathsf{W}$ and assume that $i \in I$ satisfies $w' := w s_i < w$.* By Lemma 5.15 we can define

$$\widetilde{\mathrm{E}}_i^{*\max}\colon \mathrm{Irr}(\mathfrak{B}_w) \to \mathrm{Irr}(\mathfrak{B}_{w'})$$

by

$$M \mapsto \widetilde{\mathrm{E}}_i^{*\max}(M).$$

Then it induces

$$\mathrm{Irr}(\mathscr{C}_w) \to \mathrm{Irr}(\widetilde{\mathscr{C}}_{w'})$$

by

$$M \longmapsto \mathrm{Q}_{w'}\big(\widetilde{\mathrm{E}}_i^{*\max}(M)\big).$$

Note that by the shuffle lemma (Lemma 4.22) we have for any simple $M \in \mathscr{C}_w$ and $\Lambda \in \mathsf{P}_+$,

$$\widetilde{\mathrm{E}}_i^{*\max}(\mathsf{C}_\Lambda \circ M) \simeq \widetilde{\mathrm{E}}_i^{*\max}(\mathsf{C}_\Lambda) \circ \widetilde{\mathrm{E}}_i^{*\max}(M) \tag{6.2}$$

Hence, it extends to

$$\mathbf{E}^*_{w',w}\colon \mathrm{Irr}(\widetilde{\mathscr{C}}_w) \to \mathrm{Irr}(\widetilde{\mathscr{C}}_{w'})$$

by

$$(\widetilde{\mathsf{C}}_\Lambda)^{-1} \circ \mathrm{Q}_w(M) \longmapsto \mathrm{Q}_{w'}\big(\widetilde{\mathrm{E}}_i^{*\max}(\mathsf{C}_\Lambda)\big)^{-1} \circ \mathrm{Q}_{w'}\big(\widetilde{\mathrm{E}}_i^{*\max}(M)\big) \quad \text{for } M \in \mathrm{Irr}(\mathscr{C}_w) \text{ and } \Lambda \in \mathsf{P}_+.$$

Note that $\mathrm{Q}_{w'}(\widetilde{\mathrm{E}}_i^{*\max}\mathsf{C}_\Lambda)$ is invertible by (5.6):

$$\mathrm{Q}_{w'}\big(\widetilde{\mathrm{E}}_i^{*\max}\mathsf{C}_\Lambda\big) \simeq \mathrm{Q}_{w'}\big(\mathsf{M}(w\Lambda, s_i\Lambda)\big) \simeq \mathrm{Q}_{w'}\big(\mathsf{M}_{w'}(w\Lambda, s_i\Lambda)\big) \simeq \widetilde{\mathsf{C}}^{w'}_{s_i\Lambda}.$$

Together with the map

$$\varepsilon_i^*\colon \mathrm{Irr}(\widetilde{\mathscr{C}}_w) \to \mathbb{Z},$$

we obtain

$$\mathrm{Irr}(\widetilde{\mathscr{C}}_w) \to \mathrm{Irr}(\widetilde{\mathscr{C}}_{w'}) \times \mathbb{Z} \qquad (X \mapsto (\mathbf{E}^*_{w',w}(X), \varepsilon_i^*(X)). \tag{6.3}$$

Remark 6.22. For $M \in \mathscr{C}_w$ and $i \in I$ such that $w' := ws_i < w$, the statement $\widetilde{\mathrm{E}}_i^{*\ \max}M \in \mathscr{C}_{w'}$ may not hold. For example, take $\mathfrak{g} = A_2$, $w = s_1s_2s_1$. Then, we have $\mathscr{C}_w = R$-gmod. Take $i = 1$ and $M = \langle 2\rangle$. Then $\widetilde{\mathrm{E}}_i^{*\ \max}M \simeq M \notin \mathscr{C}_{w'} = \{M \mid E_2M \simeq 0\}$.

Proposition 6.23. *Let $L \in \mathscr{C}_w$ be an affreal simple module and $M \in \mathfrak{B}_w$ a simple module. Set $L' = \mathbf{E}^*_{w',w}(L)$ and $M' = \mathbf{E}^*_{w',w}(M)$. Assume that $\varepsilon_i^*(L \nabla M) = \varepsilon_i^*(L) + \varepsilon_i^*(M)$. Then we have*
- (i) *$\mathbf{E}^*_{w',w}(L \nabla M) \simeq\ L' \nabla M'$ up to grading shifts,*
- (ii) *$\Lambda(L', M') = \Lambda(L, M) + \varepsilon_i^*(L)\big(\alpha_i, \mathrm{wt}(M)\big) - \varepsilon_i^*(M)\big(\alpha_i, \mathrm{wt}(L)\big)$.*

Proof. It follows from Proposition 4.23. □

Recall that we have assumed that $w' := ws_i < w$ in this subsection.

Proposition 6.24. *Let $M \in \mathfrak{B}_w$ and $\Lambda \in \mathsf{P}_+$. Set $\mathsf{C}''_\Lambda := \widetilde{\mathrm{E}}_i^{*\ \max}(\mathsf{C}_\Lambda) \simeq \mathsf{M}(w\Lambda, s_i\Lambda)) \simeq \mathsf{C}^{w'}_{s_i\Lambda} \in \mathscr{C}_w \cap \mathfrak{B}_{w'}$ and $M' := \widetilde{\mathrm{E}}_i^{*\ \max}(M) \in \mathfrak{B}_{w'}$. Then we have*
- (i) *$\varepsilon_i^*(\mathsf{C}_\Lambda \nabla M) = \varepsilon_i^*(\mathsf{C}_\Lambda) + \varepsilon_i^*(M)$,*
- (ii) *$\mathsf{C}''_\Lambda \nabla M' \in \mathfrak{B}_{w'}$,*
- (iii) *$\widetilde{\mathrm{E}}_i^{*\ \max}(\mathsf{C}_\Lambda \nabla M) \simeq \mathsf{C}''_\Lambda \nabla M'$,*
- (iv) *$\Lambda(\mathsf{C}''_\Lambda, M') = -\big(w\Lambda + s_i\Lambda, \mathrm{wt}(M')\big)$.*

Proof. Since $M \nabla \langle i\rangle \in \mathfrak{B}_w$, Proposition 5.10 implies that

$$\Lambda(\mathsf{C}_\Lambda, M \nabla \langle i\rangle) = -\big(w\Lambda + \Lambda, \mathrm{wt}(M \nabla \langle i\rangle)\big) = \Lambda(\mathsf{C}_\Lambda, M) + \Lambda(\mathsf{C}_\Lambda, \langle i\rangle).$$

Hence $(\mathsf{C}_\Lambda, M, \langle i\rangle)$ is a normal sequence. Hence we have

$$\Lambda(\mathsf{C}_\Lambda \nabla M, \langle i\rangle) = \Lambda(\mathsf{C}_\Lambda, \langle i\rangle) + \Lambda(M, \langle i\rangle),$$

which is nothing but (i). Hence Proposition 4.23 (iii) implies (iii). Since $\mathsf{C}_\Lambda \nabla M \in \mathfrak{B}_w$, (iii) implies (ii) by Lemma 5.15.

It remains to prove (iv). We have

$$\Lambda(\mathsf{C}''_\Lambda, M') = \Lambda\big(\mathrm{Q}_{w'}(\mathsf{C}''_\Lambda), \mathrm{Q}_{w'}(M')\big) = \Lambda\big(\widetilde{\mathsf{C}}^{w'}_{s_i\Lambda}, \mathrm{Q}_{w'}(M')\big) = -\big(w's_i\Lambda + s_i\Lambda, \mathrm{wt}(M')\big).$$

Here the first equality follows from (ii) and Lemma 5.19. $\square$

Corollary 6.25. *For any simple $M \in \mathfrak{B}_w$, we have*

$$\mathrm{Q}_{w'}(\widetilde{\mathrm{E}}_i^{*\ \max}(M)) = \mathbf{E}^*_{w',w}(\mathrm{Q}_w(M)).$$

That is, the following diagram commutes;

$$\begin{tikzcd}
\mathrm{Irr}(\mathfrak{B}_w) \arrow[d, "\mathrm{Q}_w|_{\mathrm{Irr}(\mathfrak{B}_w)}"'] \arrow[dr, "\mathrm{Q}_{w'} \circ \widetilde{\mathrm{E}}_i^{*\max}"] & \\
\mathrm{Irr}(\widetilde{\mathscr{C}}_w) \arrow[r, "\mathbf{E}^*_{w',w}"'] & \mathrm{Irr}(\widetilde{\mathscr{C}}_{w'}).
\end{tikzcd}$$

Proof. Take $\Lambda \in \mathsf{P}_+$ such that $N := \mathsf{C}_\Lambda \nabla M \in \mathscr{C}_w$. Since $\mathrm{Q}_w(M) \simeq \mathrm{Q}_w(\mathsf{C}_\Lambda)^{-1} \circ \mathrm{Q}_w(N)$, we have

$$\begin{aligned}
\mathbf{E}^*_{w',w}(\mathrm{Q}_w(M)) &\simeq \mathbf{E}^*_{w',w}(\mathrm{Q}_w(\mathsf{C}_\Lambda))^{-1} \circ \mathbf{E}^*_{w',w}(\mathrm{Q}_w(N)) \simeq \mathrm{Q}_{w'}(\mathsf{C}''_\Lambda)^{-1} \circ \mathrm{Q}_{w'}\big(\widetilde{\mathrm{E}}_i^{*\ \max}(N)\big) \\
&\simeq \mathrm{Q}_{w'}(\mathsf{C}''_\Lambda)^{-1} \circ \mathrm{Q}_{w'}(\mathsf{C}''_\Lambda \nabla \widetilde{\mathrm{E}}_i^{*\ \max}(M)\big) \\
&\simeq \mathrm{Q}_{w'}(\mathsf{C}''_\Lambda)^{-1} \circ \mathrm{Q}_{w'}(\mathsf{C}''_\Lambda) \circ \mathrm{Q}_{w'}\big(\widetilde{\mathrm{E}}_i^{*\ \max}(M)\big) \simeq \mathrm{Q}_{w'}\big(\widetilde{\mathrm{E}}_i^{*\ \max}(N)\big).
\end{aligned}$$

$\square$

Lemma 6.26. *Let $M \in \mathfrak{B}_w$ be a simple module and let X be a simple object of $\widetilde{\mathscr{C}}_w$. Then, we have*

(i) $\varepsilon_i^*\big(\mathrm{Q}_w(M)\big) = \varepsilon_i^*(M)$,
(ii) $\widetilde{\mathrm{F}}_i^* \mathrm{Q}_w(M) \simeq \mathrm{Q}_w(M \nabla \langle i \rangle)$,
(iii) $\mathbf{E}^*_{w',w}(\widetilde{\mathrm{F}}_i^*(X)) \simeq \mathbf{E}^*_{w',w}(X)$.

Proof. Since $M \nabla \langle i \rangle \in \mathfrak{B}_w$, Lemma 5.19 implies that $\Lambda\big(\mathrm{Q}_w(M), \mathrm{Q}_w(\langle i \rangle)\big) = \Lambda(M, \langle i \rangle)$ and $\mathrm{Q}_w(M) \nabla \mathrm{Q}_w(\langle i \rangle) \simeq \mathrm{Q}_w(M \nabla \langle i \rangle)$, which reads as (i) and (ii).

(iii) Let $X = (\widetilde{\mathsf{C}}_\Lambda)^{\circ -1} \circ \mathrm{Q}_w(M)$ for a simple $M \in \mathscr{C}_w$ and $\Lambda \in \mathsf{P}_+$. Then we have $\widetilde{\mathrm{F}}_i^*(X) \simeq (\widetilde{\mathsf{C}}_\Lambda)^{\circ -1} \circ \mathrm{Q}_w(\widetilde{\mathrm{F}}_i^* M)$ and

$$\begin{aligned}
\mathbf{E}^*_{w',w}(\widetilde{\mathrm{F}}_i^*(X)) &\simeq (\widetilde{\mathsf{C}}^{w'}_{s_i\Lambda})^{\circ -1} \circ \mathrm{Q}_w(\widetilde{\mathrm{E}}_i^{*\max} \widetilde{\mathrm{F}}_i^* M) \\
&\simeq (\widetilde{\mathsf{C}}^{w'}_{s_i\Lambda})^{\circ -1} \circ \mathrm{Q}_w(\widetilde{\mathrm{E}}_i^{*\max} M) \simeq \mathbf{E}^*_{w',w}(X).
\end{aligned}$$

$\square$

6.4. **Crystals.** Let us take a reduced expression $\underline{w} = s_{i_1} \cdots s_{i_\ell}$ of $w \in \mathsf{W}$. We set

$$w_{\leqslant k} = s_{i_1} \cdots s_{i_k} \quad \text{and} \quad \beta_k = w_{\leqslant k-1}\alpha_{i_k} = w_{<k}\Lambda_{i_k} - w_{\leqslant k}\Lambda_{i_k} \in \Delta_+.$$

It is well-known that $\Delta_+ \cap w\Delta_- = \{\beta_1, \dots, \beta_\ell\}$.

Recall the crystal $\mathbf{B}_{\underline{w}}$ introduced in § 2.4:

$$\mathbf{B}_{\underline{w}} := B_{i_1} \otimes \cdots \otimes B_{i_\ell}.$$

For a simple $M \in \mathfrak{B}_w$, we define $\mathrm{K}_{\underline{w}}(M) \in \mathbb{Z}^\ell$ as follows:

$$\begin{aligned} &M_\ell = M, \\ &c_k = \varepsilon^*_{i_k}(M_k) \quad \text{for } 1 \leqslant k \leqslant \ell, \\ &M_{k-1} = (\widetilde{\mathrm{E}}^*_{i_k})^{c_k}(M_k), \\ &\mathrm{K}_{\underline{w}}(M) = (c_1, \dots, c_\ell). \end{aligned}$$

We also regard $\mathrm{K}_{\underline{w}}(M)$ as an element of $\mathbf{B}_{\underline{w}}$ by $(c_1, \dots, c_\ell) \mapsto \widetilde{f}_{i_1}^{c_1}(0)_{i_1} \otimes \cdots \otimes \widetilde{f}_{i_\ell}^{c_\ell}(0)_{i_\ell} \in \mathbf{B}_{\underline{w}}$.

Then $\mathrm{K}_{\underline{w}}$ induces

$$\mathrm{Irr}(\mathscr{C}_w) \rightarrowtail \mathrm{Irr}(\mathfrak{B}_w) \rightarrowtail \mathbf{B}_{\underline{w}}.$$

Since by Proposition 6.24 (i) (see also [26, Sect.8]) we have

$$\mathrm{K}_{\underline{w}}(\mathsf{C}_\Lambda \nabla M) = \mathrm{K}_{\underline{w}}(\mathsf{C}_\Lambda) + \mathrm{K}_{\underline{w}}(M) \quad \text{for any } M \in \mathrm{Irr}(\mathscr{C}_w) \text{ and } \Lambda \in \mathsf{P}_+, \tag{6.4}$$

we can extend $\mathrm{Irr}(\mathscr{C}_w) \rightarrowtail \mathbf{B}_{\underline{w}}$ to $\mathrm{Irr}(\widetilde{\mathscr{C}}_w) \rightarrowtail \mathbf{B}_{\underline{w}}$ by

$$\widetilde{\mathsf{C}}_\Lambda^{-1} \circ \mathrm{Q}_w(M) \longmapsto \mathrm{K}_{\underline{w}}(M) - \mathrm{K}_{\underline{w}}(\mathsf{C}_\Lambda) \quad \text{for } \Lambda \in \mathsf{P}_+,$$

which we write by the same letter $\mathrm{K}_{\underline{w}}$. Then we obtain the commutative diagram:

$$\begin{tikzcd} \mathrm{Irr}(\mathscr{C}_w) \arrow[r, hook] \arrow[d, hook] & \mathrm{Irr}(\mathfrak{B}_w) \arrow[d, hook] \arrow[dl, "\mathrm{K}_{\underline{w}} \circ \mathrm{Q}_w"'] \\ \mathrm{Irr}(\widetilde{\mathscr{C}}_w) \arrow[r, "\mathrm{K}_{\underline{w}}"'] & \mathbf{B}_{\underline{w}}. \end{tikzcd}$$

By Proposition 4.16, we have that

$$\mathrm{K}_{\underline{w}}(\widetilde{\mathsf{C}}_\lambda) = \big\{-\big(\beta_k^\vee, w\lambda\big)\big\}_{1 \leqslant k \leqslant \ell} \quad \text{for any } \lambda \in \mathsf{P}.$$

The following lemma is obtained in [2] in the case when $\mathfrak{g}$ is semi-simple and w is the longest element.

Lemma 6.27. *For any $\lambda \in \mathsf{P}$, $i \in I$ and $b \in \mathbf{B}_{\underline{w}}$, we have*

(i) $\widetilde{f}_i\big(\mathrm{K}_{\underline{w}}(\widetilde{\mathsf{C}}_\lambda) + b\big) = \mathrm{K}_{\underline{w}}(\widetilde{\mathsf{C}}_\lambda) + \widetilde{f}_i(b)$,

(ii) $\varepsilon_i(\mathrm{K}_{\underline{w}}(\widetilde{\mathsf{C}}_\lambda)) = -\langle h_i, w\lambda\rangle$.

Proof. Set $\mathrm{K}_{\underline{w}}(\mathsf{C}_\lambda) = (c_1, \dots, c_\ell)$. Then we have $c_k = -(\beta_{i_k}^\vee, w\lambda)$. It is enough to show that we have $a_k = -\langle h_{i_k}, w\lambda\rangle$, where

$$a_k := c_k + \sum_{1 \leqslant s < k} \langle h_i, \alpha_{i_s}\rangle c_s = -(\beta_k^\vee, w\lambda) - \sum_{1 \leqslant s < k} \langle h_i, \alpha_{i_s}\rangle (\beta_s^\vee, w\lambda).$$

Indeed, we have seen in § 2.4 that

$$\varepsilon_i(\mathrm{K}_{\underline{w}}(\widetilde{\mathsf{C}}_\lambda)) = \max\{a_k \mid i_k = i\},$$

which implies (ii). Since

$$\tilde{m}_f^{(i)}(\mathrm{K}_{\underline{w}}(\widetilde{\mathsf{C}}_\lambda) + b) = \tilde{m}_f^{(i)}(b),$$

we obtain the assertion (i).

Set $i = i_k$, $A_k = \beta_k + \sum_{1 \leqslant s < k} (\alpha_i, \alpha_{i_s})\beta_s^\vee$. Then, we have $a_k = -(A_k, w\lambda)/\mathsf{d}_i$. Hence it is enough to show that $A_k = \alpha_i$, that is, A_k does not depend on k.

We have

$$\begin{aligned} A_k &= \beta_k + \sum_{1 \leqslant s < k} (\alpha_i, \alpha_{i_s}^\vee)\beta_s = \beta_k + \sum_{1 \leqslant s < k} w_{\leqslant s-1}(\alpha_i, \alpha_{i_s}^\vee)\alpha_s \\ &= \beta_k + \sum_{1 \leqslant s < k} w_{\leqslant s-1}(\alpha_i - s_{i_s}\alpha_i) = \beta_k + \sum_{1 \leqslant s < k} (w_{\leqslant s-1}\alpha_i - w_{\leqslant s}\alpha_i) \\ &= \beta_k + \alpha_i - w_{\leqslant k-1}\alpha_i = \alpha_i. \end{aligned}$$

□

Note that $\varepsilon_i(\mathsf{C}_\lambda) = \max(-\langle h_i, w\lambda\rangle, 0)$ for any $\lambda \in \mathsf{P}_+$ and $i \in I_w$, while we will see that $\varepsilon_i(\widetilde{\mathsf{C}}_\lambda) = -\langle h_i, w\lambda\rangle$.

The following proposition immediately follows from the definitions of $\mathbf{E}^*_{w',w}(X)$ and $\mathrm{K}_{\underline{w}}(X)$.

Proposition 6.28. *Let $X \in \mathrm{Irr}(\widetilde{\mathscr{C}_w})$. Set $X' = \mathbf{E}^*_{w',w}(X) \in \mathrm{Irr}(\widetilde{\mathscr{C}_{w'}})$, $\mathrm{K}_{\underline{w}}(X) = (c_1, \dots, c_\ell)$. Then, we have*

(i) $c_\ell = \varepsilon^*_{i_\ell}(X)$,
(ii) $\mathrm{K}_{\underline{w'}}(X') = (c_1, \dots, c_{\ell-1})$.

7. Main Theorem

Let $\underline{w} = s_{i_1} \cdots s_{i_\ell}$ (resp. $\underline{w'} = s_{i_1} \cdots s_{i_{\ell-1}}$) be a reduced expression of $w \in \mathsf{W}$ (resp. $w' := ws_{i_\ell} \in \mathsf{W}$). In this section, we shall prove the following Main Theorem 7.1, which says that $\mathrm{Irr}(\widetilde{\mathscr{C}_w}) \to \mathbf{B}_{\underline{w}}$ is a morphism of crystals.

Main Theorem 7.1. *For any simple $X \in \widetilde{\mathscr{C}}_w$ and $i \in I$, we have*

(i)
$$\varepsilon_i(X) = \begin{cases} \varepsilon_i(\mathbf{E}^*_{w',w}(X)) & \text{if } i \neq i_\ell, \\ \max\big(\varepsilon_i(\mathbf{E}^*_{w',w}(X)), -\varepsilon^*_{i_\ell}(X) - \langle h_i, \mathrm{wt}(X)\rangle\big) & \text{if } i = i_\ell. \end{cases}$$

(ii) $\mathbf{E}^*_{w',w}\big(\widetilde{\mathrm{F}}_i(X)\big) \simeq \widetilde{\mathrm{F}}_i\big(\mathbf{E}^*_{w',w}(X)\big)$ *if* $i \neq i_\ell$.

(iii) *if* $i = i_\ell$, *then*
$$\mathbf{E}^*_{w',w}\big(\widetilde{\mathrm{F}}_i(X)\big) \simeq \begin{cases} \widetilde{\mathrm{F}}_i\big(\mathbf{E}^*_{w',w}(X)\big) & \text{if } \mathfrak{d}_i(X) > 0 \text{ and } i \in I_{w'}, \\ \mathbf{E}^*_{w',w}(X) & \text{otherwise,} \end{cases}$$

(iv) $\varepsilon^*_{i_\ell}(\widetilde{\mathrm{F}}_i(X)) = \begin{cases} \varepsilon^*_{i_\ell}(X) + 1 & \text{if } i = i_\ell \text{ and } \mathfrak{d}_{i_\ell}(X) = 0, \\ \varepsilon^*_{i_\ell}(X) & \text{otherwise.} \end{cases}$

Note that

$$-\varepsilon^*_{i_\ell}(X) - \langle h_{i_\ell}, \mathrm{wt}(X)\rangle = \varepsilon^*_{i_\ell}(X) - \langle h_{i_\ell}, \mathrm{wt}(\mathbf{E}^*_{w',w}X)\rangle. \tag{7.1}$$

The proof of the main theorem will be given in the following section.

Main Theorem implies the following result.

Corollary 7.2. $\mathrm{K}_{\underline{w}} \colon \mathrm{Irr}(\widetilde{\mathscr{C}}_w) \to \mathbf{B}_{\underline{w}}$ *is a morphism of crystals.*

Proof. By induction on the length of w, it is enough to show that $\mathrm{Irr}(\widetilde{\mathscr{C}}_w) \to \mathrm{Irr}(\widetilde{\mathscr{C}}_{w'}) \otimes B_{i_\ell}$ is a morphism of crystals. It is evident $\mathrm{wt}(X) = \mathrm{wt}(\mathrm{K}_{\underline{w}}(X))$. Thus, by Main Theorem 7.1 (i), we have $\varepsilon_i(X) = \varepsilon_i(\mathrm{K}_{\underline{w}}(X))$ by (2.4) and (7.1), and also get $\varphi_i(X) = \varphi_i(\mathrm{K}_{\underline{w}}(X))$.

Next, let us see the compatibility of the actions by $\widetilde{f}_i$. By the definition, we have
$$\mathrm{K}_{\underline{w}}(X) = \mathrm{K}_{\underline{w}'}(\mathbf{E}^*_{w',w}(X)) \otimes \widetilde{f}_{i_\ell}^{\,\varepsilon^*_{i_\ell}(X)}(0)_{i_\ell}.$$
Then, if $i \neq i_\ell$, by induction on the length of w and Main Theorem 7.1 (ii), we obtain
$$\begin{aligned} \widetilde{f}_i\mathrm{K}_{\underline{w}}(X) &= \widetilde{f}_i\mathrm{K}_{\underline{w}'}(\big(\mathbf{E}^*_{w',w}(X)\big)) \otimes \widetilde{f}_{i_\ell}^{\,\varepsilon^*_{i_\ell}(X)}(0)_{i_\ell} = \mathrm{K}_{\underline{w}'}(\widetilde{\mathrm{F}}_i\big(\mathbf{E}^*_{w',w}(X)\big)) \otimes \widetilde{f}_{i_\ell}^{\,\varepsilon^*_{i_\ell}(X)}(0)_{i_\ell} \\ &= \mathrm{K}_{\underline{w}'}(\mathbf{E}^*_{w',w}(\widetilde{\mathrm{F}}_i(X))) \otimes \widetilde{f}_{i_\ell}^{\,\varepsilon^*_{i_\ell}(\widetilde{\mathrm{F}}_i(X))}(0)_{i_\ell} = \mathrm{K}_{\underline{w}}(\widetilde{\mathrm{F}}_i(X)), \end{aligned}$$
where $\varepsilon^*_{i_\ell}(\widetilde{\mathrm{F}}_i(X)) = \varepsilon^*_{i_\ell}(X)$ follows from Main Theorem 7.1 (iv) and $i \neq i_\ell$.

Assume that $i = i_\ell$. We have
$$\begin{aligned} \mathfrak{d}_{i_\ell}(X) &= \varepsilon_{i_\ell}(X) + \varepsilon^*_{i_\ell}(X) + \langle h_{i_\ell}, \mathrm{wt}(X)\rangle \\ &= \max\big(\varepsilon_{i_\ell}(\mathbf{E}^*_{w',w}(X)),\ -\varepsilon^*_{i_\ell}(X) - \langle h_{i_\ell}, \mathrm{wt}(X)\rangle\big) + \varepsilon^*_{i_\ell}(X) + \langle h_{i_\ell}, \mathrm{wt}(X)\rangle \\ &= \max\big(\varepsilon_{i_\ell}(\mathbf{E}^*_{w',w}(X)) - \varepsilon^*_{i_\ell}(X) + \langle h_{i_\ell},\ \mathrm{wt}(\mathbf{E}^*_{w',w}X)\rangle, 0\big). \end{aligned}$$

Hence $\mathfrak{d}_{i_\ell}(X) = 0$ if and only if $\varepsilon_{i_\ell}(\mathbf{E}^*_{w',w}(X)) \leqslant \varepsilon^*_{i_\ell}(X) - \langle h_{i_\ell}, \mathrm{wt}(\mathbf{E}^*_{w',w}X)\rangle$, which is equivalent to

$$\varphi_{i_\ell}(\mathrm{K}_{\underline{w'}}(\mathbf{E}^*_{w',w}(X)) - \varepsilon_{i_\ell}(\widetilde{f}_{i_\ell}^{\varepsilon^*_i(X)}(0)_{i_\ell}) = \varepsilon_{i_\ell}(\mathbf{E}^*_{w',w}(X)) - \varepsilon^*_{i_\ell}(X) + \langle h_{i_\ell}, \mathrm{wt}(\mathbf{E}^*_{w',w}X)\rangle \leqslant 0.$$

Therefore, under the condition $\mathfrak{d}_{i_\ell}(X) = 0$, (2.7) and Theorem 7.1 (iv) imply that

$$\begin{aligned}\widetilde{f}_{i_\ell}\mathrm{K}_{\underline{w}}(X) &= \widetilde{f}_{i_\ell}(\mathrm{K}_{\underline{w'}}(\mathbf{E}^*_{w',w}(X)) \otimes \widetilde{f}_{i_\ell}^{\varepsilon^*_{i_\ell}(X)}(0)_{i_\ell}) = \mathrm{K}_{\underline{w'}}(\mathbf{E}^*_{w',w}(X)) \otimes \widetilde{f}_{i_\ell}^{\varepsilon^*_{i_\ell}(X)+1}(0)_{i_\ell}\\ &= \mathrm{K}_{\underline{w'}}(\mathbf{E}^*_{w',w}(\widetilde{\mathrm{F}}_i(X))) \otimes \widetilde{f}_{i_\ell}^{\varepsilon^*_{i_\ell}(\widetilde{\mathrm{F}}_i(X))}(0)_{i_\ell} = \mathrm{K}_{\underline{w}}(\widetilde{\mathrm{F}}_{i_\ell}(X)).\end{aligned}$$

The condition $\mathfrak{d}_{i_\ell}(X) > 0$ is equivalent to the condition $\varphi_{i_\ell}\big(\mathrm{K}_{\underline{w'}}(\mathbf{E}^*_{w',w}(X)\big) > \varepsilon_{i_\ell}(\widetilde{f}_{i_\ell}^{\varepsilon^*_i(X)}(0)_{i_\ell})$. Hence, under the condition $\mathfrak{d}_{i_\ell}(X) > 0$, Main Theorem 7.1 (iv) implies that

$$\begin{aligned}\widetilde{f}_{i_\ell}\mathrm{K}_{\underline{w}}(X) &= \widetilde{f}_{i_\ell}(\mathrm{K}_{\underline{w'}}(\mathbf{E}^*_{w',w}(X)) \otimes \widetilde{f}_{i_\ell}^{\varepsilon^*_{i_\ell}(X)}(0)_{i_\ell}) = \widetilde{f}_{i_\ell}\mathrm{K}_{\underline{w'}}(\mathbf{E}^*_{w',w}(X)) \otimes \widetilde{f}_{i_\ell}^{\varepsilon^*_{i_\ell}(X)}(0)_{i_\ell}\\ &= \mathrm{K}_{\underline{w'}}(\mathbf{E}^*_{w',w}(\widetilde{\mathrm{F}}_{i_\ell}(X))) \otimes \widetilde{f}_{i_\ell}^{\varepsilon^*_{i_\ell}(\widetilde{\mathrm{F}}_i(X))}(0)_{i_\ell} = \mathrm{K}_{\underline{w}}(\widetilde{\mathrm{F}}_{i_\ell}(X)).\end{aligned}$$

Finally, since $\widetilde{\mathrm{E}}_i$ is an inverse of $\widetilde{\mathrm{F}}_i$ on $\mathrm{Irr}(\widetilde{\mathscr{C}}_w)$ and $\widetilde{e}_i$ is an inverse of $\widetilde{f}_i$ on $\mathbf{B}_{\underline{w}}$, $\mathrm{K}_{\underline{w}}\big(\widetilde{\mathrm{F}}_i(X)\big) = \widetilde{f}_i\mathrm{K}_{\underline{w}}(X)$ implies that $\mathrm{K}_{\underline{w}}\widetilde{\mathrm{E}}_i(X) = \widetilde{e}_i\mathrm{K}_{\underline{w}}(X)$. □

Corollary 7.3. *For any $i \in I_w$ and any simple $M \in \mathfrak{B}_w$, we have*

$$\varepsilon_i(M) = \max\big(\varepsilon_i\big(\mathrm{Q}_w(M)\big), 0\big).$$

8. Proof of Main Theorem

In this section, we give the proof of Main Theorem 7.1. Recall that $w' := ws_{i_\ell}$.

8.1. **Preparation.** Note that $\mathrm{Q}_w(\langle i_\ell\rangle)$ is a root object or invertible by Proposition 6.8.

In order to prove Main Theorem, we may assume that $i \in I_w$ without loss of generality. If $i \notin I_{w'}$, then $\varepsilon_i = 0$ and $\widetilde{\mathrm{F}}_i = 0$ on $\mathrm{Irr}(\widetilde{\mathscr{C}}_{w'})$ and $\mathrm{Q}_w(\langle i\rangle)$ is invertible by Lemma 6.9, and hence Main Theorem is obvious. Hence we may assume that

$$i \in I_{w'}. \tag{8.1}$$

First note that Main Theorem (iv) is an immediate consequence of Lemma 6.19 and Proposition 6.6.

For $\Lambda \in \mathsf{P}_+$, set

$$\begin{aligned}\mathsf{C}_\Lambda &:= \mathsf{M}(w\Lambda, \Lambda) \in \mathscr{C}_w,\\ \mathsf{C}'_\Lambda &:= \mathsf{C}^{w'}_\Lambda \in \mathscr{C}_{w'} \subset \mathscr{C}_w,\\ \mathsf{C}''_\Lambda &:= \widetilde{\mathrm{E}}^{*\,\max}_{i_\ell}\big(\mathsf{M}(w\Lambda, \Lambda)\big) \simeq \mathsf{M}(w\Lambda, s_{i_\ell}\Lambda) \in \mathfrak{B}_{w'} \cap \mathscr{C}_w \quad \text{and}\\ \widetilde{\mathsf{C}}_\Lambda &:= \mathrm{Q}_w(\mathsf{C}_\Lambda) \in \widetilde{\mathscr{C}}_w.\end{aligned}$$

Then, by Corollary 6.25 we have $\mathrm{Q}_{w'}(\mathsf{C}''_\Lambda) \simeq \widetilde{\mathsf{C}}^{w'}_{s_{i_\ell}\Lambda} \simeq \mathbf{E}^*_{w',w}(\widetilde{\mathsf{C}}_\Lambda)$.

By Lemma 5.23, we have

$$\varepsilon_i(\widetilde{\mathsf{C}}_\lambda) = \langle h_i, w\lambda\rangle = \varepsilon_i(\widetilde{\mathsf{C}}^{w'}_{s_{i_\ell}\lambda}) \quad \text{for any } \lambda \in \mathsf{P}.$$

Hence, we have

$$\varepsilon_i\big(\mathbf{E}^*_{w',w}(\widetilde{\mathsf{C}}_\lambda)\big) = \varepsilon_i(\widetilde{\mathsf{C}}_\lambda) \quad \text{for any } \lambda \in \mathsf{P}. \tag{8.2}$$

Proposition 8.1. *Let M be a simple module in $\mathscr{C}_w$ such that $\varepsilon^*_{i_\ell}(M) = 0$. Then we have*

$$\varepsilon_i\big(\mathrm{Q}_{w'}(M)\big) = \varepsilon_i\big(\mathrm{Q}_w(M)\big) \quad \textit{for any } i \in I_{w'}.$$

Proof. By Proposition 5.28, we can take $\Lambda \in \mathsf{P}_+$ such that $\mathsf{C}_\Lambda \nabla \langle i\rangle \in \mathscr{C}_w$ and $\langle h_{i_\ell}, \Lambda\rangle = 0$. Then, we have $\mathsf{C}_\Lambda = \mathsf{C}'_\Lambda \in \mathscr{C}_{w'} \subset \mathscr{C}_w$ and $\mathsf{C}_\Lambda \nabla \langle i\rangle \in \mathfrak{B}_{w'} \subset \mathfrak{B}_w$ by Proposition 5.20. Note that $\langle i\rangle \in \mathfrak{B}_{w'}$. Let S be a simple quotient of $(\mathsf{C}_\Lambda \nabla \langle i\rangle) \circ M$. Since $\mathsf{C}_\Lambda \nabla \langle i\rangle \in \mathscr{C}_w$ and $M \in \mathfrak{B}_w$, we have $S \in \mathfrak{B}_w$ by Lemma 5.17.

Let us first show

$$S \in \mathfrak{B}_{w'}. \tag{8.3}$$

If $\varepsilon^*_{i_\ell}(S) = 0$ then $S \in \mathfrak{B}_{w'}$ by Lemma 5.15.

Assume that $\varepsilon^*_{i_\ell}(S) > 0$. Since $\varepsilon^*_{i_\ell}(\mathsf{C}'_\Lambda \nabla \langle i\rangle) = \delta_{i,i_\ell}$ and $\varepsilon^*_{i_\ell}(M) = 0$, we have $\varepsilon^*_{i_\ell}(S) = 1$, $i = i_\ell$ and $\mathrm{E}^*_{i_\ell}\big((\mathsf{C}_\Lambda \nabla \langle i\rangle) \circ M\big) \simeq \mathsf{C}_\Lambda \circ M$ by Lemma 4.22. Applying E^*_i to $(\mathsf{C}_\Lambda \nabla \langle i\rangle) \circ M \twoheadrightarrow S$, we obtain

$$\mathsf{C}_\Lambda \circ M \xrightarrow{\ \sim\ } \mathrm{E}^*_{i_\ell}(S),$$

since $M \in \mathscr{C}_w$ implies that $\mathsf{C}_\Lambda \circ M$ is a simple module.

Hence we obtain

$$S \simeq (\mathsf{C}_\Lambda \circ M) \nabla \langle i_\ell\rangle,$$

which gives an epimorphism (note that $i = i_\ell$)

$$g\colon (\mathsf{C}_\Lambda \nabla \langle i\rangle) \circ M \twoheadrightarrow (\mathsf{C}_\Lambda \circ M) \nabla \langle i\rangle.$$

Applying Q_w to g, we obtain

$$\widetilde{\mathsf{C}}_\Lambda \circ \mathrm{Q}_w(\langle i\rangle) \circ \mathrm{Q}_w(M) \twoheadrightarrow \mathrm{Q}_w\big((\mathsf{C}_\Lambda \circ M) \nabla \langle i\rangle\big) \simeq \widetilde{\mathsf{C}}_\Lambda \circ \big(\mathrm{Q}_w(M) \nabla \mathrm{Q}_w(\langle i\rangle)\big).$$

Here the last isomorphism follows from Proposition 5.20 and $\mathsf{C}_\Lambda \circ M \in \mathscr{C}_w$, $\langle i\rangle \in \mathfrak{B}_w$ and the fact that $\mathrm{Q}_w(\langle i\rangle)$ is affreal.

Hence we have

$$\mathrm{Q}_w(\langle i\rangle) \nabla \mathrm{Q}_w(M) \simeq \mathrm{Q}_w(M) \nabla \mathrm{Q}_w(\langle i\rangle),$$

which implies that $\mathrm{Q}_w(\langle i\rangle)$ and $\mathrm{Q}_w(M)$ commute, and $\mathrm{Q}_w(\langle i\rangle)\circ\mathrm{Q}_w(M)$ is simple. Note that $\mathrm{Q}_w|_{\mathscr{C}_w}\colon \mathscr{C}_w\to\widetilde{\mathscr{C}_w}$ is fully faithful, and $\mathrm{Q}_w\big((\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle)\circ M\big)$ is simple. Since $\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle$ and M belong to $\mathscr{C}_w$, we conclude that $(\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle)\circ M$ is simple. Hence we have $S\simeq(\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle)\circ M$. Since $\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle$ and M belong to $\mathfrak{B}_{w'}$ and

$$\mathrm{Q}_{w'}(S)\simeq\mathrm{Q}_{w'}\big((\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle)\circ M\big)\simeq\mathrm{Q}_{w'}(\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle)\circ\mathrm{Q}_{w'}(M)$$

does not vanish, which means that $S\in\mathfrak{B}_{w'}$.

Thus we have obtained (8.3).

Let $f\colon(\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle)\circ M\to M\circ(\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle)$ be a morphism such that $\mathrm{Im}(f)\simeq S$ (see Lemma 4.5). Since S belongs to $\mathfrak{B}_{w'}\subset\mathfrak{B}_w$, Lemma 5.18 implies that

$$\Lambda\big(\mathrm{Q}_{w'}(\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle),\mathrm{Q}_{w'}(M)\big)=\deg(f)=\Lambda\big(\mathrm{Q}_{w}(\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle),\mathrm{Q}_{w}(M)\big).$$

Hence we obtain

$$\varepsilon_i\big(\mathrm{Q}_{w'}(M)\big)=\varepsilon_i\big(\mathrm{Q}_w(M)\big).$$

□

Remark 8.2. In the course of the proof of Proposition 8.1, we don't know if $\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle$ has an affinization. If we knew it, the proof could be much simpler. Conjecturally, any simple module in $\mathscr{C}_w$ has an affinization.

Now we divide the proof of Main Theorem into two cases: $i\neq i_\ell$ and $i=i_\ell$.

8.2. **Case $i\neq i_\ell$.** Assume that $i\in I_{w'}$ satisfies $i\neq i_\ell$.

Lemma 8.3. *For any simple $X\in\widetilde{\mathscr{C}_w}$, we have*

(i) $\varepsilon^*_{i_\ell}(\widetilde{\mathrm{F}}_iX)=\varepsilon^*_{i_\ell}(X)$,

(ii) $\mathbf{E}^*_{w',w}(\widetilde{\mathrm{F}}_i(X))\simeq\widetilde{\mathrm{F}}_i(\mathbf{E}^*_{w',w}(X))$.

Proof. The assertion (i) follows from Lemma 6.19 (ii).

(ii) We may assume that $X=\mathrm{Q}_w(M)$ for a simple $M\in\mathscr{C}_w$, since for any $X\in\widetilde{\mathscr{C}_w}$ there exists $\Lambda\in\mathsf{P}_+$ such that $\widetilde{C_\Lambda}\circ X\in\mathrm{Q}_w(\mathscr{C}_w)$ and we can reduce the assertion to $\widetilde{C_\Lambda}\circ X$.

Set $n:=\varepsilon^*_{i_\ell}(M)$ and $M':=\mathrm{E}^{*\,(n)}_{i_\ell}(M)$. Take $\Lambda\in\mathsf{P}_+$ such that $\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle\in\mathscr{C}_w$ and $\langle h_{i_\ell},\Lambda\rangle=0$ (see Proposition 5.28). Then Proposition 6.24 implies that $\varepsilon^*_{i_\ell}(\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle)=\varepsilon^*_{i_\ell}(\mathsf{C}_\Lambda)+\varepsilon^*_{i_\ell}(\langle i\rangle)=0$.

On the other hand (i) implies that

$$\varepsilon^*_{i_\ell}\big((\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle)\,\nabla\,M\big)=\varepsilon^*_{i_\ell}(\mathsf{C}_\Lambda)+\varepsilon^*_{i_\ell}(M)=\varepsilon^*_{i_\ell}(\mathsf{C}_\Lambda\,\nabla\,\langle i\rangle)+\varepsilon^*_{i_\ell}(M).$$

Hence Proposition 4.23 implies that

$$\begin{aligned}\widetilde{\mathrm{E}}_{i_\ell}^{*\,\max}\big((\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla M\big) &\simeq \widetilde{\mathrm{E}}_{i_\ell}^{*\,\max}(\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla (\widetilde{\mathrm{E}}_{i_\ell}^{*\,\max} M) \\ &\simeq (\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla M'.\end{aligned} \tag{8.4}$$

Since we have

$$\begin{aligned}\mathrm{Q}_{w'}\Big(\widetilde{\mathrm{E}}_{i_\ell}^{*\,\max}\big((\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla M\big)\Big) &\simeq \mathbf{E}^*_{w',w}(\widetilde{\mathsf{C}}_\Lambda \circ \widetilde{\mathrm{F}}_i X) \\ &\simeq \mathrm{Q}_{w'}(\mathsf{C}_\Lambda) \circ \mathbf{E}^*_{w',w}(\widetilde{\mathrm{F}}_i X)\end{aligned}$$

and

$$\mathrm{Q}_{w'}\big((\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla (\widetilde{\mathrm{E}}_{i_\ell}^{*\,\max} M)\big) \simeq \mathrm{Q}_{w'}(\mathsf{C}_\Lambda) \circ \widetilde{\mathrm{F}}_i\big(\mathbf{E}^*_{w',w}(X)\big),$$

we obtain

$$\mathbf{E}^*_{w',w}(\widetilde{\mathrm{F}}_i(X)) \simeq \widetilde{\mathrm{F}}_i(\mathbf{E}^*_{w',w}(X)).$$

□

The following lemma shows Main Theorem (i) when $i \neq i_\ell$.

Lemma 8.4. *For any simple $X \in \widetilde{\mathscr{C}_w}$ and any $i \in I_w$ such that $i \neq i_\ell$, we have*

$$\varepsilon_i(\mathbf{E}^*_{w',w}(X)) = \varepsilon_i(X).$$

Proof. We may assume that $X = \mathrm{Q}_w(M)$ for a simple $M \in \mathscr{C}_w$. Set $M' = \widetilde{\mathrm{E}}_{i_\ell}^{*\max}(M) \in \mathscr{C}_w \cap \mathfrak{B}_{w'}$. Since we have

$$\varepsilon_i(\mathbf{E}^*_{w',w}(X)) = \varepsilon_i\big(\mathrm{Q}_{w'}(M')\big) = \varepsilon_i\big(\mathrm{Q}_w(M')\big).$$

Here the last equality follows from Proposition 8.1. Set $n = \varepsilon^*_{i_\ell}(M)$. Then, since $\mathrm{Q}_w(M) \simeq \widetilde{\mathrm{F}}_i^{*n}\mathrm{Q}_w(M')$, we obtain

$$\varepsilon_i\big(\mathrm{Q}_w(M')\big) = \varepsilon_i\big(\mathrm{Q}_w(M)\big) = \varepsilon_i(X).$$

Here, the first equality follows from Lemma 6.19. □

Thus, we find that Main Theorem 7.1 holds for $i \neq i_\ell$.

8.3. **Case $i = i_\ell$.** Recall that

$$\mathfrak{d}_i(X) = \varepsilon_i(X) + \varepsilon_i^*(X) + \langle h_i, \mathrm{wt}(X)\rangle = \varepsilon_i(X) - \big(\varepsilon_i^*(X) - \langle h_i, \mathrm{wt}(\mathbf{E}^*_{w',w}X)\rangle\big), \tag{8.5}$$

where the last equality is from (7.1).

Lemma 8.5. *For any simple $X \in \widetilde{\mathscr{C}_w}$, we have*

$$\varepsilon_i\big(\mathbf{E}^*_{w',w}(X)\big) \leqslant \varepsilon_i(X).$$

Proof. By (8.2), we may assume that $X = \mathrm{Q}_w(M)$ for a simple $M \in \mathscr{C}_w$. Set $M' = \widetilde{\mathrm{E}}_i^{*\,\max}(M) \in \mathfrak{B}_{w'} \cap \mathscr{C}_w$. Then we have $\mathbf{E}^*_{w',w}(X) \simeq \mathrm{Q}_{w'}(M')$.

Let $m = \varepsilon_i^*(M)$. Then $M \simeq M' \nabla \langle i^m \rangle$. Since $M \in \mathscr{C}_w$, we obtain

$$\mathrm{Q}_w(M) \simeq \mathrm{Q}_w(M') \nabla \mathrm{Q}_w(\langle i^m \rangle) \simeq (\widetilde{\mathrm{F}}_i^*)^m(\mathrm{Q}_w(M')).$$

Since $\varepsilon_i\big(\widetilde{\mathrm{F}}_i^*(X)\big) = \varepsilon_i(X)$ or $\varepsilon_i(X)+1$ for any simple $X \in \mathscr{C}_w$ by Lemma 6.18, we obtain

$$\varepsilon_i\big(\mathrm{Q}_w(M')\big) \leqslant \varepsilon_i\big(\mathrm{Q}_w(M)\big).$$

Since Proposition 8.1 implies that $\varepsilon_i\big(\mathrm{Q}_{w'}(M')\big) = \varepsilon_i\big(\mathrm{Q}_w(M)\big)$, the assertion follows from $\varepsilon_i(\mathbf{E}^*_{w',w}X) = \varepsilon_i\big(\mathrm{Q}_{w'}(M')\big)$ and $\varepsilon_i(X) = \varepsilon_i\big(\mathrm{Q}_w(M)\big)$. □

8.3.1. Case: $\mathfrak{d}_i(X) = 0$. In this case, since $\mathrm{Q}_w(\langle i \rangle)$ and X commute with each other, we have $\widetilde{\mathrm{F}}_i(X) = \widetilde{\mathrm{F}}_i^*(X)$. Hence we obtain $\varepsilon_i^*(\widetilde{\mathrm{F}}_i X) = \varepsilon_i^*(\widetilde{\mathrm{F}}_i^* X) = \varepsilon_i^*(X)+1$, and Lemma 6.26 (iii) implies that $\mathbf{E}^*_{w',w}(\widetilde{\mathrm{F}}_i X) = \mathbf{E}^*_{w',w}(X)$. Main Theorem (i) follows from Lemma 8.5 and $\varepsilon_i(X) = -\varepsilon_{i_\ell}(X) - \langle h_i, \mathrm{wt}(X)\rangle$.

Thus Main Theorem holds in this case.

8.3.2. Case: $\mathfrak{d}_i(X) > 0$. In this case, $\mathrm{Q}_w(\langle i \rangle)$ is a root object by Proposition 6.8 and Lemma 6.1.

Let us show that

$$\mathbf{E}^*_{w',w}(\widetilde{\mathrm{F}}_i X) \simeq \widetilde{\mathrm{F}}_i\big(\mathbf{E}^*_{w',w}X\big) \quad \text{and} \quad \varepsilon_i(X) = \varepsilon_i(\mathbf{E}^*_{w',w}X). \tag{8.6}$$

They, along with (8.5), imply that

$$\varepsilon_i(\mathbf{E}^*_{w',w}X) - \varepsilon_i^*(X) + \langle h_i, \mathrm{wt}\,(\mathbf{E}^*_{w',w}\,X)\rangle = \mathfrak{d}_i(X) > 0,$$

and we obtain Main Theorem.

In order to see (8.6), we may assume that $X = \mathrm{Q}_w(M)$ for a simple $M \in \mathscr{C}_w$.

Take $\Lambda \in \mathsf{P}_+$ such that $\mathsf{C}_\Lambda \nabla \langle i \rangle \in \mathscr{C}_w$ and $\langle h_i, \Lambda \rangle = 0$. Then Proposition 5.7 implies that $(\mathsf{C}_\Lambda \nabla \langle i \rangle, M)$ is a Λ-definable pair. Then Main Theorem (iv) implies that

$$\varepsilon_i^*\big((\mathsf{C}_\Lambda \nabla \langle i \rangle) \nabla M\big) = \varepsilon_i^*(M) + \varepsilon_i^*(\mathsf{C}_\Lambda) = \varepsilon_i^*(M). \tag{8.7}$$

Set $m = \varepsilon_i^*(M)$.

Then we have an exact sequence by Lemma 4.22:

$$0 \longrightarrow \big(\mathsf{C}_\Lambda \nabla \langle i \rangle\big) \circ \mathrm{E}_i^{*\,(m)} M \longrightarrow \mathrm{E}_i^{*\,(m)}\big((\mathsf{C}_\Lambda \nabla \langle i \rangle) \circ M\big)$$
$$\longrightarrow \mathrm{E}_i^*\big(\mathsf{C}_\Lambda \nabla \langle i \rangle\big) \circ \mathrm{E}_i^{*\,(m-1)} M \longrightarrow 0.$$

Note that $\mathrm{E}_i^*\big(\mathsf{C}_\Lambda \nabla \langle i\rangle\big) \simeq \mathsf{C}_\Lambda$. If the composition

$$f\colon \big(\mathsf{C}_\Lambda \nabla \langle i\rangle\big) \circ \mathrm{E}_i^{*\,(m)} M \to \mathrm{E}_i^{*\,(m)}\big((\mathsf{C}_\Lambda \nabla \langle i\rangle) \circ M\big) \twoheadrightarrow \mathrm{E}_i^{*\,(m)}\big((\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla M\big)$$

vanishes, then we have an epimorphism

$$\begin{aligned}\mathsf{C}_\Lambda \circ \mathrm{E}_i^{*\,(m-1)} M &\simeq \mathrm{E}_i^*\big(\mathsf{C}_\Lambda \nabla \langle i\rangle\big) \circ \mathrm{E}_i^{*\,(m-1)} M \\ &\twoheadrightarrow \mathrm{E}_i^{*\,(m)}(\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla M\big) \simeq \widetilde{\mathrm{E}}_i^{*\,\max}\big((\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla M\big),\end{aligned}$$

where the last isomorphism follows from (8.7).

Hence it induces a non-zero $R(\beta)$-module homomorphism:

$$\mathrm{E}_i^{*\,(m-1)} M \to Y := \mathrm{H\small{OM}}\Big(\mathsf{C}''_\Lambda \circ R(\beta), \widetilde{\mathrm{E}}_i^{*\,\max}\big((\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla M\big)\Big),$$

where $\beta = -\mathrm{wt}\big(\mathrm{E}_i^{*\,(m-1)} M\big) \in \mathsf{Q}_+$. Since $\mathrm{E}_i^{*\,(m-1)} M$ has a simple head $(\widetilde{\mathrm{E}}_i^*)^{m-1}(M)$ with $\varepsilon_i^*\big((\widetilde{\mathrm{E}}_i^*)^{m-1}(M)\big) = 1$, and $\mathrm{E}_i^*(Y) = 0$, it is a contradiction.

Hence f does not vanish and we obtain

$$(\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla \mathrm{E}_i^{*\,(m)} M \simeq \widetilde{\mathrm{E}}_i^{*\,\max}\big((\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla M\big). \tag{8.8}$$

By(8.8), Proposition 6.24 and Corollary 6.25, we have

$$\begin{aligned}\mathbf{E}^*_{w',w}\big(\mathrm{Q}_w((\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla M)\big) &\simeq \mathrm{Q}_{w'}\Big(\widetilde{\mathrm{E}}_i^{*\,\max}\big((\mathsf{C}_\Lambda \nabla \langle i\rangle) \nabla M\big)\Big) \\ &\simeq \mathrm{hd}\Big(\mathrm{Q}_{w'}\big((\mathsf{C}_\Lambda \nabla \langle i\rangle) \circ \mathrm{E}_i^{*\,(m)} M\big)\Big) \\ &\simeq \mathrm{Q}_{w'}\big(\mathsf{C}_\Lambda \nabla \langle i\rangle\big) \nabla \mathbf{E}^*_{w',w}(\mathrm{Q}_w M).\end{aligned} \tag{8.9}$$

Hence, we obtain $\mathbf{E}^*_{w',w}(\widetilde{\mathrm{F}}_i X) \simeq \widetilde{\mathrm{F}}_i(\mathbf{E}^*_{w',w} X)$, which shows Main Theorem 7.1 (ii).

It remains to prove that

$$\varepsilon_i(X) = \varepsilon_i(\mathbf{E}^*_{w',w} X). \tag{8.10}$$

The proof of (8.10) is similar to the one of Lemma 8.5. We may assume that $i \in I_{w'}$. By (6.2) and (8.2), we may assume that $X = \mathrm{Q}_w(M)$ for a simple $M \in \mathscr{C}_w$. Set $M' = \widetilde{\mathrm{E}}_i^{*\,\max}(M) \in \mathfrak{B}_{w'} \cap \mathscr{C}_w$. Then we have $\mathbf{E}^*_{w',w}(X) \simeq \mathrm{Q}_{w'}(M')$ by Corollary 6.25. Let $m = \varepsilon_i^*(M)$. Set $X' = \mathrm{Q}_w(M')$. Then, we have $X' = \widetilde{\mathrm{E}}_i^{*\,m}(X)$. Since $\varepsilon_i(\mathbf{E}^*_{w',w}(X)) = \varepsilon_i(X')$ by Proposition 8.1 and $\varepsilon_i(X') = \varepsilon_i(X)$ follows from Lemma 6.18 (i) and $\mathfrak{d}_i(X) > 0$. Thus we obtain $\varepsilon_i\big(\mathbf{E}^*_{w',w}(X)\big) = \varepsilon_i(X)$.

Now, the proof of Main Theorem 7.1 is complete.

9. Bijectivity and connectedness

9.1. **Bijectivity of $\mathrm{K}_{\underline{w}}$.** Let $\underline{w} = s_{i_1} \cdots s_{i_\ell}$ be a reduced expression of $w \in \mathsf{W}$.

In this section, we shall prove the following proposition.

Proposition 9.1. *$\mathrm{K}_{\underline{w}} \colon \mathrm{Irr}(\widetilde{\mathscr{C}_w}) \to \mathbf{B}_{\underline{w}}$ is bijective.*

By induction on the length of w and Proposition 6.28, we can reduce this proposition to the following proposition.

Proposition 9.2. *Let $w \in \mathsf{W}$ and $i \in I$ such that $w' := ws_i < w$. Then the map*

$$\mathrm{K}'_{w',w} \colon \mathrm{Irr}(\widetilde{\mathscr{C}_w}) \to \mathrm{Irr}(\widetilde{\mathscr{C}_{w'}}) \times \mathbb{Z} \quad \textit{given by } X \mapsto \big(\mathbf{E}^*_{w',w}(X), \varepsilon^*_i(X)\big),$$

is bijective

Proof. Injectivity is obvious. Let us prove the surjectivity.

We have $\mathbf{E}^*_{w',w}\big((\widetilde{\mathrm{F}}^*_i)^n Z\big) \simeq \mathbf{E}^*_{w',w}(Z)$ by Lemma 6.26 and $\varepsilon^*_i\big((\widetilde{\mathrm{F}}^*_i)^n Z\big) = \varepsilon^*_i(Z) + n$ for any $n \in \mathbb{Z}$ and any simple $Z \in \widetilde{\mathscr{C}_w}$. Here, if $n < 0$, then $(\widetilde{\mathrm{F}}^*_i)^n$ means $(\widetilde{\mathrm{E}}^*_i)^{-n}$ by Proposition 6.12. Hence, it is enough to show that

$$\mathbf{E}^*_{w',w} \colon \mathrm{Irr}(\widetilde{\mathscr{C}_w}) \to \mathrm{Irr}(\widetilde{\mathscr{C}_{w'}}) \text{ is surjective.}$$

In order to prove this, let us recall results in [14, Theorem 4.8]):

(i) there exists a quasi-commutative diagram:

$$\begin{tikzcd} & \mathscr{C}_w \arrow[dl, "\mathrm{Q}_{w'}|_{\mathscr{C}_w}"'] \arrow[dr, "\mathrm{Q}^{\mathrm{r}}_{w,s_i}"] & \\ \widetilde{\mathscr{C}_{w'}} \arrow[rr, "\sim", "\Phi"'] & & \widetilde{\mathscr{C}_{w,s_i}} \end{tikzcd}$$

where $\mathrm{Q}^{\mathrm{r}}_{w,s_i} \colon \mathscr{C}_w \to \widetilde{\mathscr{C}_{w,s_i}}$ is the localization functor by the commuting family of right braiders $\{\mathsf{M}(w\Lambda, s_i\Lambda) \mid \Lambda \in \mathsf{P}_+\}$ in $\mathscr{C}_w$, and $\Phi \colon \widetilde{\mathscr{C}_{w'}} \xrightarrow{\sim} \widetilde{\mathscr{C}_{w,s_i}}$ is an equivalence of monoidal categories,

(ii) for any simple $Z \in \mathscr{C}_w$ such that $\mathrm{Q}^{\mathrm{r}}_{w,s_i}(Z) \not\simeq 0$, there exists $\Lambda \in \mathsf{P}_+$ such that
   (a) $Z \nabla \mathsf{M}(w\Lambda, s_i\Lambda) \in \mathscr{C}_{w,s_i}$,
   (b) $\mathrm{Q}^{\mathrm{r}}_{w,s_i}\big(Z \circ \mathsf{M}(w\Lambda, s_i\Lambda)\big) \to \mathrm{Q}^{\mathrm{r}}_{w,s_i}\big(Z \nabla \mathsf{M}(w\Lambda, s_i\Lambda)\big) \to \mathrm{Q}^{\mathrm{r}}_{w,s_i}\big(\mathsf{M}(w\Lambda, s_i\Lambda) \circ Z\big)$ are isomorphisms.

Note that $\mathscr{C}_{w,s_i}$ is the full subcategory of $\mathscr{C}_w$ consisting of objects $M \in \mathscr{C}_w$ such that $\mathrm{E}^*_i(M) \simeq 0$.

Now let us show the surjectivity of $\mathbf{E}^*_{w',w}\colon \mathrm{Irr}(\widetilde{\mathscr{C}}_w) \to \mathrm{Irr}(\widetilde{\mathscr{C}}_{w'})$. Recall $\widetilde{\mathsf{C}}^{w'}_{\lambda}$ as in (5.6). Since Proposition 6.23 implies that

$$\mathbf{E}^*_{w',w}(\mathrm{Q}_w(M) \circ \widetilde{\mathsf{C}}_{s_i\lambda}) \simeq \mathbf{E}^*_{w',w}(\mathrm{Q}_w(M)) \circ \widetilde{\mathsf{C}}^{w'}_{\lambda}$$

for any simple $M \in \mathscr{C}_w$ and $\lambda \in \mathsf{P}$, it is enough to show that

(9.1) for any simple $M \in \mathscr{C}_{w'}$, there exists $X \in \widetilde{\mathscr{C}}_w$ such that $\mathbf{E}^*_{w',w}(X) \simeq \mathrm{Q}_{w'}(M)$.

Now, since $\mathrm{Q}_{w'}(M)$ is simple, $\mathrm{Q}^{\mathrm{r}}_{w,s_i}(M)$ is simple by (i). Hence, (ii) implies that there exists $\Lambda \in \mathsf{P}_+$ such that $N{:=}M \nabla \mathsf{M}(w\Lambda, s_i\Lambda) \in \mathscr{C}_{w,s_i}$ and $\mathrm{Q}^{\mathrm{r}}_{w,s_i}(N) \xrightarrow{\sim} \mathrm{Q}^{\mathrm{r}}_{w,s_i}(\mathsf{M}(w\Lambda, s_i\Lambda) \circ M)$. Hence, we have

$$\mathrm{Q}_{w'}(N) \simeq \mathrm{Q}_{w'}\big(\mathsf{M}(w\Lambda, s_i\Lambda) \circ M\big) \simeq \widetilde{\mathsf{C}}^{w'}_{s_i\Lambda} \circ \mathrm{Q}_{w'}(M)$$

by (ii). Hence, we obtain

$$\mathrm{Q}_{w'}(M) \simeq \mathrm{Q}_{w'}(N) \circ \widetilde{\mathsf{C}}^{w'}_{-s_i\Lambda} \simeq \mathbf{E}^*_{w',w}\big(\mathrm{Q}_w(N) \circ \widetilde{\mathsf{C}}_{-\Lambda}\big).$$

Thus we obtain (9.1). □

Therefore, by Main Theorem 7.1 and Proposition 9.1 we obtain

Main Theorem 9.3. $\mathrm{K}_{\underline{w}}\colon \mathrm{Irr}(\widetilde{\mathscr{C}}_w) \to \mathbf{B}_{\underline{w}}$ *is an isomorphism of crystals.*

9.2. **Connectedness of $\mathbf{B}_{\underline{w}}$.** As an application of Main Theorem 9.3 in previous subsections, let us show the connectedness of the crystal $\mathbf{B}_{\underline{w}}$ as a crystal graph (Definition 2.2).

Theorem 9.4. *The crystal $\mathbf{B}_{\underline{w}}$ is connected.*

Note that in [2], it is shown that $\mathbf{B}_{\underline{w}}$ is connected for a semi-simple $\mathfrak{g}$.

Proof. Since $\mathbf{B}_{\underline{w}} \simeq \mathrm{Irr}(\widetilde{\mathscr{C}}_w)$, we shall show that $\mathrm{Irr}(\widetilde{\mathscr{C}}_w)$ is connected.

Since by Theorem 5.9, we have the crystal isomorphism

$$\psi_*\colon \Big(\mathrm{Irr}(\widetilde{\mathscr{C}}_w), \Big\{\widetilde{\mathrm{E}}_i, \widetilde{\mathrm{F}}_i\Big\}_{i\in I}\Big) \simeq \Big(\mathrm{Irr}(\widetilde{\mathscr{C}}_{w^{-1}}, \Big\{\widetilde{\mathrm{E}}^*_i, \widetilde{\mathrm{F}}^*_i\Big\}_{i\in I}\Big)\Big),$$

the assertion is a consequence of the following lemma. □

Lemma 9.5. *The crystal $\mathrm{Irr}(\widetilde{\mathscr{C}}_w)$ with respect to the crystal operators $\{\widetilde{\mathrm{F}}^*_i, \widetilde{\mathrm{E}}^*_i\}_{i\in I}$ is connected.*

Proof. Since as we have seen in Lemma 5.12 that $\mathrm{Irr}(\mathscr{C}_w) \sqcup \{0\}$ is stable by $\widetilde{\mathrm{E}}_i^*$, it follows from Lemma 6.15 and $\widetilde{\mathrm{E}}_{i_1}^{*\max} \cdots \widetilde{\mathrm{E}}_{i_\ell}^{*\max} M \simeq \mathbf{1}$ that $\mathrm{Q}_w(M)$ is connected to $\mathbf{1}$ for any $M \in \mathrm{Irr}(\mathscr{C}_w)$. Since $\widetilde{\mathrm{E}}_i^*(\widetilde{\mathsf{C}}_\lambda \circ X) \simeq \widetilde{\mathsf{C}}_\lambda \circ \widetilde{\mathrm{E}}_i^*(X)$ for any $X \in \mathrm{Irr}(\widetilde{\mathscr{C}_w})$ and $\lambda \in \mathsf{P}$, an object $\widetilde{\mathsf{C}}_{-\Lambda} \circ \mathrm{Q}_w(M) \in \mathrm{Irr}(\widetilde{\mathscr{C}_w})$ ($\Lambda \in \mathsf{P}_+$, $M \in \mathrm{Irr}(\mathscr{C}_w)$) is connected to $\widetilde{\mathsf{C}}_{-\Lambda} \circ \mathbf{1}$, which is connected with $\widetilde{\mathsf{C}}_{-\Lambda} \circ \widetilde{\mathsf{C}}_\Lambda \simeq \mathbf{1}$ by applying the above argument to $\widetilde{\mathsf{C}}_\Lambda = \mathrm{Q}_w(\mathsf{C}_\Lambda) \in \mathrm{Q}_w(\mathrm{Irr}(\mathscr{C}_w))$ (see (5.3)). □

Here, we give an example of $\widetilde{\mathscr{C}_w}$, as a monoidal categorification of a cluster algebra:

Example 9.6 (cf. [6])**.** Let $\mathfrak{g} = A_3$, $w = s_2 w_0 = s_1 s_2 s_3 s_2 s_1$, $\mathscr{C}_w = \{M \in R\text{-gmod} \mid \mathrm{E}_2 M \simeq 0\}$. $\mathsf{C}_{\Lambda_1} = \langle 321 \rangle$, $\mathsf{C}_{\Lambda_2} = \langle 132 \rangle$, $\mathsf{C}_{\Lambda_3} = \langle 123 \rangle$ are frozen variables. In this case, $\widetilde{\mathscr{C}_w}$ is a monoidal categorification of the cluster algebra with four clusters:

$$\begin{tikzcd} \langle 1 \rangle \arrow[r, dash, "C_4"] \arrow[d, dash, "C_1"'] & \langle 12 \rangle \arrow[d, dash, "C_3"] \\ \langle 3 \rangle \arrow[r, dash, "C_2"] & \langle 32 \rangle, \end{tikzcd}$$

$$\mathrm{K}_{\underline{w}}(\langle 1 \rangle^x \circ \langle 3 \rangle^y \circ \langle 321 \rangle^a \circ \langle 132 \rangle^b \circ \langle 123 \rangle^c) = (c+b, c, y+a+b+c, a+b, x+a),$$

$$\mathrm{K}_{\underline{w}}(\langle 3 \rangle^y \circ \langle 32 \rangle^x \circ \langle 321 \rangle^a \circ \langle 132 \rangle^b \circ \langle 123 \rangle^c) = (c+b, c, y+x+a+b+c, x+a+b, a),$$

$$\mathrm{K}_{\underline{w}}(\langle 32 \rangle^x \circ \langle 12 \rangle^y \circ \langle 321 \rangle^a \circ \langle 132 \rangle^b \circ \langle 123 \rangle^c) = (y+c+b, c, x+a+b+c, x+y+a+b, a),$$

$$\mathrm{K}_{\underline{w}}(\langle 12 \rangle^y \circ \langle 1 \rangle^x \circ \langle 321 \rangle^a \circ \langle 132 \rangle^b \circ \langle 123 \rangle^c) = (y+c+b, c, a+b+c, y+a+b, a+x).$$

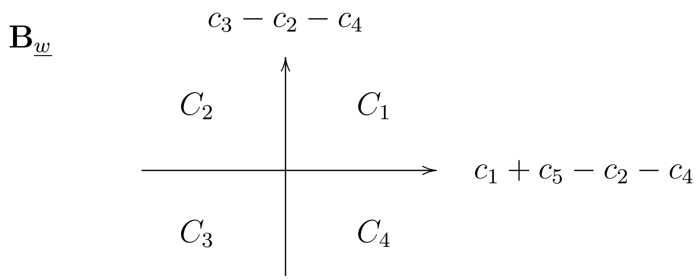
Note that the frozen variables correspond to the following elements in $\mathbf{B}_{\underline{w}}$:

$$\begin{aligned}
\mathsf{C}_{\Lambda_1} &= \langle 321 \rangle = \widetilde{\mathrm{F}}_3 \widetilde{\mathrm{F}}_2 \widetilde{\mathrm{F}}_1 \mathbf{1} \longleftrightarrow \tilde{f}_3 \tilde{f}_2 \tilde{f}_1 (0,0,0,0,0) = (0,0,1,1,1),\\
\mathsf{C}_{\Lambda_2} &= \langle 132 \rangle = \widetilde{\mathrm{F}}_1 \widetilde{\mathrm{F}}_3 \widetilde{\mathrm{F}}_2 \mathbf{1} \longleftrightarrow \tilde{f}_1 \tilde{f}_3 \tilde{f}_2 (0,0,0,0,0) = (1,0,1,1,0),\\
\mathsf{C}_{\Lambda_3} &= \langle 123 \rangle = \widetilde{\mathrm{F}}_1 \widetilde{\mathrm{F}}_2 \widetilde{\mathrm{F}}_3 \mathbf{1} \longleftrightarrow \tilde{f}_1 \tilde{f}_2 \tilde{f}_3 (0,0,0,0,0) = (1,1,1,0,0).
\end{aligned}$$

Here $x, y \in \mathbb{Z}_{\geqslant 0}$ and $a, b, c \in \mathbb{Z}$. Each cluster C_k ($1 \leqslant k \leqslant 4$) consists of a commuting family of simple modules.

Hence, the images of the four clusters are given as follows:

$$\begin{aligned}
\mathrm{K}_{\underline{w}}(C_1) &= \{(c_1, c_2, c_3, c_4, c_5) \in \mathbf{B}_{\underline{w}} \mid c_3 \geqslant c_2 + c_4,\ \ c_1 + c_5 \geqslant c_2 + c_4\},\\
\mathrm{K}_{\underline{w}}(C_2) &= \{(c_1, c_2, c_3, c_4, c_5) \in \mathbf{B}_{\underline{w}} \mid c_3 \geqslant c_2 + c_4,\ \ c_1 + c_5 \leqslant c_2 + c_4\},\\
\mathrm{K}_{\underline{w}}(C_3) &= \{(c_1, c_2, c_3, c_4, c_5) \in \mathbf{B}_{\underline{w}} \mid c_3 \leqslant c_2 + c_4,\ \ c_1 + c_5 \leqslant c_2 + c_4\},\\
\mathrm{K}_{\underline{w}}(C_4) &= \{(c_1, c_2, c_3, c_4, c_5) \in \mathbf{B}_{\underline{w}} \mid c_3 \leqslant c_2 + c_4,\ \ c_1 + c_5 \geqslant c_2 + c_4\}.
\end{aligned}$$



References

1. W.-S. Jung and E. Park, *Crystals and quantum twist automorphisms*, arXiv:2507.01306 v1.
2. Y. Kanakubo and T. Nakashima, *Half potential on geometric crystals and connectedness of cellular crystals*, Transformation Groups, **28** no.1, (2023), 327–373.
3. S.-J. Kang, M. Kashiwara, K.C. Misra, T. Miwa, T. Nakashima and A. Nakayashiki, *Perfect crystals of quantum affine Lie algebras*, Duke Math. J., **68**, (1992), 499–607.
4. ———, Affine crystals and vertex models, Int. J. Mod. Phys. A, **7**, Suppl. 1A (1992), 449–484.
5. S.-J. Kang, M. Kashiwara, M. Kim, and S.-j. Oh, *Simplicity of heads and socles of tensor products*, Compos. Math. **151** no. 2, (2015), 377–396.
6. ———, *Monoidal categorification of cluster algebras*, J. Amer. Math. Soc. **31** no. 2, (2018), 349–426.
7. M. Kashiwara, *On crystal bases of the q-analogue of universal enveloping algebras*, Duke Math. J., **63** (2), (1991), 465–516.
8. ———, *The crystal base and Littelmann's refined Demazure character formula*, Duke Math., **71** (1993), no. 3, 839–858.
9. M. Kashiwara, and M. Kim, *Laurent phenomenon and simple modules of quiver Hecke algebras*, Compos. Math. **155** no. 12, (2019), 2263–2295.
10. M. Kashiwara, M. Kim, S.-j. Oh, and E. Park, *Monoidal categories associated with strata of flag manifolds*, Adv. Math. **328** (2018), 959–1009.
11. ———, *Localizations for quiver Hecke algebras*, Pure Appl. Math. Q. **17** no. 4, (2021), 1465–1548.
12. ———, *Localizations for quiver Hecke algebras II*, Proc. Lond. Math. Soc. (3) **127**, no. 4, (2023), 1134–1184.
13. ———, *PBW theory for quantum affine algebra*, J. Eur. Math. Soc. **26** no. 7, (2024), 2679–2743.
14. ———, *Localizations for quiver Hecke algebras III*, arXiv:2308.09243v1.
15. ———, *Affinizations, R-matrices and reflection functors*, arXiv:2304.00238v3.
16. ———, *Reflection functors on quiver Hecke algebras*, arXiv:2506.12681v1.
17. M. Kashiwara and T. Nakashima, *Crystal graph for representations of the q-analogue of classical Lie algebras*, J. Algebra **165**, no. 2, (1994), 295–345.
18. M. Kashiwara and E. Park, *Affinizations and R-matrices for quiver Hecke algebras*, J. Eur. Math. Soc. **20**, (2024), 1161–1193.
19. M. Kashiwara and Y. Saito, *Geometric construction of crystal bases*, Duke Math.J., **89** no.1, (2003), 9–36.

20. M. Kashiwara and P. Schapira, *Categories and Sheaves*, Grundlehren der Mathematischen Wissenschaften, vol. **332**, Springer-Verlag, Berlin, (2006).
21. M. Khovanov and A. Lauda, *A diagrammatic approach to categorification of quantum groups I*, Represent. Theory **13** (2009), 309–347.
22. Y. Kimura and H. Oya, *Twist automorphisms on quantum unipotent cells and dual canonical bases*, Int. Math. Res. Not. no.9, (2021), 6772–6847.
23. A, Kleshchev and A, Ram, *Homogeneous representations of Khovanov-Lauda algebras*, J. Eur. Math. Soc. **12** no. 5, (2010), 1293–1306.
24. A. Lauda and M. Vazirani, *Crystals from categorified quantum groups*, Adv. Math. **228** no. 2, (2011), 803–861.
25. T. Nakashima, *Geometric crystals on Schubert varieties*, J. Geometry and Physics, **53** (2), (2005), 197–225.
26. ———, *Categorified crystal structure on localized quantum coordinate rings*, arXiv:2208.08396v2.
27. ———, *Polytopes for crystallized Demazure modules and extremal vectors*, Communications in Algebra, **30** (3), (2002), 1349–1367.
28. ———, *Polyhedral realizations of crystal bases and braid-type isomorphisms*, Contemporary Mathematics **248**, (1999), 419–435.
29. P. Tingley and B. Webster, *Mirković-Vilonen polytopes and Khovanov-Lauda-Rouquier algebras*, Compos. Math. 152 no. 8, (2016), 1648–1696.

(M. Kashiwara) Kyoto University Institute for Advanced Study, Research Institute for Mathematical Sciences, Kyoto University, Kyoto 606-8502, Japan

Email address: masaki@kurims.kyoto-u.ac.jp

(T.Nakashima) Division of Mathematics, Sophia University, Kioicho 7-1, Chiyoda-ku, Tokyo 102-8554, Japan

Email address: toshiki@sophia.ac.jp